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(54) **TRANSMISSIVE METASURFACE LENS
INTEGRATION**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

3,877,034 A 4/1975 Nelson
4,777,116 A 10/1988 Kawatsuki et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CA 3006173 A1 6/2017
CA 3020261 A1 10/2017
(Continued)

OTHER PUBLICATIONS

Arbabi et al., Miniature optical planar camera based on a wide-angle metasurface doublet corrected for monochromatic aberrations. *Nat Commun.* Nov. 28, 2016;7:13682. doi: 10.1038/ncomms13682. PMID: 27892454; PMCID: PMC5133709. (Year: 2016).*

(Continued)

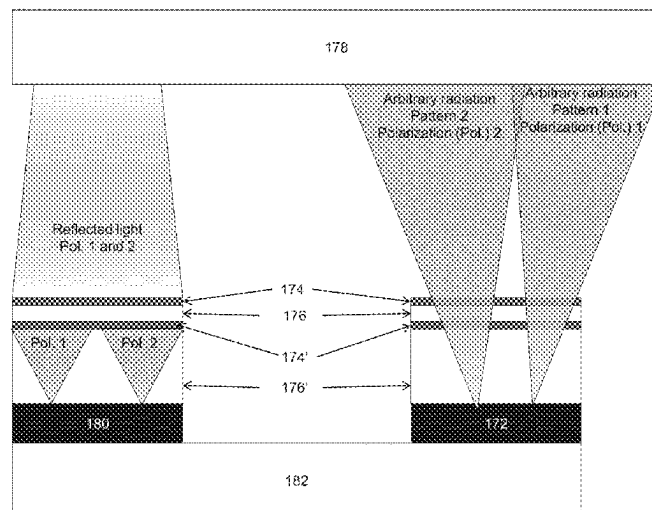
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(57) **ABSTRACT**

Metasurface elements, integrated systems incorporating such metasurface elements with light sources and/or detectors, and methods of the manufacture and operation of such optical arrangements and integrated systems are provided. Systems and methods for integrating transmissive metasurfaces with other semiconductor devices or additional metasurface elements, and more particularly to the integration of such metasurfaces with substrates, illumination sources and sensors are also provided. The metasurface elements provided may be used to shape output light from an illumination source or collect light reflected from a scene to form two unique patterns using the polarization of light. In such embodiments, shaped-emission and collection may be combined into a single co-designed probing and sensing optical system.

16 Claims, 29 Drawing Sheets



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(56)

References Cited

U.S. PATENT DOCUMENTS

4,856,899 A 8/1989 Iwaoka et al.
 5,085,496 A 2/1992 Yoshida et al.
 5,245,466 A 9/1993 Burns et al.
 5,337,146 A 8/1994 Azzam
 5,452,126 A 9/1995 Johnson
 5,620,792 A 4/1997 Challener, IV
 6,097,856 A 8/2000 Hammond, Jr.
 6,731,839 B2 5/2004 Bhagavatula et al.
 6,825,986 B2 11/2004 Ashkinazy et al.
 6,834,027 B1 12/2004 Sakaguchi et al.
 6,924,457 B2 8/2005 Koyama et al.
 6,927,922 B2 8/2005 George et al.
 6,958,207 B1 10/2005 Khusnatdinov et al.
 7,057,151 B2 6/2006 Lezec et al.
 7,061,612 B2 6/2006 Johnston
 7,061,693 B2 6/2006 Zalevsky
 7,171,078 B2 1/2007 Sasaki et al.
 7,171,084 B2 1/2007 Izumi et al.
 7,186,969 B2 3/2007 Altendorf et al.
 7,241,988 B2 7/2007 Gruber et al.
 7,324,210 B2 1/2008 De et al.
 7,327,468 B2* 2/2008 Maznev G01N 21/1717

7,402,131 B2 7/2008 Mueth et al.
 7,450,618 B2 11/2008 Dantus et al.
 7,547,874 B2 6/2009 Liang
 7,561,264 B2 7/2009 Treado et al.
 7,576,899 B2 8/2009 Kanesaka et al.
 7,679,830 B2 3/2010 Dowski, Jr.
 7,684,097 B2* 3/2010 Fukumoto G11B 7/0065
 359/21
 7,773,307 B2 8/2010 Shih
 7,800,683 B2 9/2010 Zalevsky et al.
 7,812,295 B2 10/2010 Zalevsky et al.
 7,928,900 B2 4/2011 Fuller et al.
 7,929,220 B2 4/2011 Sayag
 7,965,607 B2 6/2011 Fukumoto et al.
 8,009,358 B2 8/2011 Zalevsky et al.
 8,040,604 B2 10/2011 Zalevsky et al.
 8,107,705 B2 1/2012 Dowski, Jr. et al.
 8,152,307 B2 4/2012 Duelli et al.
 8,169,703 B1 5/2012 Mossberg et al.
 8,192,022 B2 6/2012 Zalevsky
 8,212,866 B2 7/2012 Lemmer et al.
 8,318,386 B2 11/2012 Kobrin
 8,328,396 B2 12/2012 Capasso et al.
 8,351,048 B2 1/2013 Millerd
 8,351,120 B2 1/2013 Deng et al.
 8,390,932 B2 3/2013 Jia et al.
 8,400,494 B2* 3/2013 Zalevsky H10B 41/35
 348/46
 8,430,513 B2 4/2013 Chang et al.
 8,451,368 B2 5/2013 Sung et al.
 8,472,797 B2 6/2013 Ok et al.
 8,481,948 B2 7/2013 Frach et al.
 8,558,873 B2* 10/2013 McEldowney G01B 11/2513
 382/128
 8,587,474 B2 11/2013 Fuller et al.
 8,649,631 B2 2/2014 Islam et al.
 8,681,428 B1 3/2014 Brown
 8,687,040 B2 4/2014 Silveira
 8,716,677 B2 5/2014 Cui
 8,734,033 B2 5/2014 Walters et al.
 8,743,923 B2 6/2014 Geske et al.
 8,816,460 B2 8/2014 Kalevo et al.
 8,848,273 B2 9/2014 Yu et al.
 8,876,289 B2 11/2014 Dorronsoro Diaz et al.
 8,908,149 B2 12/2014 Freimann
 8,912,973 B2 12/2014 Werner et al.
 8,981,337 B1 3/2015 Burckel et al.
 9,007,451 B2 4/2015 Rogers et al.
 9,116,302 B2 8/2015 Mccarthy et al.
 9,151,891 B2 10/2015 Ma et al.
 9,212,899 B2 12/2015 Johnson et al.
 9,298,060 B2 3/2016 Shen et al.
 9,309,274 B2 4/2016 Van et al.
 9,310,535 B1 4/2016 Greiner et al.
 9,329,484 B1 5/2016 Markle et al.
 9,330,704 B2 5/2016 Nishimura et al.
 9,367,036 B2 6/2016 Pyun et al.
 9,369,621 B2 6/2016 Malone et al.
 9,391,700 B1 7/2016 Bruce et al.
 9,392,153 B2 7/2016 Myhre et al.
 9,411,103 B2 8/2016 Astratov
 9,482,796 B2* 11/2016 Arbabi G02B 5/1842
 9,500,771 B2 11/2016 Liu et al.
 9,553,423 B2* 1/2017 Chen H04N 13/204
 9,557,585 B1 1/2017 Yap et al.
 9,606,415 B2 3/2017 Zheludev et al.
 9,609,190 B2* 3/2017 Lee H04N 23/69
 9,739,918 B2* 8/2017 Arbabi G02B 5/021
 9,766,463 B2 9/2017 Border et al.
 9,778,404 B2 10/2017 Divliansky et al.
 9,825,074 B2 11/2017 Tian et al.
 9,829,700 B2 11/2017 Parent et al.
 9,835,870 B2 12/2017 Astratov et al.
 9,836,122 B2 12/2017 Border
 9,869,580 B2* 1/2018 Grossinger G01S 17/46
 9,880,377 B1 1/2018 Safrani et al.
 9,885,859 B2 2/2018 Harris
 9,891,393 B2 2/2018 Reece
 9,939,129 B2 4/2018 Byrnes et al.

References Cited

2005/0151698	A1	7/2005	Mohamadi	
2005/0161589	A1	7/2005	Kim et al.	
2005/0211665	A1	9/2005	Gao et al.	
2005/0220162	A1	10/2005	Nakamura	
2005/0239003	A1	10/2005	Chiodini et al.	
2006/0042322	A1	3/2006	Mendoza et al.	
2006/0127829	A1	6/2006	Deng et al.	
2007/0024975	A1	2/2007	McGrew	
2007/0026585	A1	2/2007	Wong et al.	
2007/0030870	A1	2/2007	Bour et al.	
2007/0273957	A1	11/2007	Zalevsky et al.	
2008/0014632	A1	1/2008	Cunningham et al.	
2009/0128908	A1	5/2009	Nakazawa et al.	
2009/0135086	A1	5/2009	Fuller et al.	
2009/0230333	A1	9/2009	Eleftheriades	
2009/0296223	A1	12/2009	Werner et al.	
2010/0033701	A1	2/2010	Lee et al.	
2010/0055621	A1	3/2010	Hatakeyama et al.	
2010/0072170	A1	3/2010	Wu et al.	
2010/0091224	A1	4/2010	Cho et al.	
2010/0110430	A1	5/2010	Ebbesen et al.	
2010/0134869	A1	6/2010	Bernet et al.	
2010/0177164	A1*	7/2010	Zalevsky	H04N 13/20 348/46
2010/0187658	A1	7/2010	Wei	
2010/0226134	A1	9/2010	Capasso et al.	
2010/0232017	A1	9/2010	Mccarthy et al.	
2010/0255428	A1	10/2010	Chen et al.	
2011/0012807	A1	1/2011	Sorvala	
2011/0019180	A1	1/2011	Kruglick	
2011/0187577	A1	8/2011	Fuller et al.	
2012/0068347	A1	3/2012	Isobayashi et al.	
2012/0140235	A1	6/2012	Lee et al.	
2012/0258407	A1	10/2012	Sirat	
2012/0269483	A1	10/2012	Mossberg et al.	
2012/0293854	A1	11/2012	Zheludev et al.	
2012/0327666	A1	12/2012	Liu et al.	
2012/0328240	A1	12/2012	Ma et al.	
2013/0016030	A1	1/2013	Liu et al.	
2013/0032949	A1	2/2013	Lin et al.	
2013/0037873	A1	2/2013	Suzuki et al.	
2013/0050285	A1	2/2013	Takahashi et al.	
2013/0058071	A1	3/2013	Ben	
2013/0194787	A1	8/2013	Geske et al.	
2013/0208332	A1	8/2013	Yu et al.	
2014/0009823	A1	1/2014	Park et al.	
2014/0043846	A1	2/2014	Yang et al.	
2014/0210835	A1	7/2014	Hong et al.	
2014/0263982	A1	9/2014	Shkunov et al.	
2015/0011073	A1	1/2015	Lei et al.	
2015/0017466	A1	1/2015	Ayon et al.	
2015/0018500	A1	1/2015	Gerber et al.	
2015/0055745	A1	2/2015	Holzner et al.	
2015/0068599	A1	3/2015	Chou et al.	
2015/0090862	A1	4/2015	Matsui et al.	
2015/0092139	A1	4/2015	Eguchi	
2015/0098002	A1	4/2015	Wang	
2015/0116721	A1	4/2015	Kats et al.	
2015/0125111	A1	5/2015	Orcutt et al.	
2015/0185413	A1	7/2015	Greiner et al.	
2015/0219806	A1*	8/2015	Arbabi	G02B 5/1842 29/436
2015/0241608	A1	8/2015	Shian et al.	
2015/0316717	A1	11/2015	Astratov	
2016/0025914	A1	1/2016	Brongersma et al.	
2016/0037146	A1	2/2016	McGrew	
2016/0077261	A1*	3/2016	Arbabi	G02B 5/021 359/493.01
2016/0133762	A1	5/2016	Blasco et al.	
2016/0161826	A1	6/2016	Shen et al.	
2016/0195705	A1	7/2016	Betzig et al.	
2016/0254638	A1*	9/2016	Chen	H04N 13/254 362/11
2016/0276979	A1*	9/2016	Shaver	H01Q 15/148
2016/0299337	A1	10/2016	Arbabi et al.	
2016/0299426	A1	10/2016	Gates et al.	
2016/0306079	A1	10/2016	Arbabi et al.	
2016/0306157	A1	10/2016	Rho et al.	
2016/0318067	A1	11/2016	Banerjee et al.	

References Cited

2019/0170655	A1	6/2019	Smith
2019/0191144	A1	6/2019	Arbabi et al.
2019/0196068	A1	6/2019	Tsai et al.
2019/0206136	A1	7/2019	West et al.
2019/0219835	A1	7/2019	Skinner et al.
2019/0235139	A1	8/2019	Chen et al.
2019/0250107	A1	8/2019	Sreenivasan et al.
2019/0369401	A1	12/2019	Rolland et al.
2019/0377084	A1	12/2019	Seasman et al.
2019/0391378	A1	12/2019	Eichelkraut et al.
2020/0025888	A1	1/2020	Jang et al.
2020/0052027	A1	2/2020	Arbabi et al.
2020/0096672	A1	3/2020	Yu et al.
2020/0150311	A1	5/2020	Zhang et al.
2020/0249429	A1	8/2020	Han et al.
2020/0271941	A1	8/2020	Riley, Jr. et al.
2020/0284960	A1	9/2020	Ellenbogen et al.
2020/0355913	A1	11/2020	Park et al.
2020/0404183	A1	12/2020	Kimura
2021/0028215	A1	1/2021	Devlin et al.
2021/0048569	A1	2/2021	Rubin et al.
2021/0109364	A1	4/2021	Aieta et al.
2021/0149081	A1	5/2021	Groever et al.
2021/0149082	A1	5/2021	Chen et al.
2021/0190593	A1	6/2021	Yao
2021/0208469	A1	7/2021	Didomenico
2021/0263329	A1	8/2021	Latawiec
2021/0286188	A1	9/2021	Rubin et al.
2021/0288095	A1	9/2021	Delga et al.
2021/0302763	A1	9/2021	Yao et al.
2021/0311588	A1	10/2021	Han et al.
2022/0050294	A1	2/2022	Fermigier et al.
2022/0052093	A1	2/2022	Devlin et al.
2022/0091428	A1	3/2022	Riley, Jr. et al.
2022/0103753	A1	3/2022	Kojima et al.
2022/0107263	A1	4/2022	Biesinger et al.
2022/0196480	A1	6/2022	Ang et al.
2022/0206186	A1	6/2022	Chen et al.
2022/0206205	A1	6/2022	Rubin et al.
2022/0214219	A1	7/2022	Faraon et al.
2022/0244442	A1	8/2022	Rubin et al.
2022/0283411	A1	9/2022	Devlin et al.
2024/0234461	A1	7/2024	Devlin et al.

FOREIGN PATENT DOCUMENTS

CA	3064764	A1	11/2018
CN	101158727	A	4/2008
CN	101164147	A	4/2008
CN	100476504	C	4/2009
CN	101546002	A	9/2009
CN	101681095	A	3/2010
CN	101510013	B	6/2010
CN	101510012	B	8/2010
CN	101510011	B	9/2010
CN	101241173	B	8/2011
CN	202854395	U	4/2013
CN	103092049	A	5/2013
CN	203799117	U	8/2014
CN	104067171	A	9/2014
CN	104374745	A	2/2015
CN	204422813	U	6/2015
CN	104932043	A	9/2015
CN	104956491	A	9/2015
CN	204719330	U	10/2015
CN	105068396	A	11/2015
CN	103869484	B	1/2016
CN	105223689	A	1/2016
CN	105278026	A	1/2016
CN	105278309	A	1/2016
CN	105655286	A	6/2016
CN	105676314	A	6/2016
CN	105917277	A	8/2016
CN	103257441	B	10/2016
CN	205620619	U	10/2016
CN	104834079	B	4/2017
CN	106611699	A	5/2017
CN	104834089	B	6/2017
CN	106848555	A	6/2017

(56)	References Cited			CN	114373825	A	4/2022
	FOREIGN PATENT DOCUMENTS			CN	114384612	A	4/2022
				CN	114397092	A	4/2022
				CN	114397718	A	4/2022
CN	106200276	B	10/2017	CN	114415386	A	4/2022
CN	107272212	A	10/2017	CN	216345776	U	4/2022
CN	104834088	B	12/2017	CN	216351311	U	4/2022
CN	105676314	B	1/2018	CN	216351591	U	4/2022
CN	107561857	A	1/2018	CN	216355281	U	4/2022
CN	107884850	A	4/2018	CN	216361353	U	4/2022
CN	108089325	A	5/2018	CN	111316138	B	5/2022
CN	108291983	A	7/2018	CN	114488365	A	5/2022
CN	207623619	U	7/2018	CN	114543993	A	5/2022
CN	106199997	B	8/2018	CN	114545367	A	5/2022
CN	108474869	A	8/2018	CN	114545370	A	5/2022
CN	108507542	A	9/2018	CN	114554062	A	5/2022
CN	207923075	U	9/2018	CN	114561266	A	5/2022
CN	108680544	A	10/2018	CN	216593224	U	5/2022
CN	108761779	A	11/2018	CN	216605227	U	5/2022
CN	109000692	A	12/2018	CN	216622749	U	5/2022
CN	208270846	U	12/2018	CN	114578642	A	6/2022
CN	109196387	A	1/2019	CN	114593689	A	6/2022
CN	208421387	U	1/2019	CN	114623960	A	6/2022
CN	106199956	B	2/2019	CN	114624878	A	6/2022
CN	109360139	A	2/2019	CN	114660683	A	6/2022
CN	106950195	B	5/2019	CN	114660780	A	6/2022
CN	106324832	B	7/2019	CN	114690387	A	7/2022
CN	106526730	B	7/2019	CN	114740631	A	7/2022
CN	106485761	B	8/2019	CN	114743714	A	7/2022
CN	110160685	A	8/2019	CN	114779437	A	7/2022
CN	110678773	A	1/2020	CN	216896898	U	7/2022
CN	108474869	B	6/2020	CN	216900930	U	7/2022
CN	111316138	A	6/2020	CN	216901121	U	7/2022
CN	111580190	A	8/2020	CN	216901165	U	7/2022
CN	111656707	A	9/2020	CN	216901317	U	7/2022
CN	111819489	A	10/2020	CN	216901952	U	7/2022
CN	213092332	U	4/2021	CN	216903719	U	7/2022
CN	113050295	A	6/2021	CN	216933177	U	7/2022
CN	113168022	A	7/2021	CN	217034311	U	7/2022
CN	110376665	B	8/2021	CN	217034418	U	7/2022
CN	213902664	U	8/2021	CN	217034466	U	7/2022
CN	213903843	U	8/2021	CN	114859446	A	8/2022
CN	214098104	U	8/2021	CN	114859447	A	8/2022
CN	113703080	A	11/2021	CN	114859570	A	8/2022
CN	111580190	B	12/2021	CN	114935741	A	8/2022
CN	113791524	A	12/2021	CN	217276608	U	8/2022
CN	113807312	A	12/2021	CN	217278911	U	8/2022
CN	113820839	A	12/2021	CN	217278915	U	8/2022
CN	113834568	A	12/2021	CN	217278989	U	8/2022
CN	113835227	A	12/2021	CN	217279003	U	8/2022
CN	113851573	A	12/2021	CN	217279087	U	8/2022
CN	215005942	U	12/2021	CN	217279110	U	8/2022
CN	215010478	U	12/2021	CN	217279168	U	8/2022
CN	110494771	B	1/2022	CN	217279244	U	8/2022
CN	113885106	A	1/2022	CN	217280797	U	8/2022
CN	113899451	A	1/2022	CN	217280851	U	8/2022
CN	113900078	A	1/2022	CN	217281621	U	8/2022
CN	113900162	A	1/2022	CN	217281623	U	8/2022
CN	113906320	A	1/2022	CN	114995038	A	9/2022
CN	113917574	A	1/2022	CN	115016099	A	9/2022
CN	113917578	A	1/2022	CN	115016150	A	9/2022
CN	113934004	A	1/2022	CN	115032766	A	9/2022
CN	113934005	A	1/2022	CN	115047432	A	9/2022
CN	113959984	A	1/2022	CN	115047548	A	9/2022
CN	114002707	A	2/2022	CN	115047653	A	9/2022
CN	114019589	A	2/2022	CN	115061114	A	9/2022
CN	114047632	A	2/2022	CN	115079415	A	9/2022
CN	114047637	A	2/2022	CN	115113174	A	9/2022
CN	114112058	A	3/2022	CN	217456368	U	9/2022
CN	114156168	A	3/2022	CN	217465697	U	9/2022
CN	114176492	A	3/2022	CN	217466052	U	9/2022
CN	215932365	U	3/2022	CN	217466667	U	9/2022
CN	114280704	A	4/2022	CN	217467162	U	9/2022
CN	114280716	A	4/2022	CN	217467176	U	9/2022
CN	114286953	A	4/2022	CN	217467177	U	9/2022
CN	114296180	A	4/2022	CN	217467226	U	9/2022
CN	114325886	A	4/2022	CN	217467326	U	9/2022
CN	114326163	A	4/2022	CN	217467327	U	9/2022
CN	114354141	A	4/2022	CN	217467336	U	9/2022

(56)	References Cited			CN	217821110	U	11/2022
	FOREIGN PATENT DOCUMENTS			CN	217821111	U	11/2022
				CN	217821113	U	11/2022
				CN	217821122	U	11/2022
				CN	217821160	U	11/2022
CN	217467338	U	9/2022	CN	217821236	U	11/2022
CN	217467351	U	9/2022	CN	217821680	U	11/2022
CN	217467352	U	9/2022	CN	217821696	U	11/2022
CN	217467353	U	9/2022	CN	217822825	U	11/2022
CN	217467355	U	9/2022	CN	217823690	U	11/2022
CN	217467357	U	9/2022	CN	217825178	U	11/2022
CN	217467358	U	9/2022	CN	217885960	U	11/2022
CN	217467363	U	9/2022	CN	217902220	U	11/2022
CN	217467364	U	9/2022	CN	217902222	U	11/2022
CN	217467367	U	9/2022	CN	115421295	A	12/2022
CN	217467368	U	9/2022	CN	115453754	A	12/2022
CN	217467395	U	9/2022	CN	115524768	A	12/2022
CN	217467396	U	9/2022	CN	115524775	A	12/2022
CN	217467399	U	9/2022	CN	115524874	A	12/2022
CN	217467439	U	9/2022	CN	217981833	U	12/2022
CN	115164714	A	10/2022	CN	217981857	U	12/2022
CN	115166876	A	10/2022	CN	217981991	U	12/2022
CN	115166958	A	10/2022	CN	217981992	U	12/2022
CN	115185082	A	10/2022	CN	217982020	U	12/2022
CN	115211799	A	10/2022	CN	217982038	U	12/2022
CN	115236795	A	10/2022	CN	217982089	U	12/2022
CN	217639515	U	10/2022	CN	217982120	U	12/2022
CN	217639519	U	10/2022	CN	217983382	U	12/2022
CN	217639539	U	10/2022	CN	217984044	U	12/2022
CN	217639544	U	10/2022	CN	107039885	B	4/2023
CN	217639611	U	10/2022	DE	102007058558	A1	6/2009
CN	217639612	U	10/2022	DE	102009037629	A1	2/2011
CN	217639613	U	10/2022	DE	102012212753	A1	1/2014
CN	217639715	U	10/2022	DE	102015221985	A1	5/2017
CN	217639718	U	10/2022	DE	102016218996	A1	9/2017
CN	217639719	U	10/2022	DE	112018002811	T5	2/2020
CN	217639720	U	10/2022	DE	112018002670	T5	3/2020
CN	217639722	U	10/2022	EP	1251397	A2	10/2002
CN	217639723	U	10/2022	EP	1252623	B1	10/2004
CN	217639724	U	10/2022	EP	2631740	A2	8/2013
CN	217639725	U	10/2022	EP	2763519	A2	8/2014
CN	217639726	U	10/2022	EP	2338114	B1	3/2017
CN	217639763	U	10/2022	EP	3226042	A1	10/2017
CN	217639765	U	10/2022	EP	3353578	A1	8/2018
CN	217639767	U	10/2022	EP	3380876	A1	10/2018
CN	217639768	U	10/2022	EP	3385770	A1	10/2018
CN	217639769	U	10/2022	EP	3440484	A1	2/2019
CN	217639770	U	10/2022	EP	3504566	A2	7/2019
CN	217639771	U	10/2022	EP	3564747	A1	11/2019
CN	217639772	U	10/2022	EP	3631533	A1	4/2020
CN	217639773	U	10/2022	EP	3676973	A1	7/2020
CN	217639774	U	10/2022	EP	3743764	A1	12/2020
CN	217639776	U	10/2022	EP	3353578	B1	1/2021
CN	217639777	U	10/2022	EP	3799626	A1	4/2021
CN	217639778	U	10/2022	EP	3956702	A1	2/2022
CN	217639903	U	10/2022	EP	4004608	A1	6/2022
CN	217639920	U	10/2022	EP	3440484	B1	7/2024
CN	115268058	A	11/2022	GB	2490895	A	11/2012
CN	115327865	A	11/2022	GB	2499869	B	3/2018
CN	115332917	A	11/2022	GB	2578049	A	4/2020
CN	115343795	A	11/2022	GB	2578233	A	4/2020
CN	115390176	A	11/2022	GB	2578236	A	4/2020
CN	217809433	U	11/2022	GB	2578236	B	11/2022
CN	217818613	U	11/2022	HK	40010538		7/2020
CN	217819022	U	11/2022	JP	2004302457	A	10/2004
CN	217820828	U	11/2022	JP	2005017408	A	1/2005
CN	217820829	U	11/2022	JP	2005274847	A	10/2005
CN	217820831	U	11/2022	JP	2008046428	A	2/2008
CN	217820834	U	11/2022	JP	2008299084	A	12/2008
CN	217820838	U	11/2022	JP	2010085977	A	4/2010
CN	217820839	U	11/2022	JP	2015502581	A	1/2015
CN	217820840	U	11/2022	JP	2015092234	A	5/2015
CN	217820943	U	11/2022	JP	2015170638	A	9/2015
CN	217820944	U	11/2022	JP	2016511936	A	4/2016
CN	217820945	U	11/2022	JP	2017062373	A	3/2017
CN	217820971	U	11/2022	JP	2018536204	A	12/2018
CN	217821058	U	11/2022	JP	2018537804	A	12/2018
CN	217821068	U	11/2022	JP	2019516128	A	6/2019
CN	217821071	U	11/2022	JP	2020522009	A	7/2020
CN	217821091	U	11/2022				

(56)

References Cited

FOREIGN PATENT DOCUMENTS

JP	2021511553	A	5/2021	
JP	6925358	B2	8/2021	
JP	2022542172	A	9/2022	
JP	7461294	B2	3/2024	
JP	2024045435	A	4/2024	
KR	20080099452	A	11/2008	
KR	20080103149	A	11/2008	
KR	20090002583	A	1/2009	
KR	20100027995	A	3/2010	
KR	101493928	B1	3/2015	
KR	20150113041	A	10/2015	
KR	20170015109	A	2/2017	
KR	20180083885	A	7/2018	
KR	20180121309	A	11/2018	
KR	20180124106	A	11/2018	
KR	101905444	B1	12/2018	
KR	20190038221	A	4/2019	
KR	102036640	B1	10/2019	
KR	10-2020-0008630	A	1/2020	
KR	20200008630	A	1/2020	
KR	10-2020-0108901	A	9/2020	
KR	20210088520	A	7/2021	
KR	10-2363805	B1	2/2022	
KR	1020220035971	A	3/2022	
KR	102601597	B1	11/2023	
KR	102718309	B1	10/2024	
KR	20240153619	A	10/2024	
SG	11201808772	W	11/2018	
SG	11201804346	P	11/2021	
SG	11202001717		2/2023	
TW	201017338	A	5/2010	
TW	201438242	A	10/2014	
TW	201908232	A	3/2019	
WO	0043750	A2	7/2000	
WO	2006101064	A1	9/2006	
WO	2007141788	A2	12/2007	
WO	2008020899	A2	2/2008	
WO	2009067540	A1	5/2009	
WO	2009124181	A2	10/2009	
WO	2011106553	A2	9/2011	
WO	2011106553	A3	1/2012	
WO	2012139634	A1	10/2012	
WO	2012144997	A1	10/2012	
WO	2012172366	A1	12/2012	
WO	2013033591	A1	3/2013	
WO	2013109265	A1	7/2013	
WO	2014116500	A1	7/2014	
WO	2015021255	A1	2/2015	
WO	2015077926	A1	6/2015	
WO	2015112939	A1	7/2015	
WO	2015129753	A1	9/2015	
WO	2015160412	A2	10/2015	
WO	2016044104	A1	3/2016	
WO	WO-2016049629	A1 *	3/2016	 A61B 5/0031
WO	2016051325	A1	4/2016	
WO	2016086204	A1	6/2016	
WO	2016140720	A2	9/2016	
WO	2016140720	A3	10/2016	
WO	2016168173	A1	10/2016	
WO	2016178740	A2	11/2016	
WO	2016191142	A2	12/2016	
WO	2017005709	A1	1/2017	
WO	2017034995	A1	3/2017	
WO	2017040854	A1	3/2017	
WO	2017053309	A1	3/2017	
WO	2017069954	A1	4/2017	
WO	2017079480	A1	5/2017	
WO	2017091738	A1	6/2017	
WO	2017176921	A1	10/2017	
WO	2017182771	A1	10/2017	
WO	2018063455	A1	4/2018	
WO	2018067246	A2	4/2018	
WO	2018063455	A9	5/2018	
WO	2018118984	A1	6/2018	
WO	2018134215	A1	7/2018	

WO	2018067246	A3	8/2018
WO	2018142339	A1	8/2018
WO	2018204856	A1	11/2018
WO	2018218063	A1	11/2018
WO	2018219710	A1	12/2018
WO	2018222944	A1	12/2018
WO	2019015735	A1	1/2019
WO	2019039241	A1	2/2019
WO	2019043016	A1	3/2019
WO	2019046827	A1	3/2019
WO	2019057907	A1	3/2019
WO	2019075335	A1	4/2019
WO	2019101750	A2	5/2019
WO	2019103762	A2	5/2019
WO	2019108290	A1	6/2019
WO	2019113106	A1	6/2019
WO	2019116364	A1	6/2019
WO	2019118646	A1	6/2019
WO	2019119025	A1	6/2019
WO	2019103762	A3	7/2019
WO	2019136166	A1	7/2019
WO	2019103762	A9	8/2019
WO	2019147828	A1	8/2019
WO	2019148200	A1	8/2019
WO	2019164542	A1	8/2019
WO	2019164849	A1	8/2019
WO	2019173357	A1	9/2019
WO	2019198568	A1	10/2019
WO	2019203876	A2	10/2019
WO	2019204667	A1	10/2019
WO	2019206430	A1	10/2019
WO	2020001938	A1	1/2020
WO	2020010084	A1	1/2020
WO	2020101568	A1	5/2020
WO	2020146029	A2	7/2020
WO	2020176227	A1	9/2020
WO	2020214615	A1	10/2020
WO	2020214617	A1	10/2020
WO	2021021671	A1	2/2021
WO	2021130085	A1	7/2021
WO	2021230868	A1	11/2021
WO	2022150816	A1	7/2022

OTHER PUBLICATIONS

- Roy et al., Sub-wavelength focusing meta-lens, *Opt. Express* 21, 7577-7582 (2013) (Year: 2013).*
- Wen et al., Metasurface for characterization of the polarization state of light, *Opt. Express* 23, 10272-10281 (2015) (Year: 2015).*
- Birch et al., 3D Imaging with Structured Illumination for Advanced Security Applications. United States: N. p., 2015. Web. doi: 10.2172/1221516. <https://www.osti.gov/biblio/1221516> (Year: 2015).*
- Yah Leng Lim, Milan Nikolic, Karl Bertling, Russell Kliese, and Aleksandar D. Rakić, "Self-mixing imaging sensor using a monolithic VCSEL array with parallel readout," *Opt. Express* 17, 5517-5525 (2009) (Year: 2009).*
- Luyao Xu, Daguang Chen, Christopher A. Curwen, Mohammad Memarian, John L. Reno, Tatsuo Itoh, and Benjamin S. Williams, "Metasurface quantum-cascade laser with electrically switchable polarization," *Optica* 4, 468-475 (2017) (Year: 2017).*
- J. Martin-Regalado, F. Prati, M. San Miguel and N. B. Abraham, "Polarization properties of vertical-cavity surface-emitting lasers," in *IEEE Journal of Quantum Electronics*, vol. 33, No. 5, pp. 765-783, May 1997, doi: 10.1109/3.572151. (Year: 1997).*
- Zhouping Su, Donglin Xue, and Zhicheng Ji, "Designing LED array for uniform illumination distribution by simulated annealing algorithm," *Opt. Express* 20, A843-A855 (2012) (Year: 2012).*
- Peinado et al., Polarization imaging with enhanced spatial resolution, *Optics Communications* 338 (2015) 95-100 (Year: 2015).*
- Zewei Cai, Xiaoli Liu, Xiang Peng, Yongkai Yin, Ameng Li, Jiachen Wu, and Bruce Z. Gao, "Structured light field 3D imaging," *Opt. Express* 24, 20324-20334 (2016) (Year: 2016).*
- Yanqin Chen, Xin Jin, and Qionghai Dai, "Distance measurement based on light field geometry and ray tracing," *Opt. Express* 25, 59-76 (2017) (Year: 2017).*

(56)

References Cited**OTHER PUBLICATIONS**

- Xiao Huang, Jian Bai, Kaiwei Wang, Qun Liu, Yujie Luo, Kailun Yang, and Xianjing Zhang. Target enhanced 3D reconstruction based on polarization-coded structured light. *Opt. Express*, 25(2):1173-1184, 2017a (Year: 2017).*
- Huang et al. [2017b] ↑ Xiao Huang, Yujie Luo, Jian Bai, Ruiqi Cheng, Kejing He, Kaiwei Wang, Qun Liu, Yupeng Luo, and Juan Du. Polarimetric target depth sensing in ambient illumination based on polarization-coded structured light. *Appl. Opt.*, 56(27):7741-7748, 2017b (Year: 2017).*
- Communication Pursuant to Article 94(3) EPC for European Application No. 16869282.0, issued Jan. 13, 2022, 6 pages.
- Extended European Search Report for European Application No. 19744012.6, Search completed Sep. 3, 2021, Mailed Dec. 16, 2021, 10 pages.
- Extended European Search Report for European Application No. 19830958.5, Search completed Feb. 17, 2022, Mailed Feb. 25, 2022, 8 pages.
- Extended European Search Report for European Application No. 20790964.9, Search completed Nov. 22, 2022, Mailed Dec. 2, 2022, 10 pages.
- Extended European Search Report for European Application No. 20847649.9, Search completed Jul. 20, 2023, Mailed Aug. 1, 2023, 11 Pgs.
- International Preliminary Report on Patentability for International Application No. PCT/US2020/028157, Report issued Sep. 28, 2021, Mailed Oct. 28, 2021, 7 pages.
- International Search Report and Written Opinion for International Application No. PCT/US2020/028157, Search completed Jun. 16, 2020, Mailed Sep. 4, 2020, 9 pages.
- International Search Report and Written Opinion for International Application No. PCT/US2022/070043, Search completed May 5, 2022, Mailed Jun. 9, 2022, 16 pages.
- "Elliptical Polarization", Wikipedia, XP055893535, Jan. 11, 2022, retrieved from the Internet URL: <https://en.wikipedia.org/wiki/Elliptical_polarization>, retrieved on Feb. 18, 2022, 4 pages.
- Arbabi et al., "Full Stokes Imaging Polarimetry Using Dielectric Metasurfaces", *ArXiv:1803.03384v1*, Mar. 9, 2018, retrieved from the internet: URL: <<https://arxiv.org/pdf/1803.03384.pdf>>, retrieved on Jun. 15, 2020, 20 pages.
- Arbabi et al., "Supplementary Figures of Miniature Optical Planar Camera Based on a Wide-Angle Metasurface Doublet Corrected for Monochromatic Aberrations", *Nature Communications*, 2016, vol. 7, Article No. 13682, 9 pages.
- Chen et al., "Engineering the Phase Front of Light with Phase-Change Material Based Planar Lenses", *Scientific Reports*, vol. 5, No. 8660, Mar. 2, 2015, pp. 1-7.
- Chen et al., "Supplementary Information of Engineering the Phase Front of Light with Phase-Change Material Based Planar Lenses", *Scientific Reports*, 2015, 4 pages.
- Cumme et al., "From Regular Periodic Micro-Lens Arrays to Randomized Continuous Phase Profiles", *Advanced Optical Technologies*, vol. 4, No. 1, 2015, pp. 47-61.
- Horie et al., "Reflective Optical Phase Modulator Based on High-Contrast Grating Mirrors", *Optical Society of America, IEEE*, 2014, 2 pages.
- Li et al., "Metalens-Based Miniaturized Optical Systems", *Micro-machines*, vol. 10, No. 310, May 8, 2019, pp. 1-21, doi: 10.3390/mi10050310.
- Mirsalehi, Mir M., "Optical Information Processing", *Encyclopedia of Physical Science and Technology*, 3rd edition, 2001, pp. 335-340.
- Orazbayev et al., "Tunable Beam Steering Enabled by Graphene Metamaterials", *Optics Express*, vol. 24, No. 8, pp. 8848-8861, doi: 10.1364/OE.24.008848.
- Sayyah et al., "Two-Dimensional Pseudo-Random Optical Phased Array Based on Tandem Optical Injection Locking of Vertical Cavity Surface Emitting Lasers", *Optics Express* vol. 23, No. 15, Jul. 27, 2015, pp. 19405-19416, doi: 10.1364/OE.23.0194051.
- Seurin et al., "High-Efficiency VCSEL Arrays for Illumination and Sensing in Consumer Applications", *Proceedings of SPIE*, vol. 9766, 2016, pp. 97660D-1-97660D-9, doi: 10.1117/12.2213295.
- Shim et al., "Hard-Tip, Soft-Spring Lithography", *Nature*, vol. 469, Jan. 27, 2011, pp. 516-521.
- Silvestri et al., "Robust Design Procedure for Dielectric Resonator Metasurface Lens Array", *Optics Express*, vol. 24, No. 25, Dec. 12, 2016, pp. 29153-29169.
- Voelkel et al., "Laser Beam Homogenizing: Limitations and Constraints", *DPIE, Europe, Optical Systems Design*, 2008, 12 pages.
- Zhao et al., "Tailoring the Dispersion of Plasmonic Nanorods to Realize Broadband Optical Meta-Waveplates", *Nano Letter*, vol. 13, Feb. 5, 2013, pp. 1086-1091, doi: dx.doi.org/10.1021/nl304392b.
- Zhou et al., "Progress on Vertical-Cavity Surface-Emitting Laser Arrays for Infrared Illumination Applications", *Proc. SPIE 9001, Vertical-Cavity Surface-Emitting Lasers XVIII*, 90010E, Feb. 27, 2014, 11 pages.
- Extended European Search Report for European Application No. 16869282.0, Search completed Nov. 8, 2019, Mailed Nov. 20, 2019, 15 pages.
- Extended European Search Report for European Application No. 17779772.7, Search completed Oct. 15, 2019, Mailed Oct. 25, 2019, 10 pages.
- Extended European Search Report for European Application No. 17858861.2, Search completed Mar. 13, 2020, Mailed Mar. 23, 2020, 9 pages.
- Extended European Search Report for European Application No. 18805669.1, Search completed Feb. 9, 2021, Mailed Feb. 18, 2021, 13 pages.
- Extended European Search Report for European Application No. 18852460.7, Search completed Mar. 25, 2021, Mailed Apr. 6, 2021, 13 pages.
- First Office Action and Search Report for Chinese Application No. 201680077924.9, issued Aug. 30, 2019, 10 pages.
- First Office Action and Search Report for Chinese Patent Application No. 201780031669.9, mailed on Mar. 4, 2020, 29 pages.
- International Preliminary Report on Patentability for International Application No. PCT/US2018/038357, Report issued Dec. 24, 2019, Mailed Jan. 2, 2020, 7 pages.
- International Preliminary Report on Patentability for International Application No. PCT/US2008/084068, Report issued May 25, 2010, 5 pages.
- International Preliminary Report on Patentability for International Application No. PCT/US2012/053434, Report issued Mar. 4, 2014, Mailed Mar. 13, 2014, 9 pages.
- International Preliminary Report on Patentability for International Application No. PCT/US2015/064930, Report issued Jun. 13, 2017, Mailed Jun. 22, 2017, 9 pages.
- International Preliminary Report on Patentability for International Application No. PCT/US2016/052685, Report issued Mar. 27, 2018, Mailed Apr. 5, 2018, 9 pages.
- International Preliminary Report on Patentability for International Application No. PCT/US2016/063617, Report issued May 29, 2018, Mailed Jun. 7, 2018, 7 pages.
- International Preliminary Report on Patentability for International Application No. PCT/US2017/026206, Report issued Oct. 9, 2018, Mailed Oct. 18, 2018, 9 pages.
- International Preliminary Report on Patentability for International Application No. PCT/US2017/036897, Report issued Dec. 11, 2018, Mailed Dec. 20, 2018, 9 pages.
- International Preliminary Report on Patentability for International Application No. PCT/US2017/048469, Report issued Feb. 26, 2019, Mailed Mar. 7, 2019, 6 pages.
- International Preliminary Report on Patentability for International Application No. PCT/US2018/031204, Report issued Nov. 5, 2019, Mailed Nov. 14, 2019, 9 pages.
- International Preliminary Report on Patentability for International Application No. PCT/US2018/034460, Report issued Nov. 26, 2019, Mailed Dec. 5, 2019, 7 pages.
- International Preliminary Report on Patentability for International Application No. PCT/US2018/035502, Report issued Dec. 3, 2019, Mailed Dec. 12, 2019, 8 pages.

(56)

References Cited**OTHER PUBLICATIONS**

International Preliminary Report on Patentability for International Application No. PCT/US2018/046947, Report issued Feb. 18, 2020, Mailed on Feb. 27, 2020, 7 pages.

International Preliminary Report on Patentability for International Application No. PCT/US2018/049276, Report issued on Mar. 3, 2020, Mailed on Mar. 12, 2020, 9 pages.

International Preliminary Report on Patentability for International Application No. PCT/US2019/040302, Report issued Jan. 5, 2021, Mailed Jan. 14, 2021, 6 pages.

International Preliminary Report on Patentability for International Application No. PCT/US2020/028159, Report issued Sep. 28, 2021, Mailed on Oct. 28, 2021, 7 pages.

International Preliminary Report on Patentability for International Application No. PCT/US2020/043600, Report issued Feb. 1, 2022, Mailed on Feb. 10, 2022, 9 pages.

International Search Report and Written Opinion for International Application No. PCT/US2008/084068, Search completed Jan. 13, 2009, Mailed Feb. 2, 2009, 6 pages.

International Search Report and Written Opinion for International Application No. PCT/US2012/053434, Search completed Oct. 17, 2012, Mailed Dec. 17, 2012, 7 pages.

International Search Report and Written Opinion for International Application No. PCT/US2015/064930, Search completed Sep. 9, 2016, Mailed Sep. 20, 2016, 11 pages.

International Search Report and Written Opinion for International Application No. PCT/US2016/052685, Search completed Nov. 30, 2016, Mailed Dec. 9, 2016, 12 pages.

International Search Report and Written Opinion for International Application No. PCT/US2016/063617, Search completed Jan. 19, 2017, Mailed Mar. 31, 2017, 9 pages.

International Search Report and Written Opinion for International Application No. PCT/US2017/026206, Search completed Jun. 10, 2017, Mailed Jun. 28, 2017, 15 pages.

International Search Report and Written Opinion for International Application No. PCT/US2017/036897, Search completed Jan. 31, 2018, Mailed Feb. 21, 2018, 9 pages.

International Search Report and Written Opinion for International Application No. PCT/US2017/048469, Search completed Apr. 20, 2018, Mailed May 4, 2018, 9 pages.

International Search Report and Written Opinion for International Application No. PCT/US2018/031204, Search completed Jun. 29, 2018, Mailed Jul. 23, 2018, 14 pages.

International Search Report and Written Opinion for International Application No. PCT/US2018/034460, Search completed Jul. 29, 2018, Mailed Aug. 24, 2018, 10 pages.

International Search Report and Written Opinion for International Application No. PCT/US2018/035502, Search completed Jul. 31, 2018, Mailed Aug. 24, 2018, 13 pages.

International Search Report and Written Opinion for International Application No. PCT/US2018/038357, Search completed Apr. 9, 2019, Mailed May 13, 2019, 12 pages.

International Search Report and Written Opinion for International Application No. PCT/US2018/046947, Search completed Oct. 14, 2019, Mailed Oct. 25, 2019, 10 pages.

International Search Report and Written Opinion for International Application No. PCT/US2018/049276, Search completed Oct. 26, 2018, Mailed Jan. 15, 2019, 12 pages.

International Search Report and Written Opinion for International Application No. PCT/US2019/014975, Search completed Jun. 17, 2019, Mailed Jul. 8, 2019, 10 pages.

International Search Report and Written Opinion for International Application No. PCT/US2019/018615, Search completed Apr. 12, 2019, Mailed May 6, 2019, 12 pages.

International Search Report and Written Opinion for International Application No. PCT/US2019/040302, Search completed Aug. 29, 2019, Mailed Oct. 17, 2019, 6 pages.

International Search Report and Written Opinion for International Application No. PCT/US2020/028159, Search completed Jun. 15, 2020, Mailed Aug. 11, 2020, 7 pages.

International Search Report and Written Opinion for International Application No. PCT/US2020/043600, Search completed Sep. 29, 2020, Mailed Nov. 24, 2020, 11 pages.

Partial European Search Report for European Application No. 19744012.6, Search completed Sep. 3, 2021, Mailed Sep. 15, 2021, 12 Pages.

Search Report and Written Opinion for Singapore Patent Application No. SG11201808772W, Search completed Jan. 20, 2020, Mailed Jan. 28, 2020, 12 pages.

Supplementary Partial European Search Report for European Application No. 16869282.0, Search completed Jun. 19, 2019, Mailed Jul. 2, 2019, 12 pages.

"Materials for High and Low Refractive Index Coatings", Sigma-Aldrich Inc., retrieved from the Internet URL: <www.sigmaaldrich.com/materials-science/organic-electronics/ri-coatings.html>, 2019, 3 pages.

"These Tiny, Incredible 'Metalenses' are the Next Giant Leap in Optics", PetaPixel, retrieved from the Internet URL: <<https://petapixel.com/2016/06/03/incredible-flat-metalenses-may-future-optics/>>, Jun. 3, 2016, 21 pages.

Aieta et al., "Aberration-Free Ultrathin Flat Lenses and Axicons at Telecom Wavelengths Based on Plasmonic Metasurfaces", Nano Letters, vol. 12, Aug. 15, 2012, pp. 4932-4936, doi: dx.doi.org/10.1021/nl302516v.

Aieta et al., "Aberrations of Flat Lenses and Aplanatic Metasurfaces", Optics Express, vol. 21, No. 25, Dec. 16, 2013, pp. 31530-31539, doi: 10.1364/OE.21.031530.

Aieta et al., "Multiwavelength Achromatic Metasurfaces by Dispersive Phase Compensation", Sciences, 2015, 15 pages, doi: 10.1126/science.aaa2494.

Aieta et al., "Out-of-Plane Reflection and Refraction of Light by Anisotropic Optical Antenna Metasurfaces with Phase Discontinuities", Nano Letters, Feb. 13, 2012, pp. A-E, doi: dx.doi.org/10.1021/nl300204s.

Arbabi et al., "Dielectric Metasurfaces for Complete Control of Phase and Polarization with Subwavelength Spatial Resolution and High Transmission", Nature Nanotechnology, vol. 10, Nov. 2015, pp. 937-943, doi: 10.1038/NNANO.2015.186.

Arbabi et al., "Efficient Dielectric Metasurface Collimating Lenses for Mid-Infrared Quantum Cascade Lasers", Optics Express, vol. 23, No. 26, Dec. 28, 2015, pp. 33310-33317, doi: 10.1364/OE.23.033310.

Arbabi et al., "Subwavelength-Thick Lenses with High Numerical Apertures and Large Efficiency Based on High-Contrast Transmittarrays", Nature Communications, vol. 6, No. 7069, May 7, 2015, pp. 1-6, doi: 10.1038/ncomms8069.

Azadegan et al., "A Novel Approach for Miniaturization of Slot Antennas", IEEE Transactions on Antennas and Propagation, vol. 51, No. 3, Mar. 2003, pp. 421-429, doi: 10.1109/TAP.2003.809853.

Blanchard et al., "Modeling Nanoscale, V-Shaped Antennas for the Design of Optical Phased Arrays", Physical Review B, vol. 85, 2012, pp. 155457-1-155457-11, doi: 10.1103/physRevB.85.155457.

Buralli et al., "Optical Performance of Holographic Kinoforms", Applied Optics, vol. 28, No. 5, Mar. 1, 1989, pp. 976-983, doi: 10.1364/AO.28.000976.

Byrnes et al., "Designing Large, High-Efficiency, High-Numerical-Aperture, Transmissive Meta-Lenses for Visible Light", Optics Express, vol. 24, No. 5, Mar. 7, 2016, pp. 5110-5124, doi: 10.1364/OE.24.005110.

Campione et al., "Tailoring Dielectric Resonator Geometries for Directional Scattering and Huygens' Metasurfaces", Optics Express, vol. 23, No. 3, Feb. 9, 2015, pp. 2293-2307, doi: 10.1364/OE.23.002293.

Chen et al., "A Broadband Achromatic Metalens for Focusing and Imaging in the Visible", Nature Nanotechnology, 2018, 8 pages, doi: 10.1038/s41565-017-0034-6.

Chen et al., "A Review of Metasurfaces: Physics and Applications", Reports on Progress in Physics, vol. 79, No. 076401, Jun. 16, 2016, 40 pages, doi: 10.1088/0034-4885/79/7/076401.

Chen et al., "Broadband Achromatic Metasurface-Refractive Optics", Nano Letters, vol. 18, Nov. 13, 2018, pp. 7801-7808.

(56)

References Cited

OTHER PUBLICATIONS

- Chen et al., "Dual-Polarity Plasmonic Metalens for Visible Light", *Nature Communications*, vol. 3, No. 1198, Nov. 13, 2012, pp. 1-6, doi: 10.1038/ncomms2207.
- Chen et al., "High-Efficiency Broadband Meta-Hologram with Polarization-Controlled Dual Images", *Nano Letters*, 2013, pp. A-F, doi: dx.doi.org/10.1021/nl403811d.
- Chen et al., "Immersion Meta-Lenses at Visible Wavelengths for Nanoscale Imaging", *Nano Letters*, Apr. 7, 2017, pp. A-G, doi: 10.1021/acs.nanolett.7b00717.
- Chen et al., "Integrated Plasmonic Metasurfaces for Spectropolarimetry", *Nanotechnology*, vol. 27, Apr. 26, 2016, pp. 1-7, doi:10.1088/0957-4484/27/22/224002.
- Chen et al., "Phase and Dispersion Engineering of Metalenses: Broadband Achromatic Focusing and Imaging in the Visible", *Arxiv*, 2017, retrieved from the Internet: URL: <https://arxiv.org/abs/1711.09343v1>, 30 pages.
- Chou et al., "Imprint Lithography with 25-Nanometer Resolution", *Science*, vol. 272, Apr. 5, 1996, pp. 85-87.
- Dayal et al., "Polarization Control of 0.85 μm Vertical-cavity Surface-Emitting Lasers Integrated with Gold Nanorod Arrays", *Applied Physics Letters*, vol. 91, Sep. 12, 2007, pp. 111107-1-111107-3, doi: 10.1063/1.2783281.
- Decker et al., "High-Efficiency Light-Wave Control with All-Dielectric Optical Huygens' Metasurfaces", *Advanced Optical Materials*, arXiv:1405.5038, 2014, 17 pages, doi: 10.1002/adom.201400584.
- Devlin et al., "Arbitrary Spin-to-Orbital Angular Momentum Conversion of Light", *Science*, vol. 358, Nov. 17, 2017, pp. 896-901, doi: 10.1126/science.aao5392.
- Devlin et al., "Broadband High-Efficiency Dielectric Metasurfaces for the Visible Spectrum", *PNAS*, vol. 113, No. 38, Sep. 20, 2016, pp. 10473-10478, doi: 10.1073/pnas.1611740113.
- Devlin et al., "High Efficiency Dielectric Metasurfaces at Visible Wavelengths", *Arxiv*, retrieved from the Internet: URL: <https://arxiv.org/ftp/arxiv/papers/1603/1603.02735.pdf>, Mar. 8, 2016, pp. 1-18.
- Ding et al., "Gradient Metasurfaces: Fundamentals and Applications", *ArXiv:1704.03032v1 [physics.optics]*, Apr. 10, 2017, 83 pages.
- Dong et al., "Zero-Index Photonic Crystal as Low-Aberration Optical Lens (Conference Presentation)", *Proceedings of SPIE 9918, Metamaterials, Metadevices, and Metasystems*, Nov. 9, 2016, 2 pages, doi.org/10.1117/12.2237137.
- Engelberg et al., "Near-IR Wide Field-of-View Huygens Metalens for Outdoor Imaging Applications", *IEEE, CLEO*, 2019, 2 pages.
- Evlyukhin et al., "Optical Response Features of Si-Nanoparticle Arrays", *Physical Review B*, vol. 82, 2010, pp. 045404-1-045404-11, doi: 10.1103/PhysRevB.82.045404.
- Fattal et al., "Flat Dielectric Grating Reflectors with Focusing Abilities", *Nature Photonics*, May 2, 2010, pp. 1-5, doi: 10.1038/nphoton.2010.116.
- Genevet et al., "Breakthroughs in Photonics 2013: Flat Optics: Wavefronts Control with Huygens' Interfaces", *IEEE Photonics Journal*, vol. 6, No. 2, Apr. 2014, 4 pages, doi: 10.1109/JPHOT.2014.2308194.
- Genevet et al., "Recent Advances in Planar Optics: From Plasmonic to Dielectric Metasurfaces", *Optica*, vol. 4, No. 1, Jan. 2017, pp. 139-152, doi: 10.1364/Optica.4.000139.
- Goldberg, D. E. "Genetic Algorithms in Search, Optimization, and Machine Learning", Addison Wesley, 1989, 432 pages (presented in 2 parts).
- Groever et al., "Meta-Lens Doublet in the Visible Region", *Nano Letters*, Jun. 29, 2017, pp. A-F, doi: 10.1021/acs.nanolett.7b01888.
- Hartwig et al., "Challenges for Reducing the Size of Laser Activated Remote Phosphor Light Engines for DLP Projection", *Proceedings of SPIE, International Optical Design Conference*, vol. 9293, Dec. 17, 2014, pp. 929313-1-929313-6, doi: 10.1117/12.2073275.
- Herrera-Fernandez et al., "Double Diffractive Optical Element System for Near-Field Shaping", *Applied Optics*, vol. 50, No. 23, Aug. 10, 2011, pp. 4587-4593.
- Hidber et al., "Microcontact Printing of Palladium Colloids: Micron-Scale Patterning by Electroless Deposition of Copper", *Langmuir*, vol. 12, No. 5, 1996, pp. 1375-1380.
- Hsiao et al., "Fundamentals and Applications of Metasurfaces", *Small Methods*, vol. 1, Mar. 24, 2017, pp. 1600064-1-1600064-20.
- Jin et al., "Waveforms for Optimal Sub-keV High-Order Harmonics with Synthesized Two- or Three-Colour Laser Fields", *Nature Communications*, vol. 5, No. 4003, May 30, 2014, pp. 1-6.
- Kats et al., "Giant Birefringence in Optical Antenna Arrays with Widely Tailorable Optical Anisotropy", *PNAS*, vol. 109, No. 31, Jul. 31, 2012, pp. 12364-12368, doi: www.pnas.org/dgi/doi/10.1073/pnas.1210686109.
- Khorasaninejad et al., "Achromatic Metalens over 60 nm Bandwidth in the Visible and Metalens with Reverse Chromatic Dispersion", *Nano Letters*, Jan. 26, 2017, pp. A-F, doi: 10.1021/acs.nanolett.6b05137.
- Khorasaninejad et al., "Achromatic Metasurface Lens at Telecommunication Wavelengths", *Nano Letters*, Jul. 2, 2015, pp. A-E, doi: 10.1021/acs.nanolett.5b01727.
- Khorasaninejad et al., "Broadband and Chiral Binary Dielectric Meta-Holograms", *Science Advances*, vol. 2, No. e1501258, May 13, 2016, pp. 1-6, doi:10.1126/sciadv.1501258.
- Khorasaninejad et al., "Broadband Multifunctional Efficient Meta-Gratings Based on Dielectric Waveguide Phase Shifters", *Nano Letters*, Sep. 11, 2015, pp. A-G, doi: 10.1021/acs.nanolett.5b02524.
- Khorasaninejad et al., "Metalenses at Visible Wavelengths: Diffraction-Limited Focusing and Subwavelength Resolution Imaging", *Science*, vol. 352, No. 6290, Jun. 3, 2016, pp. 1190-1194, doi: 10.1126/science.aaf6644.
- Khorasaninejad et al., "Multispectral Chiral Imaging with a Metalens", *Nano Letters*, vol. 16, Jun. 7, 2016, pp. 4595-4600, doi: 10.1021/acs.nanolett.6b01897.
- Khorasaninejad et al., "Planar Lenses at Visible Wavelengths", *Arxiv*, May 7, 2016, 17 pages.
- Khorasaninejad et al., "Polarization-Insensitive Metalenses at Visible Wavelengths", *Nano Letters*, Oct. 24, 2016, pp. A-F, doi: 10.1021/acs.nanolett.6b03626.
- Khorasaninejad et al., "Silicon Nanofin Grating as a Miniature Chirality-Distinguishing Beam-Splitter", *Nature Communications*, vol. 5, No. 5386, Nov. 12, 2014, pp. 1-6, doi: 10.1038/ncomms6386.
- Khorasaninejad et al., "Super-Dispersive Off-Axis Meta-Lenses for Compact High Resolution Spectroscopy", *Nano Letters*, Apr. 19, 2016, pp. A-E, doi: 10.1021/acs.nanolett.6b01097.
- Khorasaninejad et al., "Visible Wavelength Planar Metalenses Based on Titanium Dioxide", *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 23, No. 3, May/Jun. 2017, pp. 16 pages.
- Kildishev et al., "Planar Photonics with Metasurfaces", *Science*, vol. 339, No. 6125, Mar. 15, 2013, pp. 1232009-1-1232009-6, DOI: 10.1126/science.1232009.
- Kokkoris et al., "Nanoscale Roughness Effects at the Interface of Lithography and Plasma Etching: Modeling of Line-Edge-Roughness Transfer During Plasma Etching", *IEEE Transactions on Plasma Science*, vol. 37, No. 9, Sep. 2009, pp. 1705-1714, doi: 10.1109/TPS.2009.2024117.
- Kominami et al., "Dipole and Slot Elements and Arrays on Semi-Infinite Substrates", *IEEE Transactions on Antennas and Propagation*, vol. AP33, No. 6, Jun. 1985, pp. 600-607.
- Krasnok et al., "All-Dielectric Optical Nanoantennas", *Optics Express*, vol. 20, No. 18, Aug. 27, 2012, pp. 20599-20604.
- Kress et al., "Applied Digital Optics from Micro-Optics to Nanophotonics", *Applied Digital Optics*, Wiley, 2009, 30 pages.
- Lalanne et al., "Interaction Between Optical Nano-objects at Metallo-dielectric Interfaces", *Nature Physics*, vol. 2, Aug. 2006, pp. 551-556, doi:10.1038/nphys364.
- Lee et al., "Giant Nonlinear Response from Plasmonic Metasurfaces Coupled to Intersubband Transitions", *Nature*, vol. 511, Jul. 3, 2014, pp. 65-69.

(56)

References Cited

OTHER PUBLICATIONS

- Leveque et al., "Transient Behavior of Surface Plasmon Polaritons Scattered at a Subwavelength Groove", *Physical Reviews B*, vol. 76, Oct. 18, 2007, pp. 155418-1-155418-8, doi: 10.1103/PhysRevB.76.155418.
- Lezec et al., "Beaming Light from a Subwavelength Aperture", *Science*, Aug. 2, 2002, vol. 297, pp. 820-822, doi:10.1126/science.1071895.
- Li et al., "Achromatic Flat Optical Components via Compensation between Structure and Material Dispersions.", *Scientific Reports*, vol. 6, No. 19885, Jan. 22, 2016, pp. 1-7, doi:10.1038/srep19885.
- Li et al., "All-Silicon Nanorod-Based Dammann Gratings", *Optics Letters*, vol. 40, No. 18, Sep. 15, 2015, pp. 4285-4288.
- Li et al., "Broadband Diodelike Asymmetric Transmission of Linearly Polarized Light in Ultrathin Hybrid Metamaterial", *Applied Physics Letters*, vol. 105, Nov. 19, 2014, pp. 201103-1-201103-5, doi: 10.1063/1.4902162.
- Li et al., "Dispersion Controlling Meta-Lens at Visible Frequency", *Optics Express*, vol. 25, No. 18, Sep. 4, 2017, pp. 21419-21427.
- Li et al., "Flat Metasurfaces to Focus Electromagnetic Waves in Reflection Geometry", *Optics Letters*, vol. 37, No. 23, Dec. 1, 2012, pp. 4940-4942.
- Lin et al., "Dielectric Gradient Metasurface Optical Elements", *Science*, vol. 345, No. 6194, Jul. 18, 2014, pp. 298-302, doi:10.1126/science.1253213.
- Liu et al., "Realization of Polarization Evolution on Higher-Order Poincare Sphere with Metasurface", *Applied Physics Letters*, vol. 104, 2014, pp. 191110-1-191101-4, doi: <http://dx.doi.org/10.1063/1.4878409>.
- Lo et al., "New Architecture for Space Telescopes Uses Fresnel Lenses", *SPIE Newsroom*, 2006, 2 pages, doi: 10.1117/2.1200608.0333.
- Lu et al., "Planar High-Numerical-Aperture Low-Loss Focusing Reflectors and Lenses Using Subwavelength High Contrast Gratings", *Optics Express*, vol. 18, No. 12, Jun. 7, 2010, pp. 12606-12614, doi: 10.1364/OE.18.012606.
- Luk et al., "Dielectric Resonator Antennas", *Research Studies Press Ltd.*, Hertfordshire, 2003, 404 pages (presented in 2 parts).
- Mackus et al., "The Use of Atomic Layer Deposition in Advanced Nanopatterning", *Nanoscale*, vol. 6, Jul. 25, 2014, 10941-10960.
- Maguid et al., "Multifunctional Interleaved Geometric-Phase Dielectric Metasurfaces", *Light: Science & Applications*, vol. 6, No. E17027, Aug. 11, 2017, pp. 1-7, doi: 10.1038/lsa.2017.27.
- Mao et al., "Nanopatterning Using a Simple Bi-Layer Lift-Off Process for the Fabrication of a Photonic Crystal Nanostructure", *Nanotechnology*, vol. 24, No. 085302, Feb. 1, 2013, pp. 1-6, doi:10.1088/0957-4484/24/8/085302.
- Mao et al., "Surface Patterning of Nonscattering Phosphors for Light Extraction", *Optics Letters*, vol. 38, No. 15, Aug. 1, 2013, pp. 2796-2799, doi: 10.1364/OL.38.002796.
- Martin-Moreno, L., "Theory of Highly Directional Emission from a Single Subwavelength Aperture Surrounded by Surface Corrugations", *Physical Review Letters*, vol. 90, No. 16, Apr. 25, 2003, pp. 167401-1-167401-4, doi: 10.1103/PhysRevLett.9.167401.
- McLeod, J., "Thin-Film Optical Filters", Third Edition, *Institute of Physics Publishing*, Adam Hilger, 1986, 667 pages (presented in 2 parts).
- Meng et al., "A Novel Nanofabrication Technique of Silicon-Based Nanostructures", *Nanoscale Research Letters* vol. 11, No. 504, pp. 1-9, doi:10.1186/s11671-016-1702-4.
- Miyazaki et al., "Ultraviolet-Nanoimprinted Packaged Metasurface Thermal Emitters for Infrared CO₂ Sensing", *Science and Technology of Advanced Materials*, vol. 16, No. 035005, May 20, 2015, pp. 1-5, doi: 10.1088/1468-6996/16/3/035005.
- Monticone et al., "Full Control of Nanoscale Optical Transmission with a Composite Metascreen", *Physical Review Letters*, vol. 110, May 17, 2013, pp. 203903-1-203903-5, doi: 10.1103/PhysRevLett.110.203903.
- Mueller et al., "Metasurface Polarization Optics: Independent Phase Control of Arbitrary Orthogonal States of Polarization", *Physical Review Letters*, vol. 118, Mar. 17, 2017, pp. 113901-1-113901-5.
- Mueller et al., "Ultracompact Metasurface In-Line Polarimeter", *Optica*, vol. 3, No. 1, Jan. 8, 2016, pp. 42-47.
- Ni et al., "Broadband Light Bending with Plasmonic Nanoantennas", *Scienceexpress*, Dec. 22, 2011, 3 pages, doi: 10.1126/science.1214686.
- Ni et al., "Ultra-Thin, Planar, Babinet-Inverted Plasmonic Metalenses", *Light: Science & Applications*, vol. 2, No. e72, Apr. 26, 2013, pp. 1-6, doi: 10.1038/lsa.2013.28.
- Okaya et al., "The Dielectric Microwave Resonator", *Proceedings of the IRE*, Oct. 1962, pp. 2081-2092, doi: 10.1109/JRPROC.1962.288245.
- Pacheco-Peña et al., "Epsilon-Near-Zero Metalenses Operating in the Visible", *Optics & Laser Technology*, 2016, pp. 1-18.
- Peinado et al., "Optimization and Performance Criteria of a Stokes Polarimeter Based on Two Variable Retarders", *Optics Express*, vol. 18, No. 10, May 10, 2010, pp. 9815-9530.
- Petosa et al., "An Overview of Tuning Techniques for Frequency-Agile Antennas" *IEEE Antennas and Propagation Magazine*, vol. 54, No. 5, Oct. 2012, pp. 271-296.
- Pfeiffer et al., "Cascaded Metasurfaces for Complete Phase and Polarization Control", *Applied Physics Letters*, vol. 102, Jun. 11, 2013, pp. 231116-1-231116-4, doi: 10.1063/1.4810873.
- Pfeiffer et al., "Metamaterial Huygens' Surface: Tailoring Wave Fronts with Reflectionless Sheets", *Physical Review Letters*, vol. 110, May 10, 2013, pp. 197401-1-197401-5. DOI: 10.1103/PhysRevLett.110.197401.
- Pors et al., "Broadband Focusing Flat Mirrors Based on Plasmonic Gradient Metasurfaces", *Nano Letters*, vol. 13, Jan. 23, 2013, pp. 829-834, doi: <https://doi.org/10.1021/nl304761m>.
- Pors et al., "Plasmonic Metagratings for Simultaneous Determination of Strokes Parameters", *Arxiv:1609.04691v1 [physics.optics]*, Sep. 15, 2016, 21 pages, doi: <http://dx.doi.org/10.1364/Optica.2.000716>.
- Reichelt et al., "Capabilities of Diffractive Optical Elements for Real-Time Holographic Displays", *Proceedings of SPIE*, vol. 6912, Feb. 2008, pp. 69120-69130, doi: <http://dx.doi.org/10.1117/12.762887>.
- Rubin et al., "Polarization State Generation and Measurement with a Single Metasurface", *Optics Express*, vol. 26, No. 17, Aug. 20, 2018, pp. 21455-21478, doi: 10.1364/OE.26.021455.
- Saeidi et al., "Wideband Plasmonic Focusing Metasurfaces", *Applied Physics Letters*, vol. 105, Aug. 2014, pp. 053107-1-053107-4, doi: <http://dx.doi.org/10.1063/1.4892560>.
- Sales et al., "Diffractive-Refractive Behavior of Kinoform Lenses", *Applied Optics*, vol. 36, No. 1, Jan. 1, 1997, pp. 253-257, doi: 10.1364/AO.36.000253.
- Sancho-Parramon et al., "Optical Characterization of HfO₂ by Spectroscopic Ellipsometry: Dispersion Models and Direct Data Inversion", *Thin Solid Films*, vol. 516, 2008, pp. 7990-7995, doi:10.1016/j.tsf.2008.04.007.
- Schulz et al., "Quantifying the Impact of Proximity Error Correction on Plasmonic Metasurfaces", *Optical Materials Express*, vol. 5, No. 12, Dec. 1, 2015, pp. 2798-2803, doi: 10.1364/OME.5.002798.
- She et al., "Large Area Metalenses: Design, Characterization, and Mass Manufacturing", *Optics Express*, vol. 26, No. 2, Jan. 22, 2018, pp. 1573-1585, doi: 10.1364/OE.26.001573.
- Song et al., "Vividly-Colored Silicon Metasurface Based on Collective Electric and Magnetic Resonances", *IEEE*, Jan. 11, 2016, 2 pages.
- Sun et al., "High-Efficiency Broadband Anomalous Reflection by Gradient Meta-Surfaces", *Nano Letters*, vol. 12, Nov. 28, 2012, pp. 6223-6229, doi: dx.doi.org/10.1021/nl3032668.
- Vo et al., "Sub-Wavelength Grating Lenses with a Twist", *IEEE Photonics Technology Letters*, vol. 26, No. 13, Jul. 1, 2014, pp. 1375-1378, DOI: 10.1109/LPT.2014.2325947.
- Voelz, "Chapter 6: Transmittance Functions, Lenses, and Gratings", pp. 89-111.
- Walther et al., "Spatial and Spectral Light Shaping with Metamaterials", *Advanced Materials*, vol. 24, 2012, pp. 6300-6304, doi: 10.1002/adma.201202540.

(56)

References Cited

OTHER PUBLICATIONS

- Wang et al., "Broadband Achromatic Optical Metasurface Devices", *Nature Communications*, vol. 8, No. 187, Aug. 4, 2017, pp. 1-9, doi: 10.1038/s41467-017-00166-7.
- Wang et al., "Generation of Steep Phase Anisotropy with Zero-Backscattering by Arrays of Coupled Dielectric Nano-Resonators", *Applied Physics Letters*, vol. 105, Sep. 25, 2014, pp. 121112-1-121112-5, doi:https://doi.org/10.1063/1.4896631.
- Wei et al., "Design of Ultracompact Polarimeters based on Dielectric Metasurfaces", *Optics Letters*, vol. 42, No. 8, Apr. 11, 2017, pp. 1580-1583, doi: https://doi.org/10.1364/OL.42.001580.
- Wu et al., "Spectrally selective chiral silicon metasurfaces based on infrared Fano resonances", *Nature Communications*, 2014, vol. 5, No. 3892, published online May 27, 2014, 9 pgs., DOI: 10.1038/ncomms4892.
- Wu et al., "Spectrally Selective Chiral Silicon Metasurfaces based on infrared Fano Resonances", *Nature Communications*, vol. 5, No. 3892, May 27, 2014, pp. 1-9, doi: doi: 10.1038/ncomms4892.
- Xu et al., "Metasurface External Cavity Laser", *Applied Physics Letters*, vol. 107, No. 221105, 2015, pp. 221105-1-221105-5, doi: 10.1063/1.4936887.
- Yang et al., "Design of Ultrathin Plasmonic Quarter-Wave Plate based on Period Coupling", *Optics Letters*, 2013, vol. 38, No. 5, pp. 679-681, doi: https://doi.org/10.1364/OL.38.000679.
- Yao et al., "Wide Wavelength Tuning of Optical Antennas on Graphene with Nanosecond Response Time", *Nano Letters*, 2014, vol. 14, No. 1, Dec. 3, 2013, pp. A-F, doi: 10.1021/nl403751p.
- Yu et al., "A Broadband, Background-Free Quarter-Wave Plate Based on Plasmonic Metasurfaces", *Nano Letters*, vol. 12, No. 12, Nov. 3, 2012, pp. A-F, doi: dx.doi.org/10.1021/nl303445u.
- Yu et al., "Flat Optics with Designer Metasurfaces", *Nature Materials*, vol. 13, Feb. 2014, pp. 139-150, doi :10.1038/NMAT3839.
- Yu et al., "Flat optics: Controlling Wavefronts with Optical Antenna Metasurfaces", *IEEE Journal of Selected Topics*, vol. 19, No. 3, May/Jun. 2013, 23 pages.
- Yu et al., "High-Transmission Dielectric Metasurface with 2π Phase Control at Visible Wavelengths", *Laser & Photonics Reviews*, Jun. 26, 2015, No. 4, pp. 412-418, doi: 10.1002/lpor.201500041.

- Yu et al., "Light Propagation with Phase Discontinuities: Generalized Laws of Reflection and Refraction", *Science*, vol. 334, No. 6054, Oct. 21, 2011, pp. 333-337, doi: 10.1126/science.1210713.
- Yu et al., "Optical Metasurfaces and Prospect of their Applications Including Fiber Optics", *Journal of Lightwave Technology*, 2015, vol. 33, No. 12, pp. 2344-2358.
- Yu et al., "Quantum Cascade Lasers with Integrated Plasmonic Antenna-Array Collimators", *Optics Express*, vol. 16, No. 24, Nov. 24, 2008, pp. 19447-19461.
- Yu et al., "Small Divergence Edge-Emitting Semiconductor Lasers with Two-Dimensional Plasmonic Collimators", *Applied Physics Letters*, vol. 93, 2008, pp. 181101-1-181101-3, doi: 10.1063/1.3009599.
- Yu et al., "Small-Divergence Semiconductor Lasers by Plasmonic Collimation", *Nature Photonics*, Sep. 2008, vol. 2, pp. 564-570, doi: 10.1038/nphoton.2008.052.
- Zhan et al., "Low-Contrast Dielectric Metasurface Optics", *ACS Photonics*, vol. 3, 2016, pp. A-F, doi: 10.1021/acsp Photonics.5b00660.
- Zhao et al., "Mie Resonance-Based Dielectric Metamaterials", *Materials Today*, vol. 12, No. 12, Dec. 2009, pp. 60-69.
- Zhao et al., "Recent Advances on Optical Metasurfaces", *Journal of Optics*, Institute of Physics Publishing, vol. 16, Issue 12, 2014, 14 pages, doi: 10.1088/2040489781/16/12/123001.
- Zhao et al., "Twisted Optical metamaterials or planarized ultrathin broadband circular polarizers", *Nature Communications*, 2012, vol. 3, No. 870, pp. 1-7, DOI: 10.1038/ncomms1877.
- Zhou et al., "Characteristic Analysis of Compact Spectrometer Based on Off-Axis Meta-Lens", *Applied Sciences*, vol. 8, No. 321, 2018, 11 pages, doi: 10.3390/app8030321.
- Zhou et al., "Plasmonic Holographic Imaging with V-shaped Nanoantenna Array", *Optics Express*, vol. 21, No. 4, Feb. 25, 2013, pp. 4348-4354.
- Zhu et al., "Ultra-Compact Visible Chiral Spectrometer with Meta-Lenses", *APL Photonics*, vol. 2, Feb. 7, 2017, pp. 036103-1-036103-12, 13 pages, doi: https://doi.org/10.1063/1.4974259.
- Zou et al., "Dielectric Resonator Nanoantennas at Visible Frequencies", *Optics Express*, vol. 21, No. 1, Jan. 14, 2013, pp. 1344-1352.

* cited by examiner

FIG. 1A

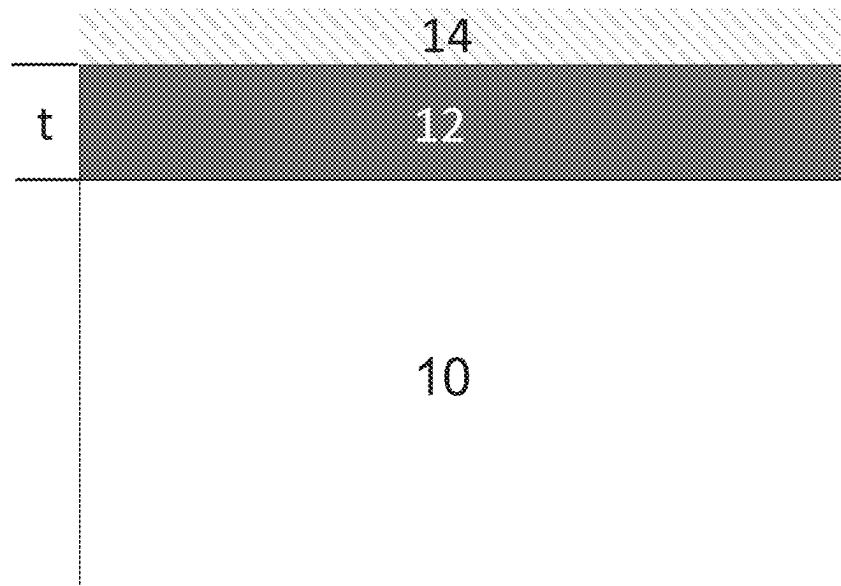


FIG. 1B

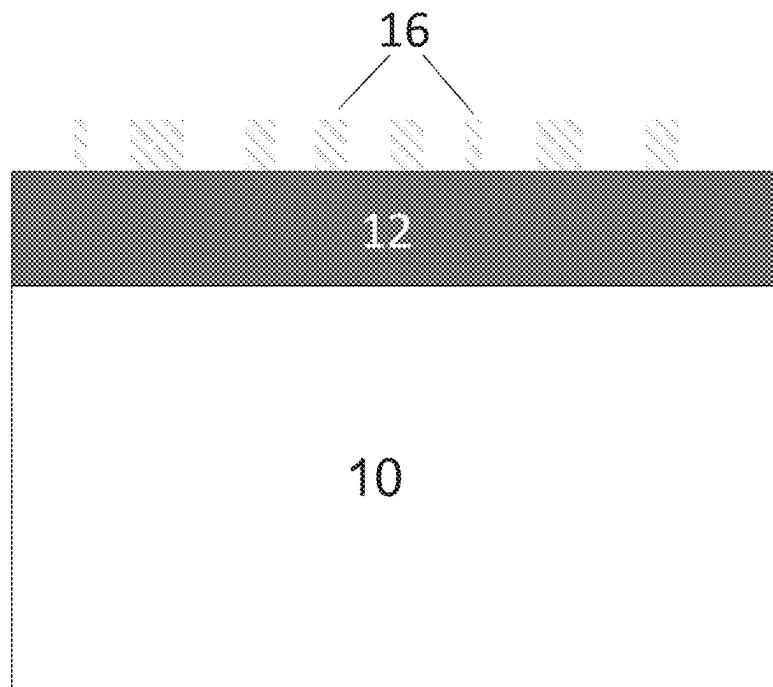


FIG. 1C

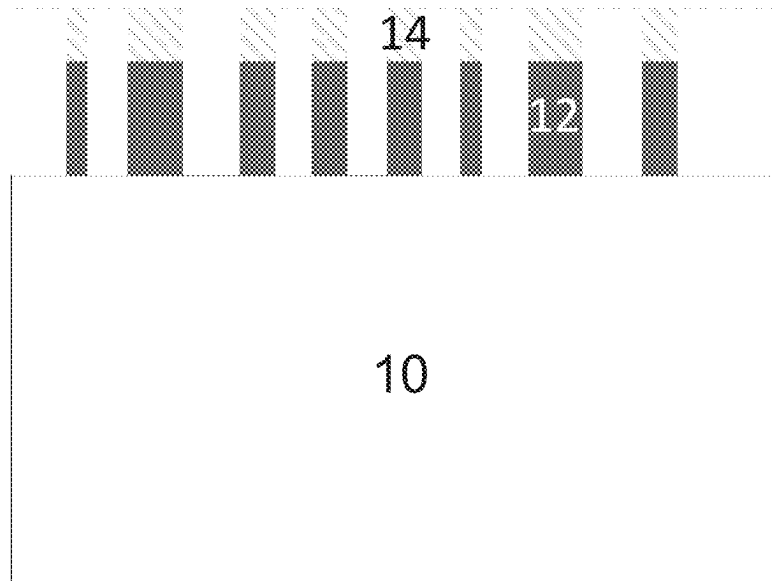


FIG. 1D

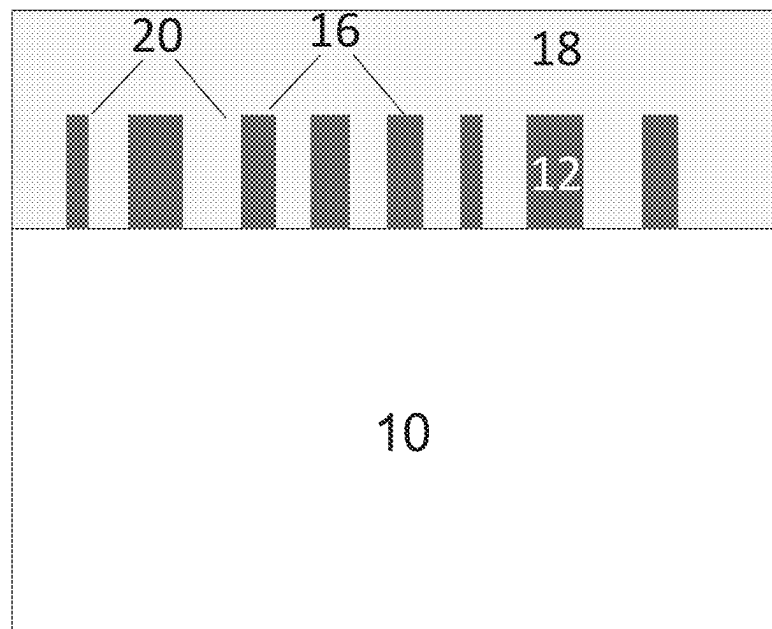


FIG. 1E

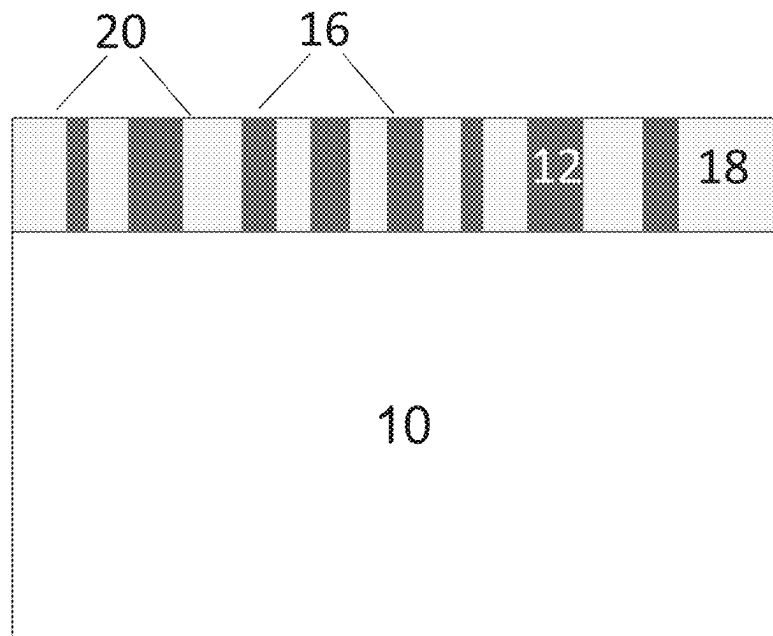


FIG. 1F

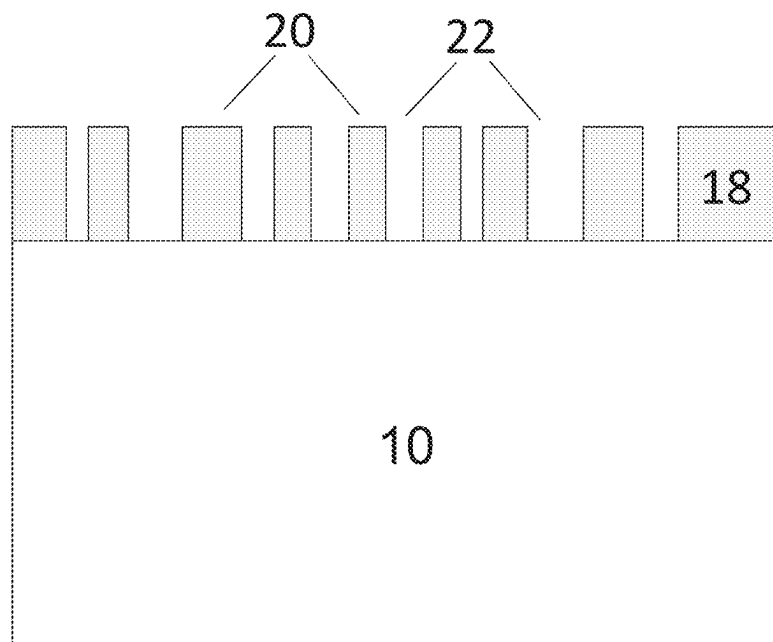


FIG. 1G

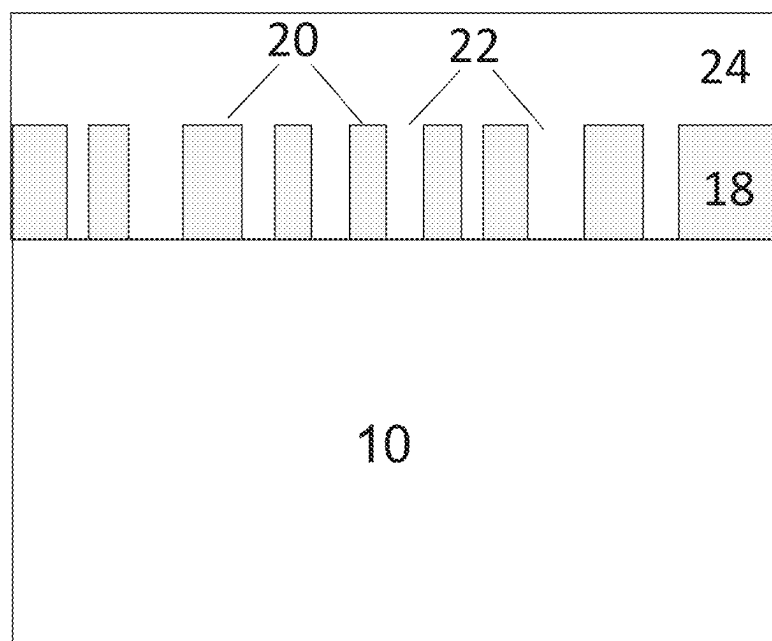


FIG. 2B

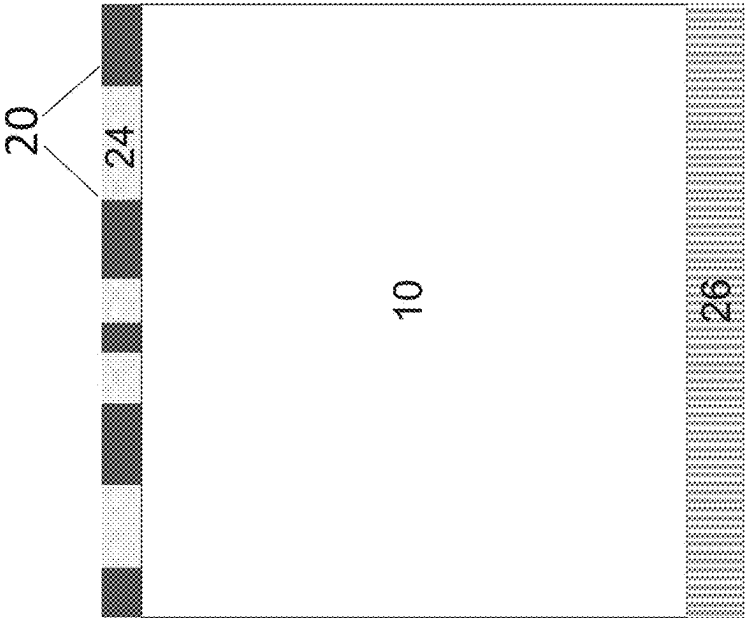


FIG. 2A

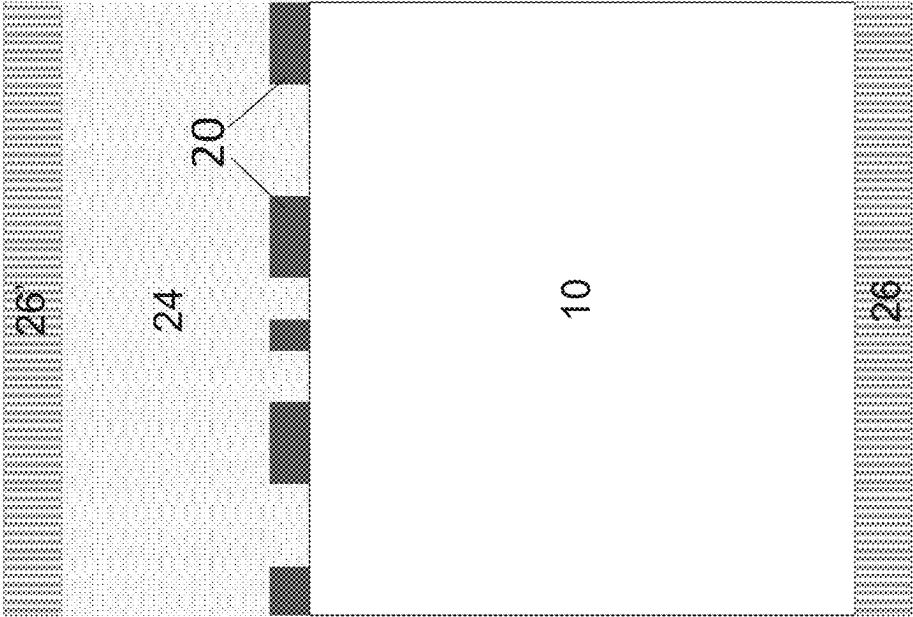


FIG. 3

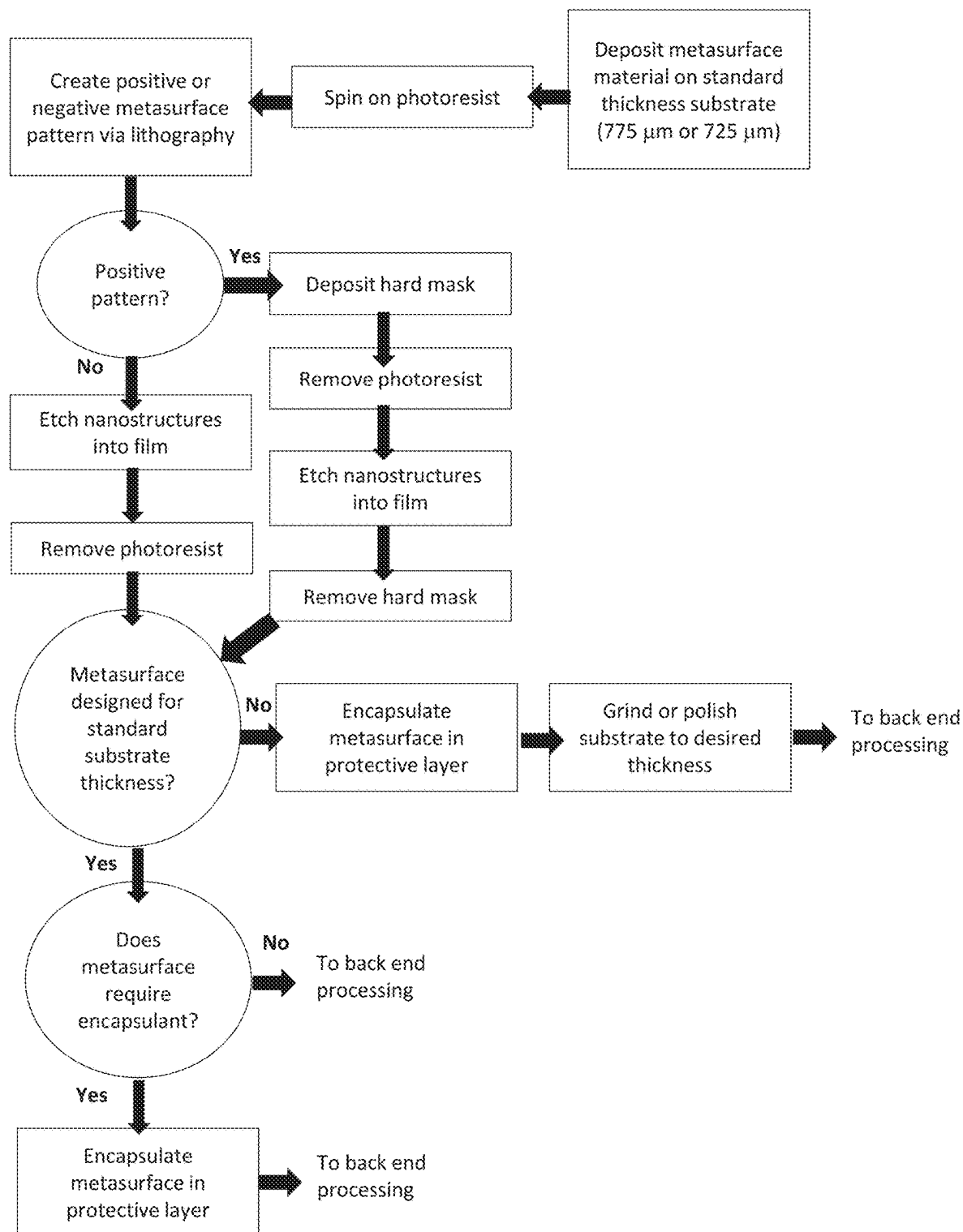


FIG. 4A

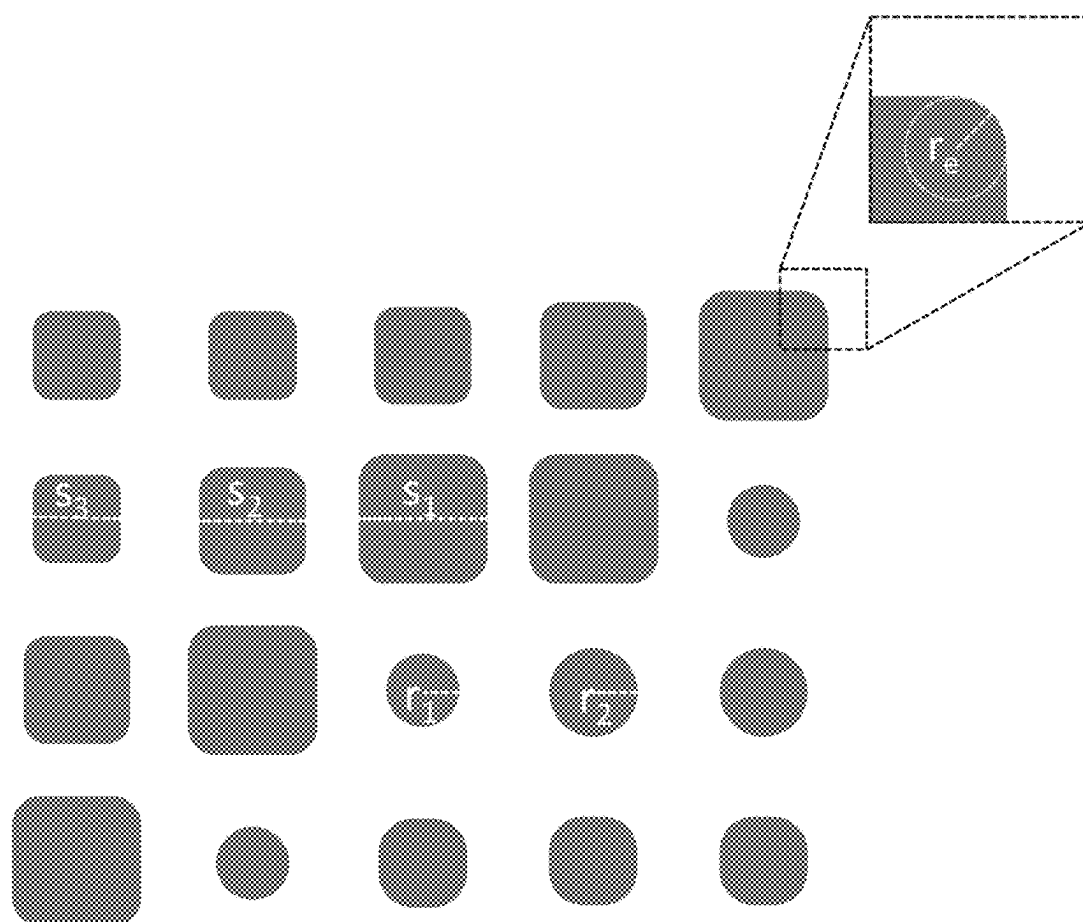


FIG. 4B

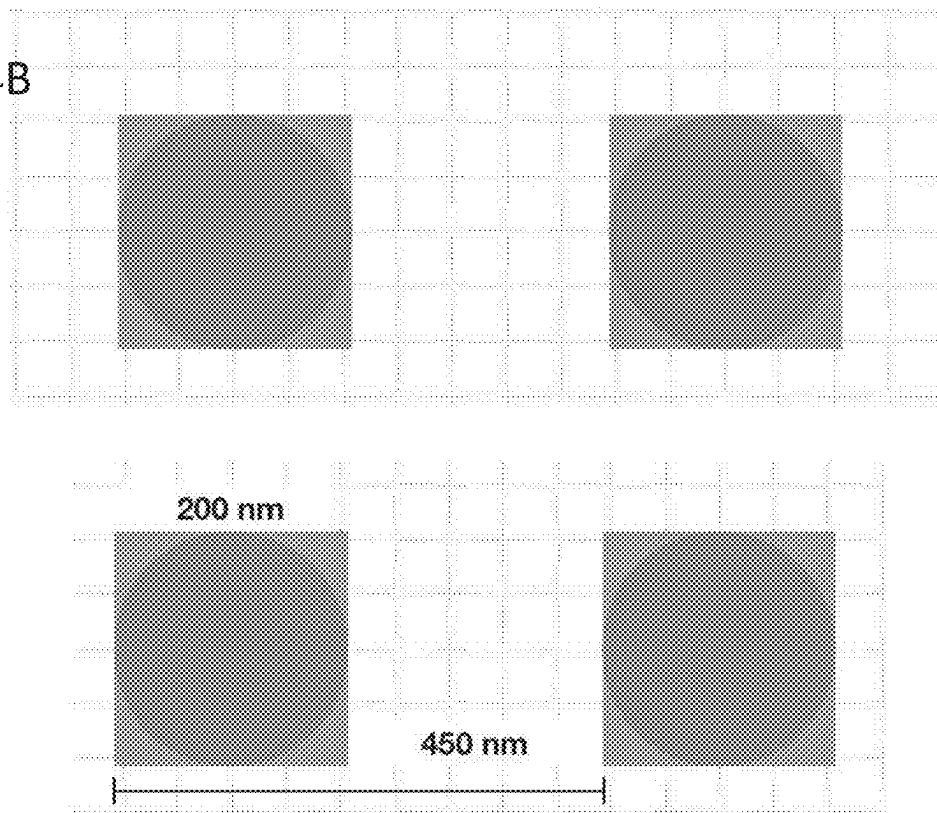


FIG. 4C

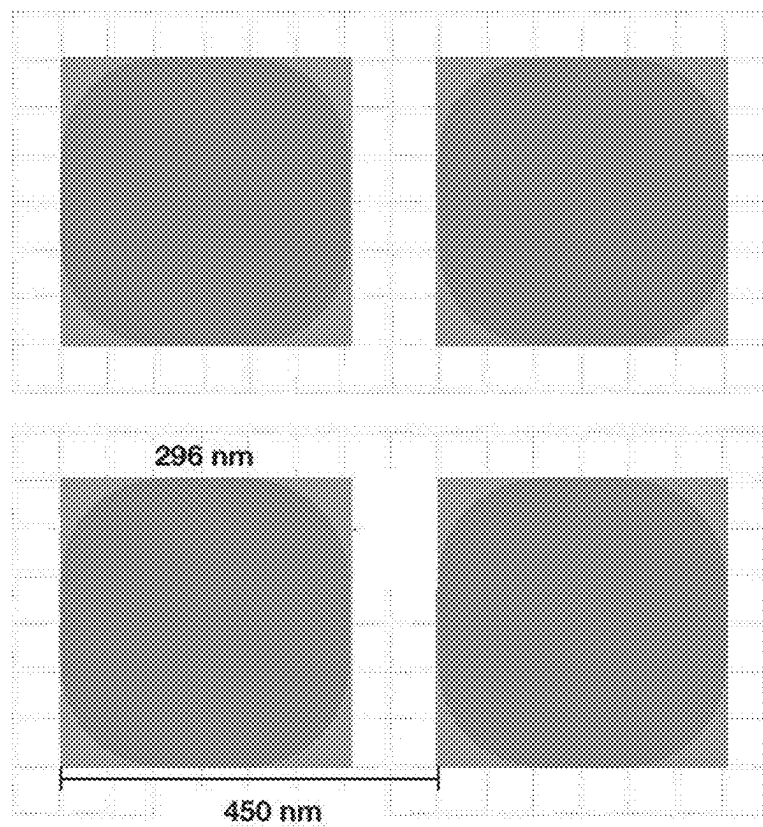


FIG. 5

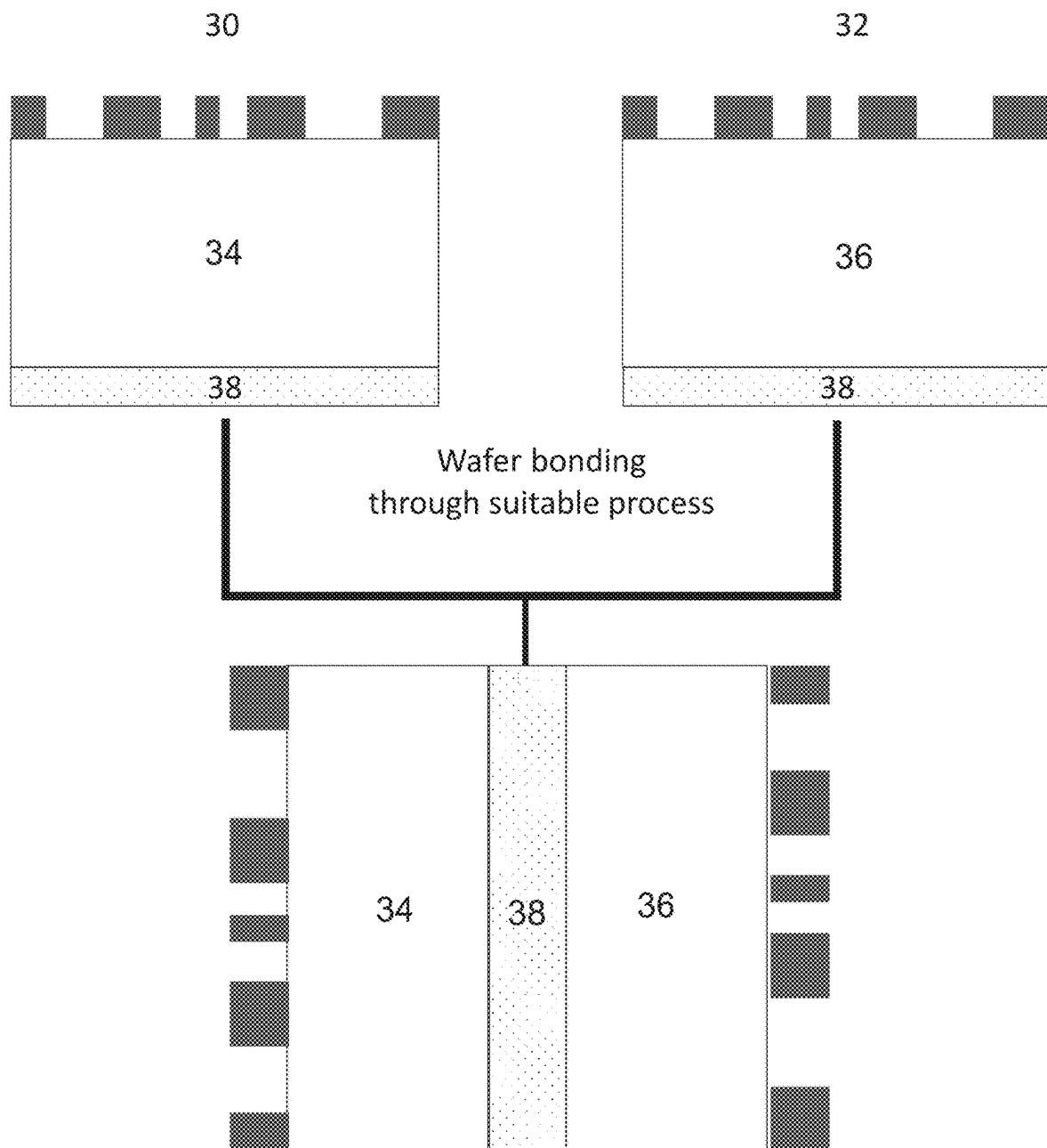


FIG. 6

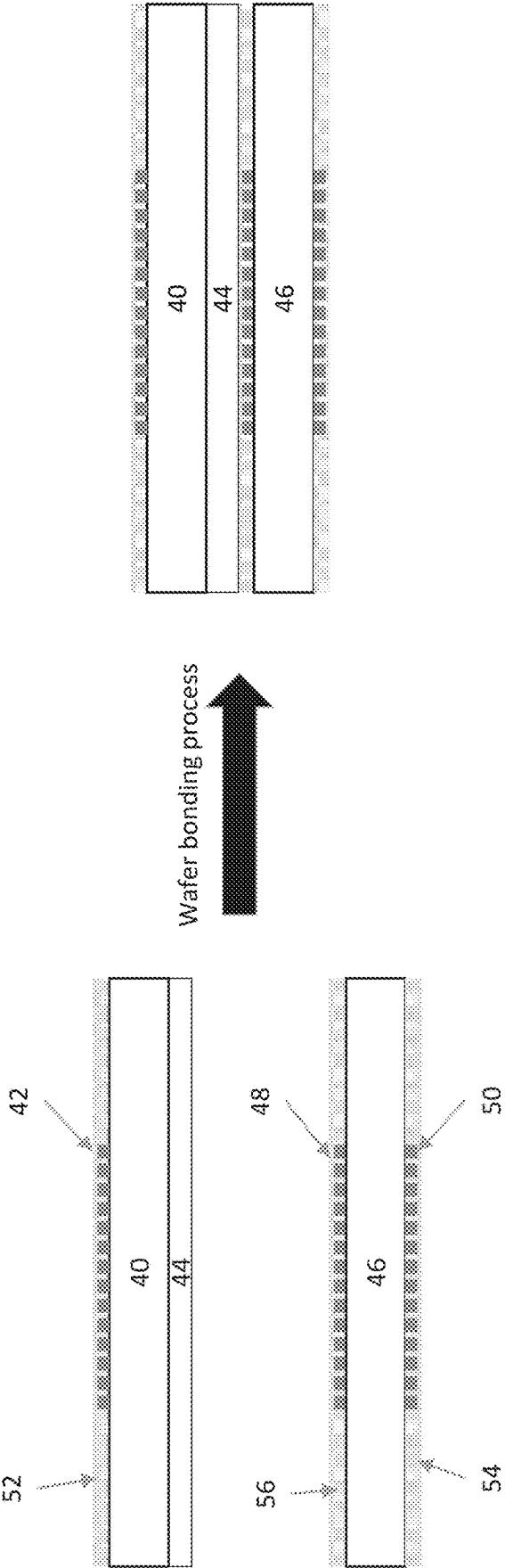


FIG. 7

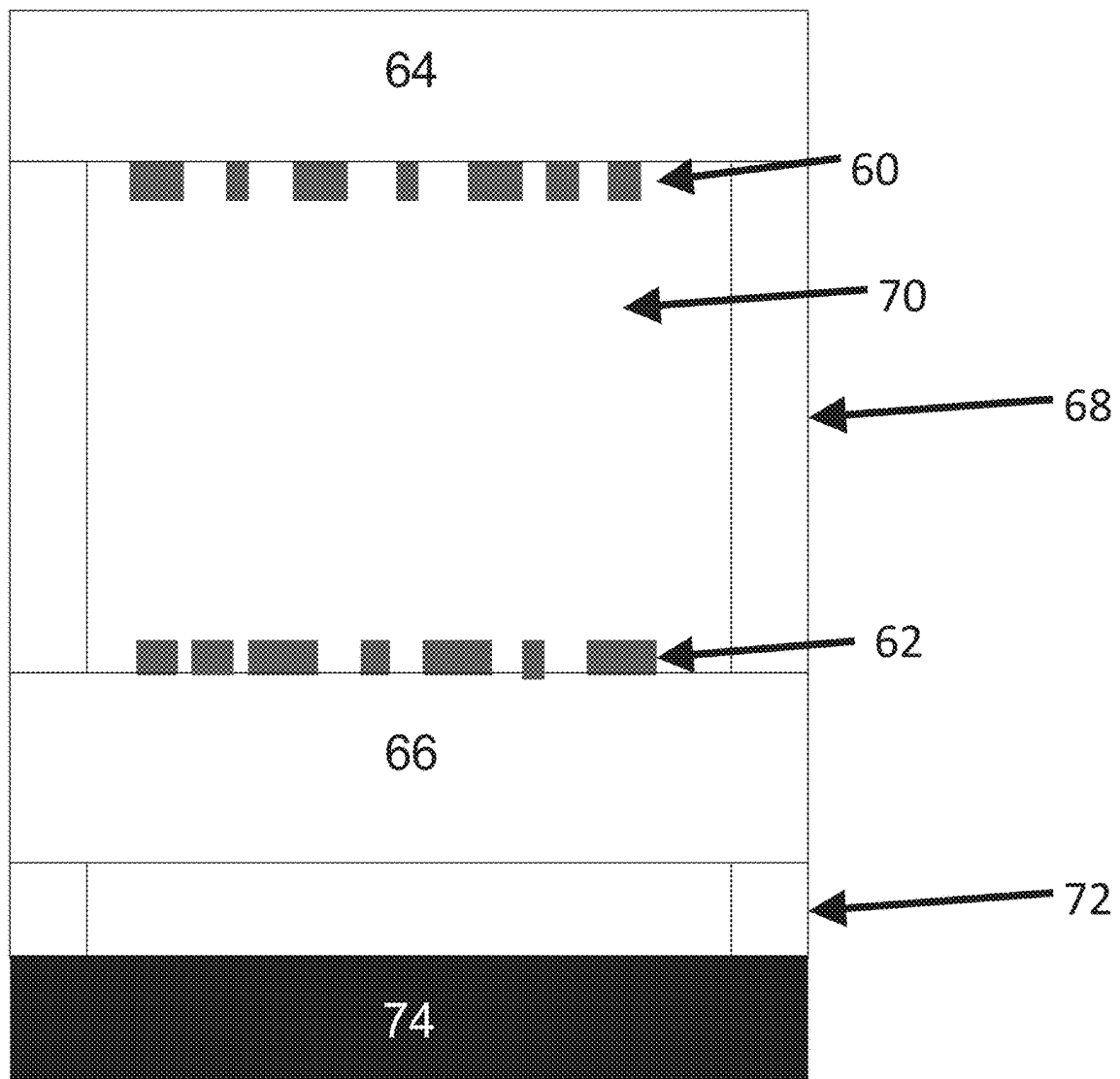


FIG. 8A

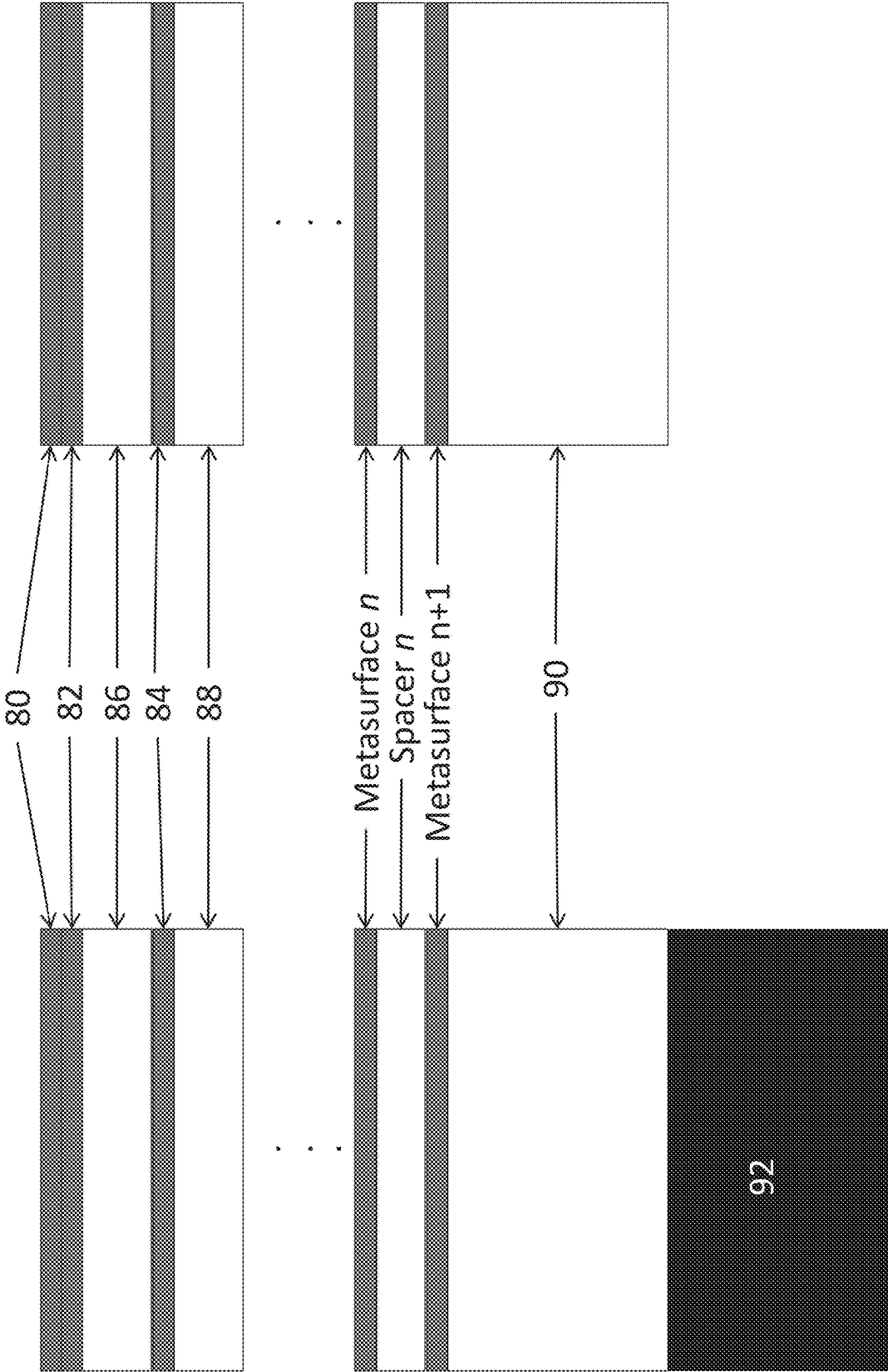


FIG. 8B

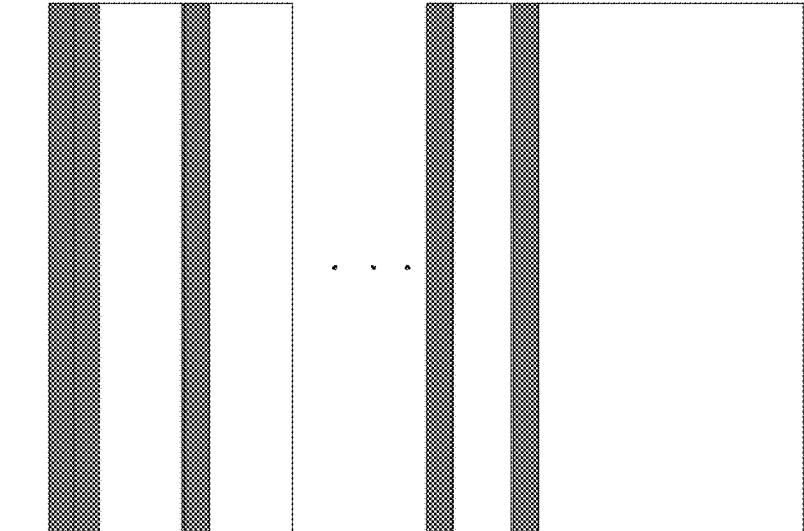


FIG. 9

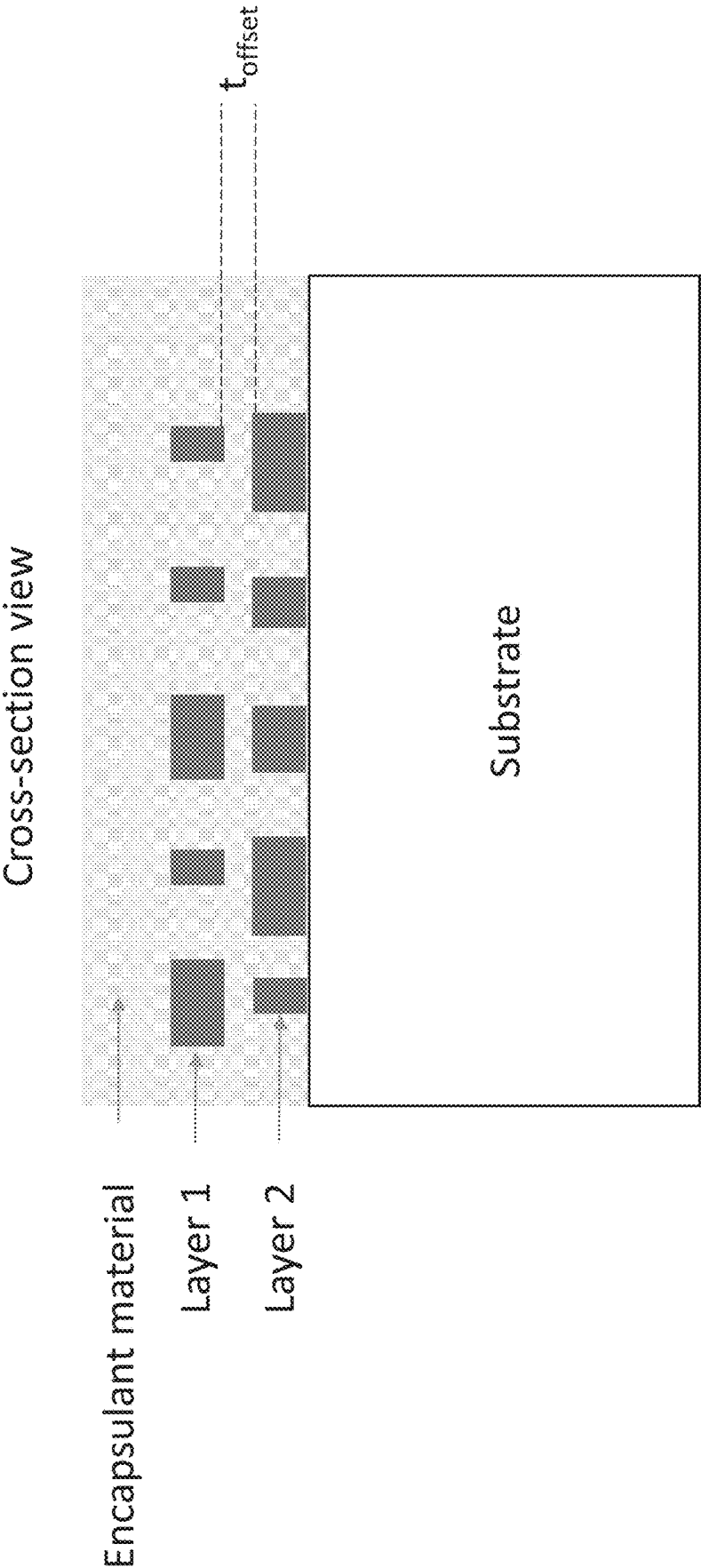


FIG. 10A

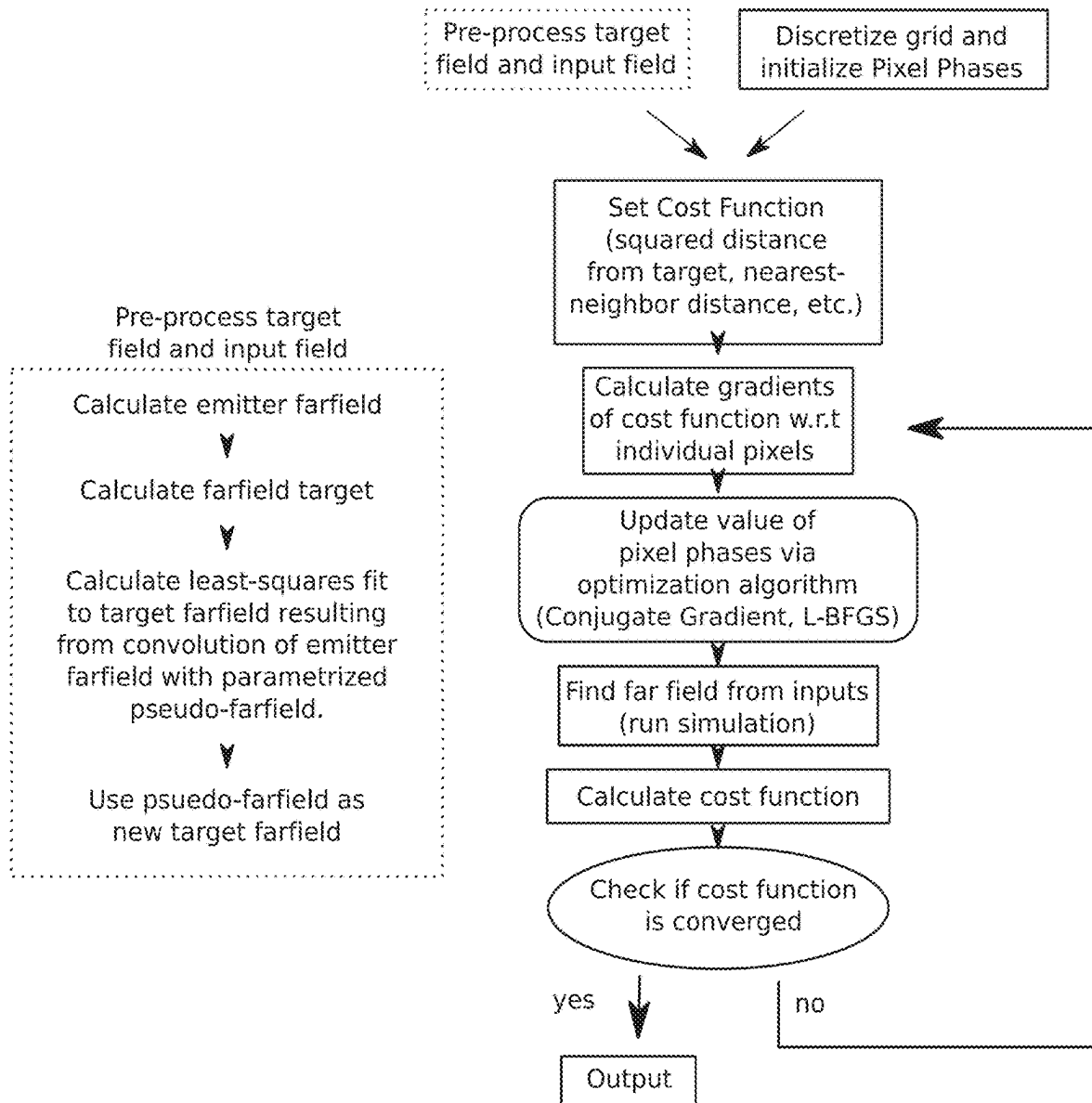


FIG. 10B

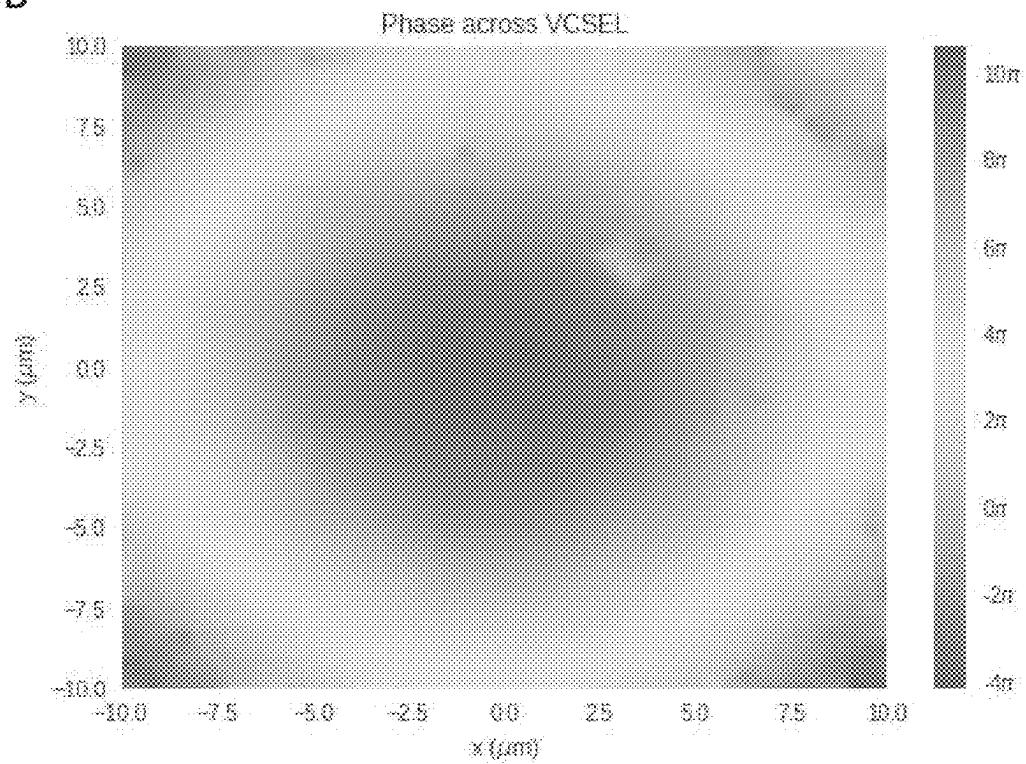


FIG. 10C

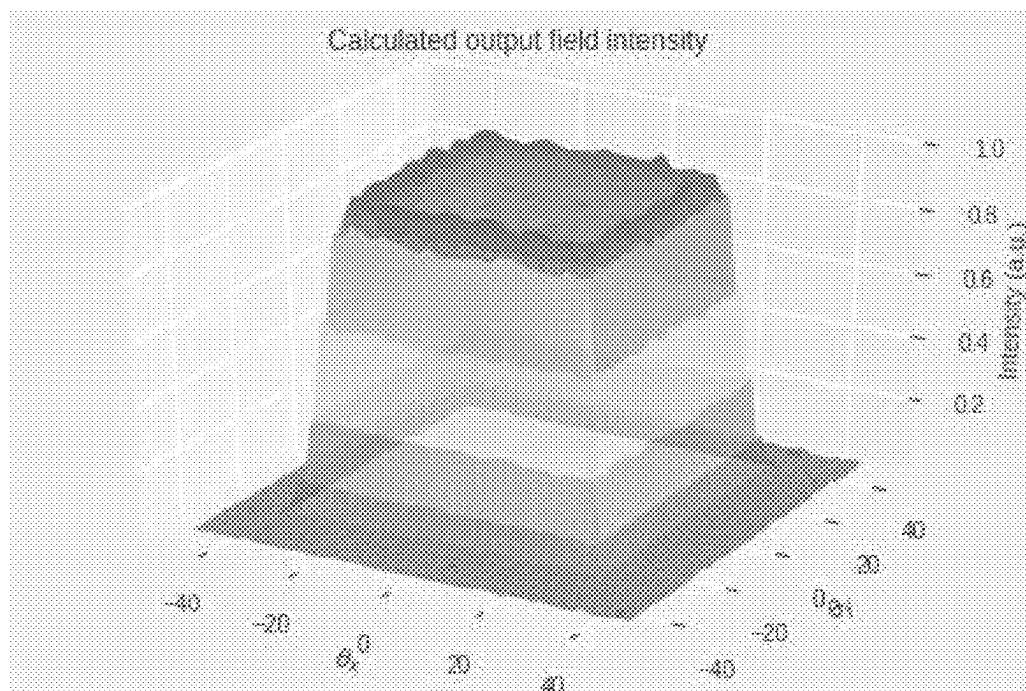


FIG. 11

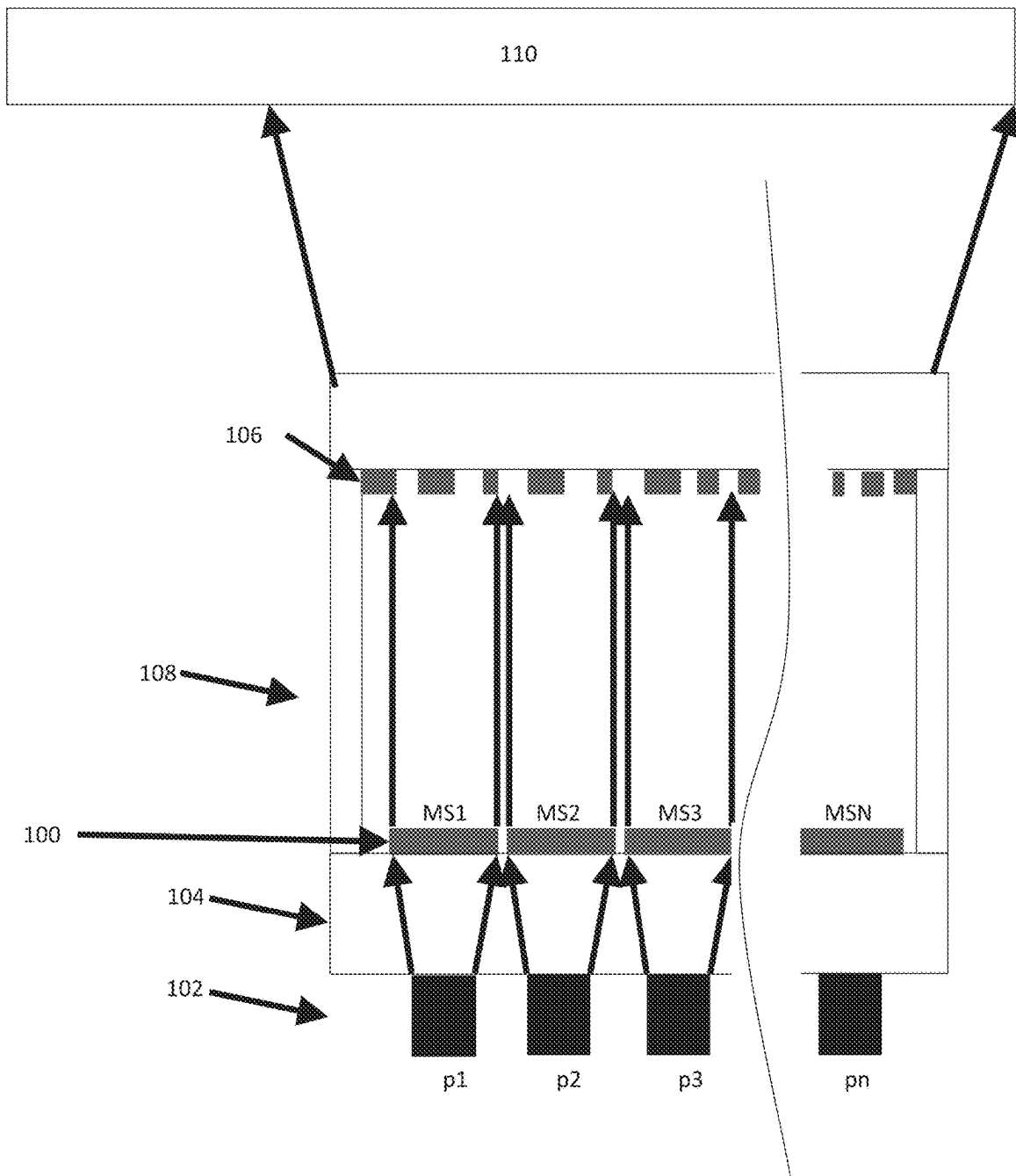


FIG. 12A

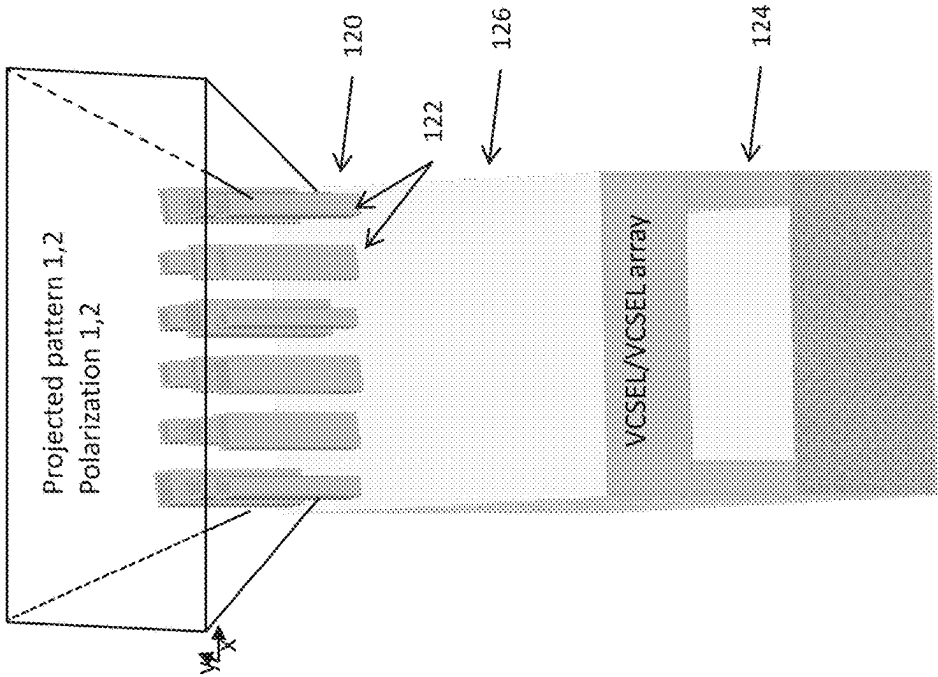


FIG. 12B

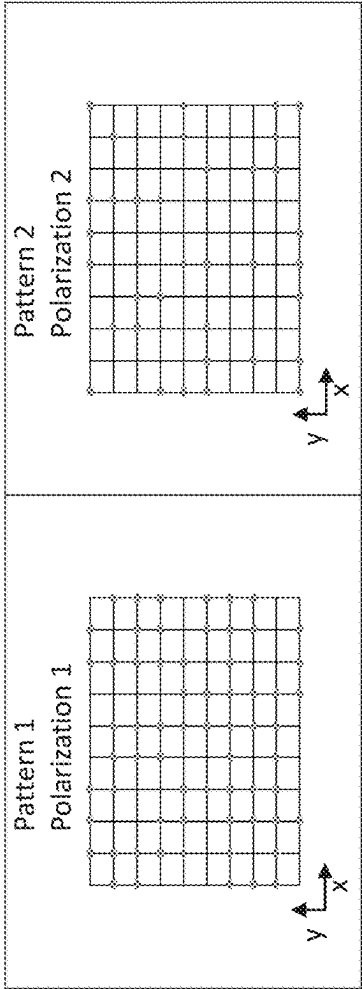


FIG. 12C

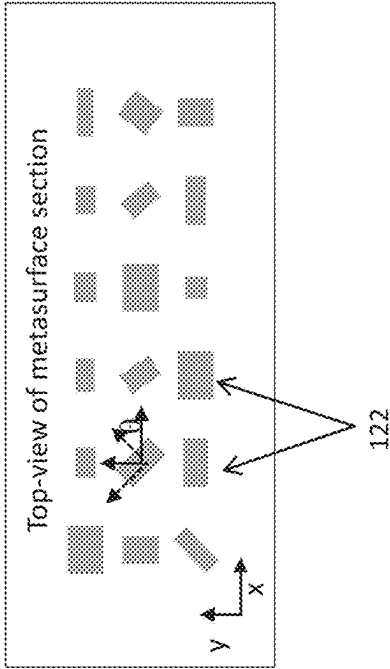


FIG. 13

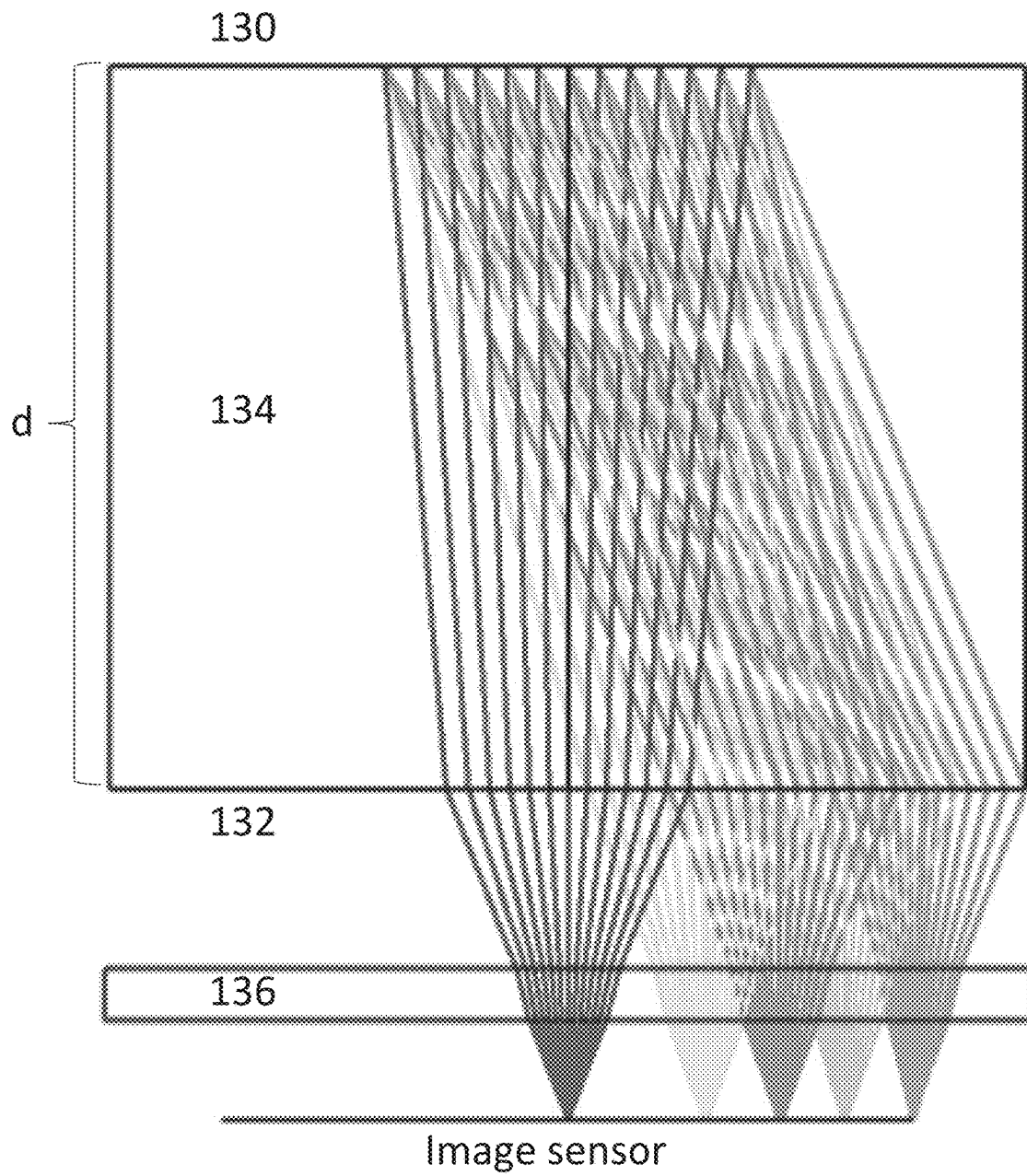
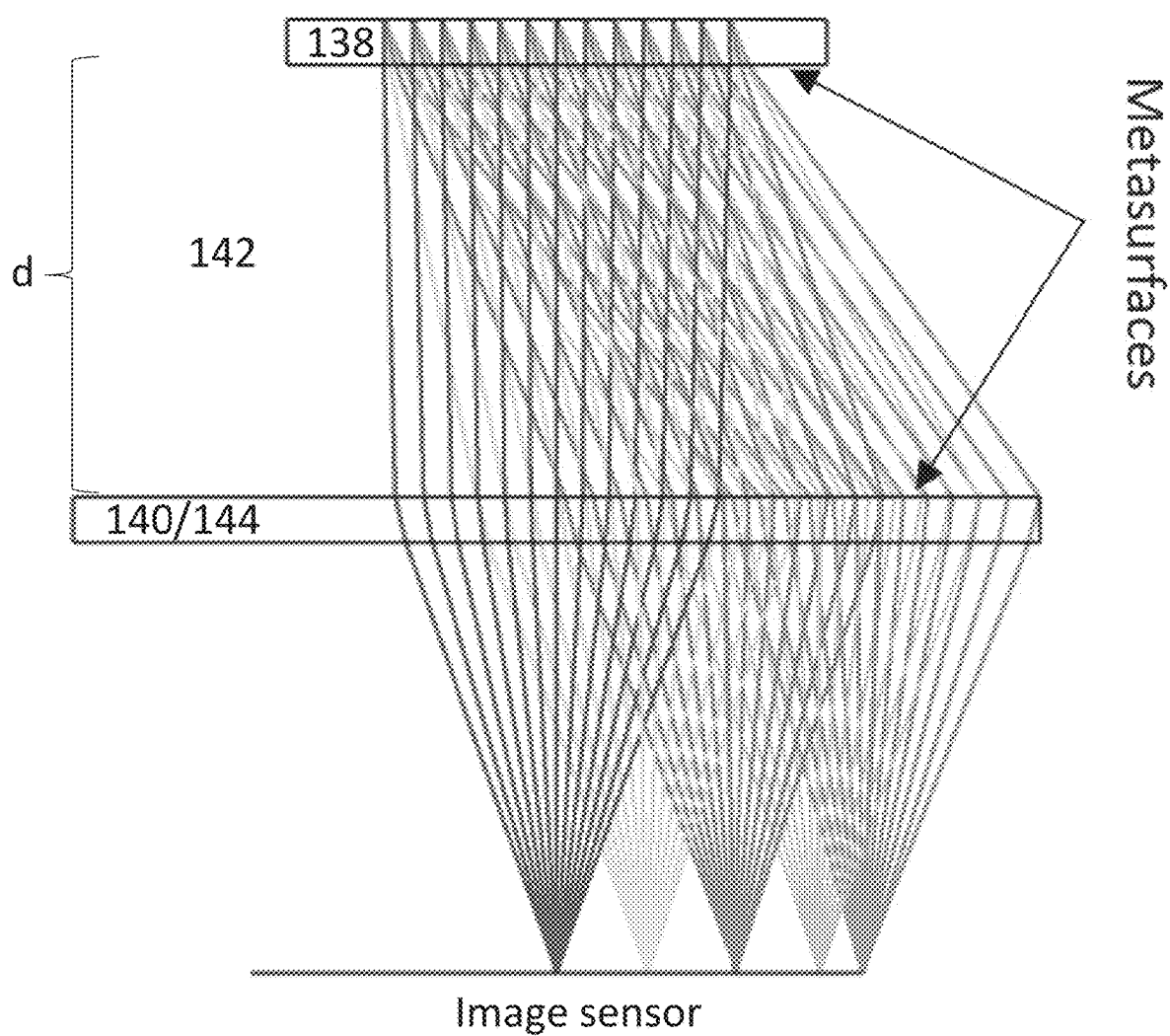


FIG. 14



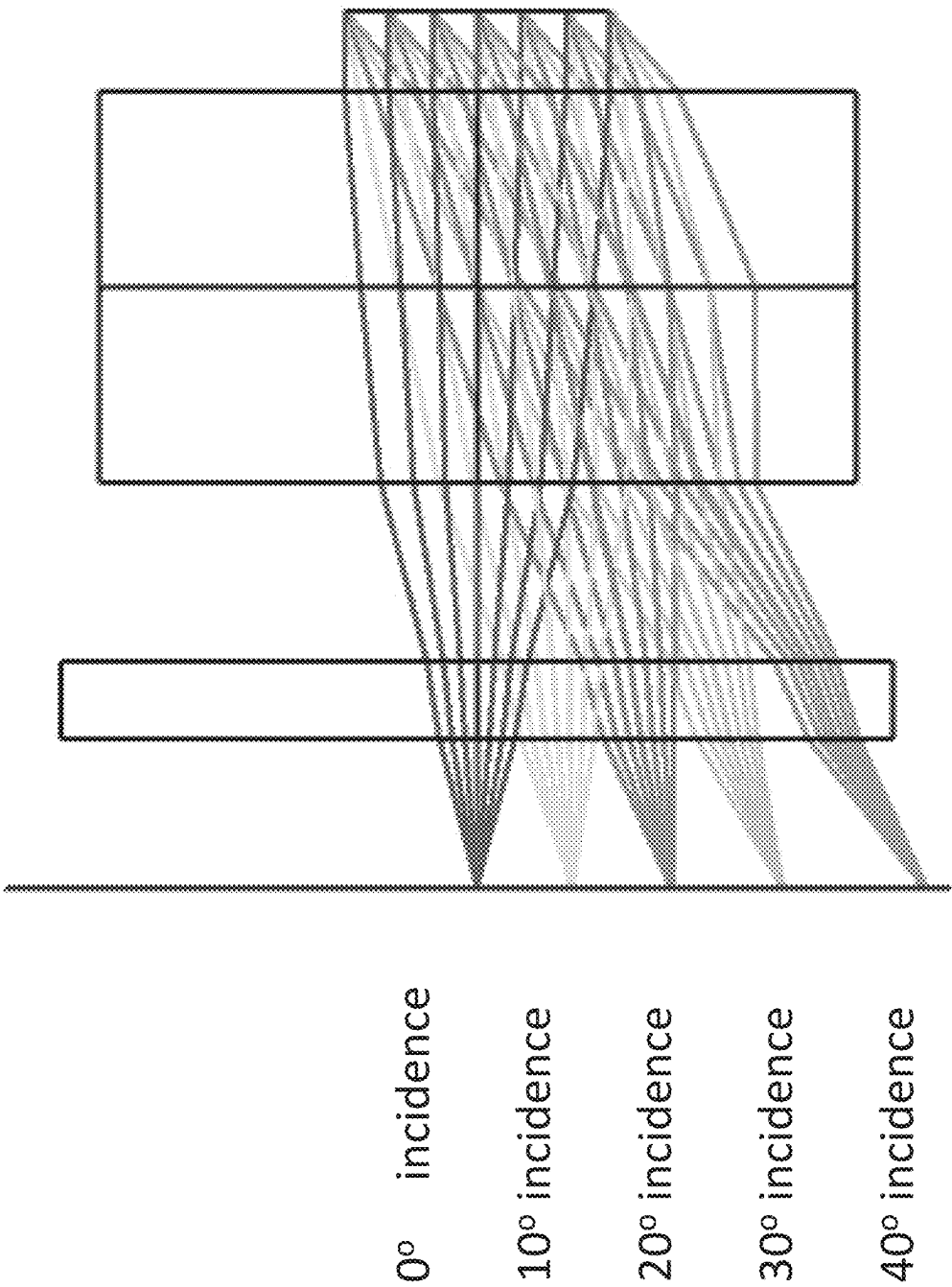


FIG. 15

FIG. 16

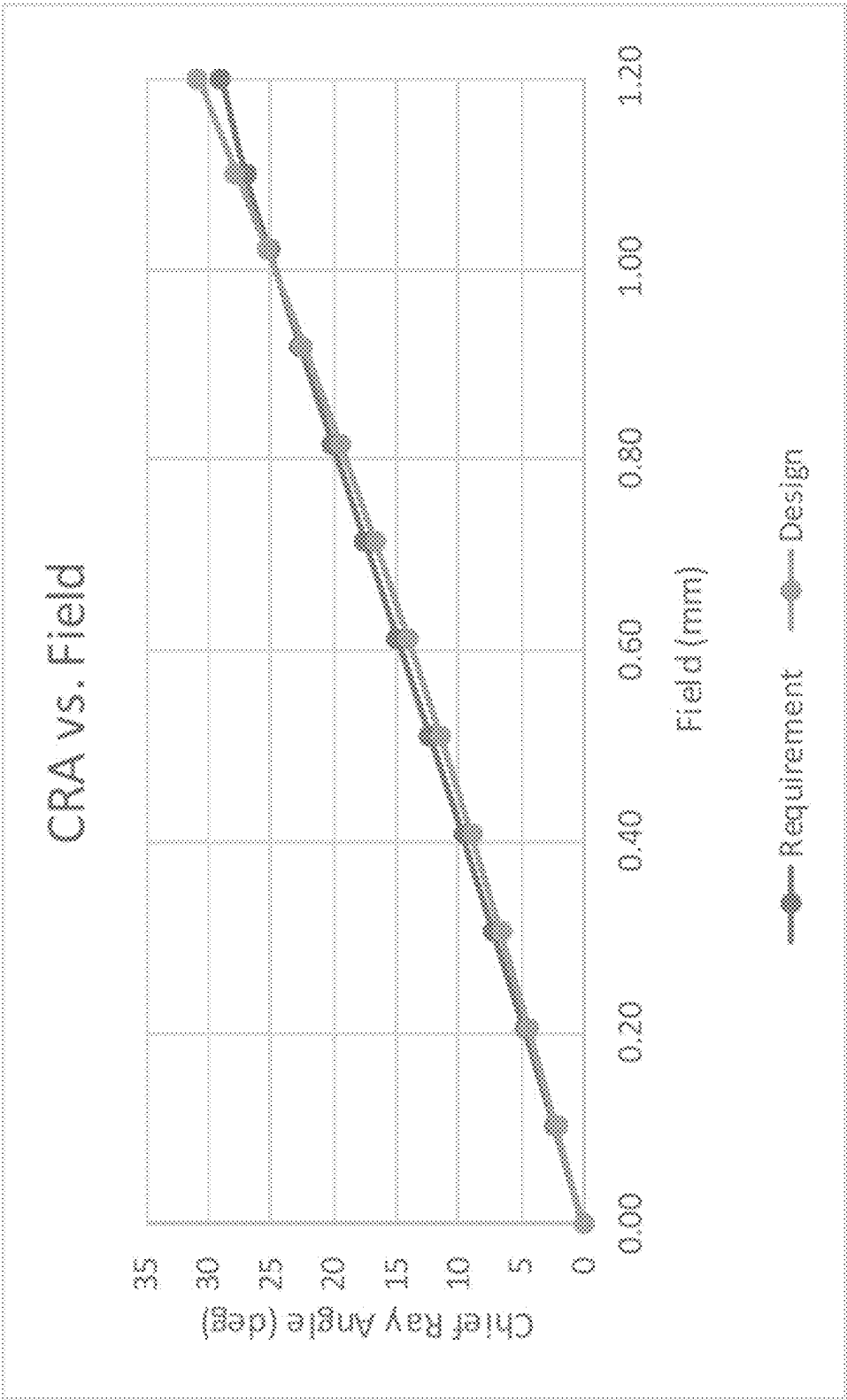


FIG. 17

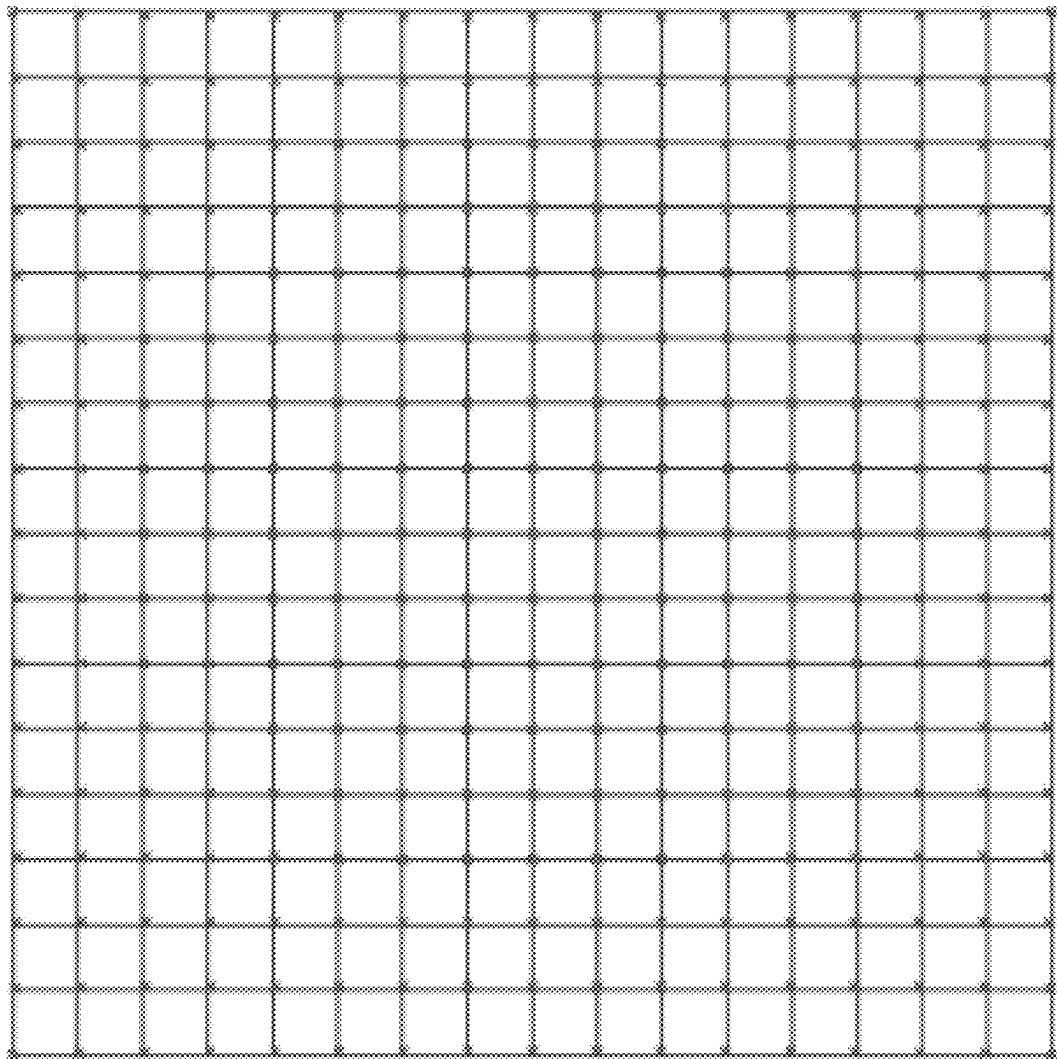


FIG. 18

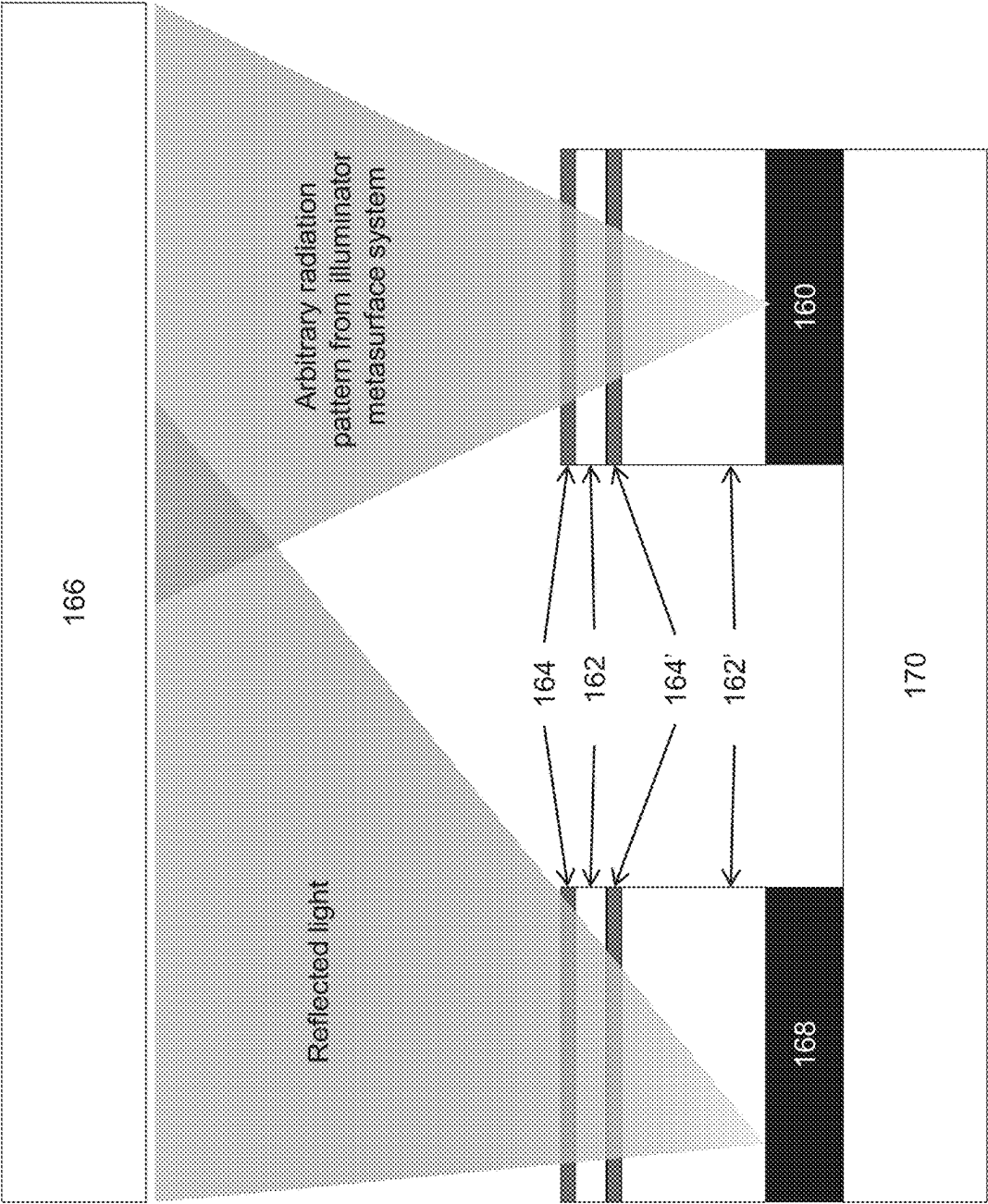


FIG. 19

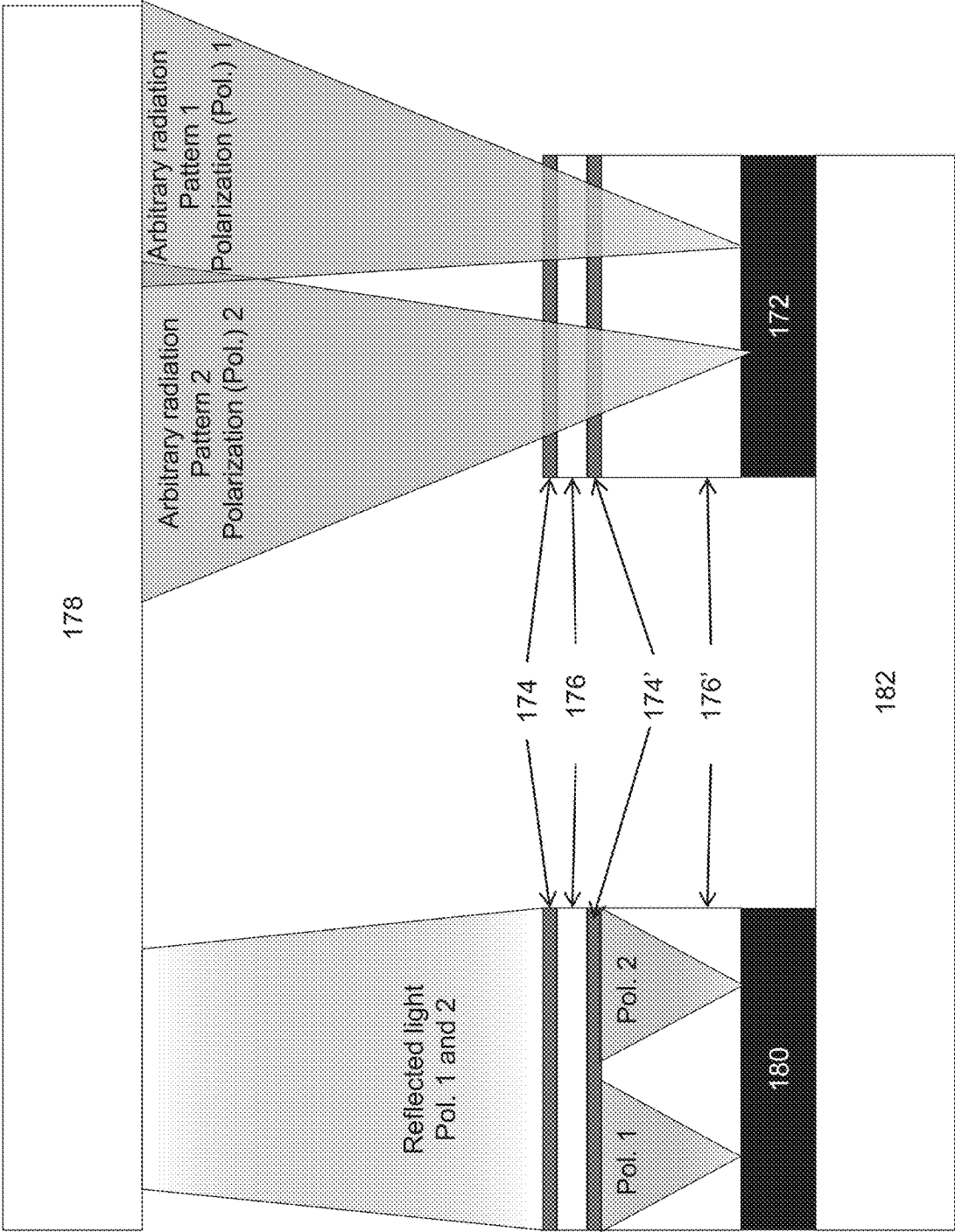


FIG. 20

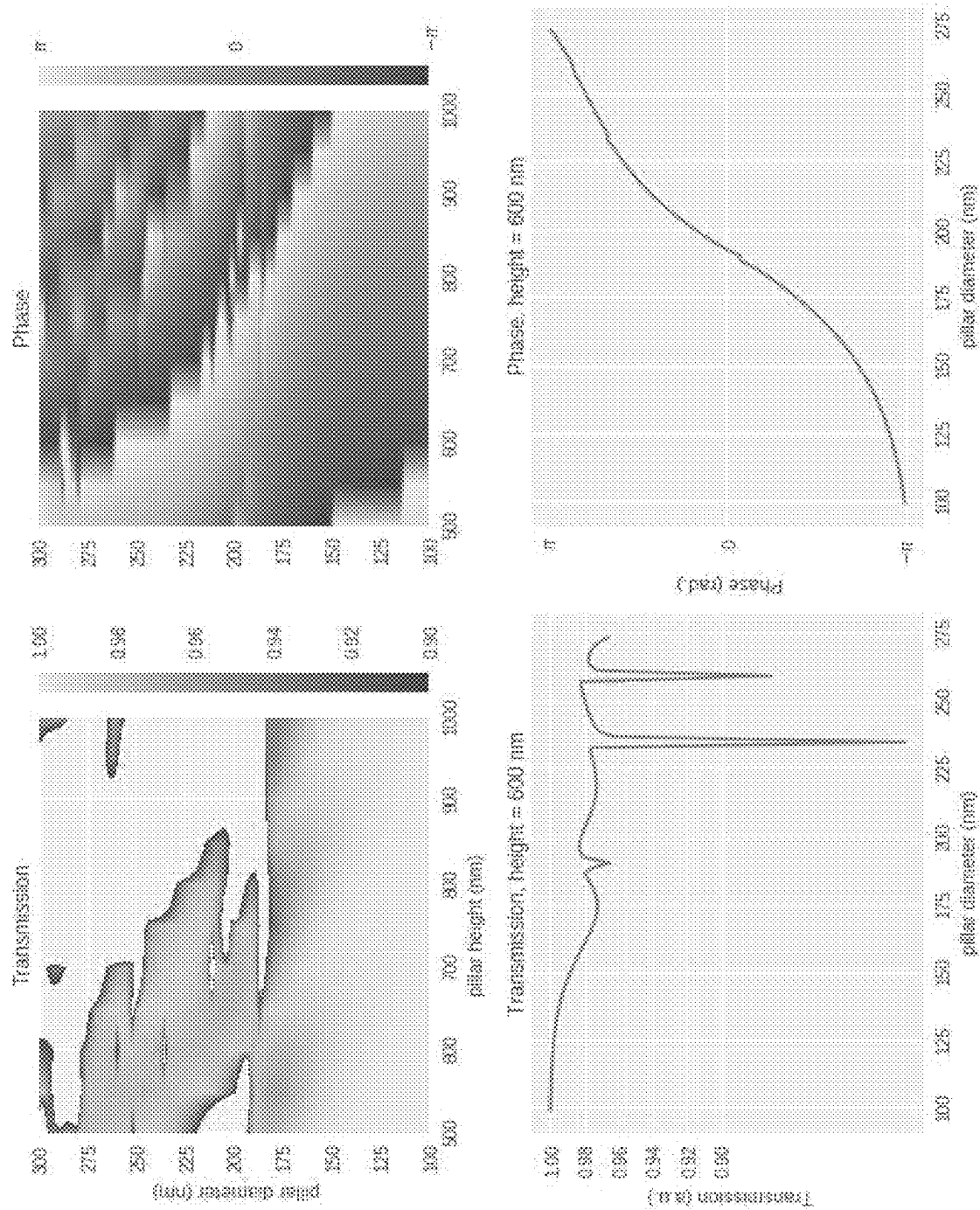


FIG. 21

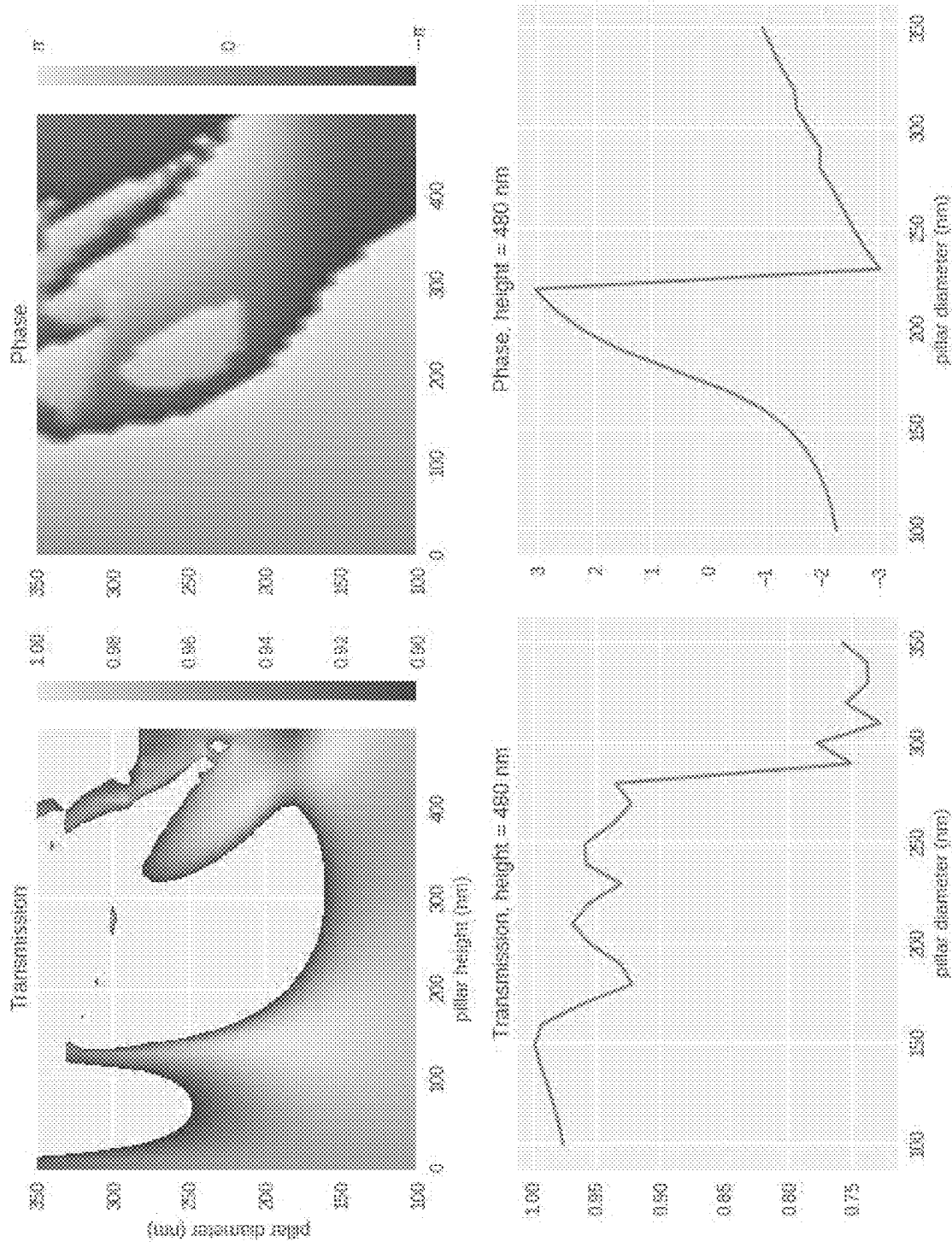


FIG. 22

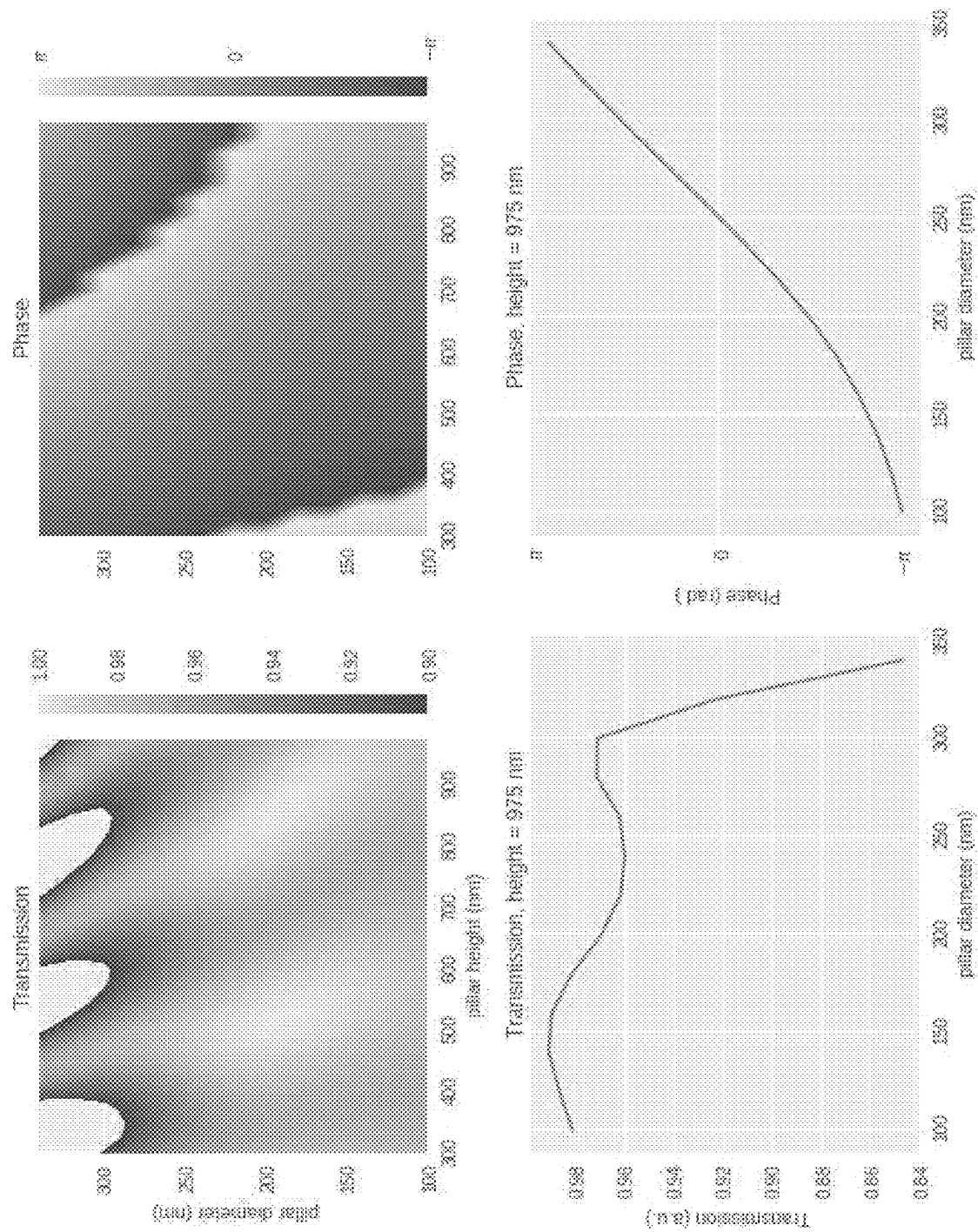


FIG. 23

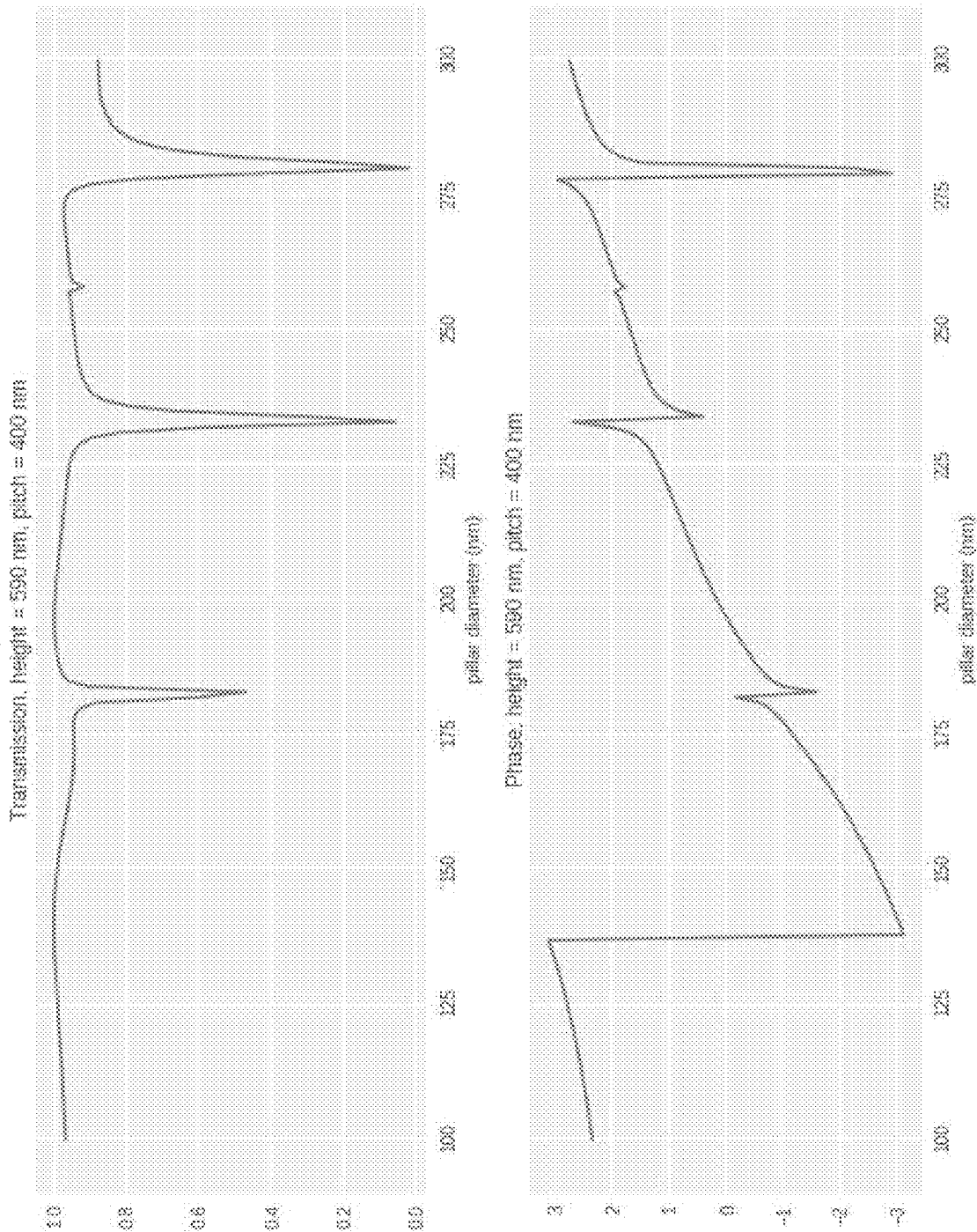
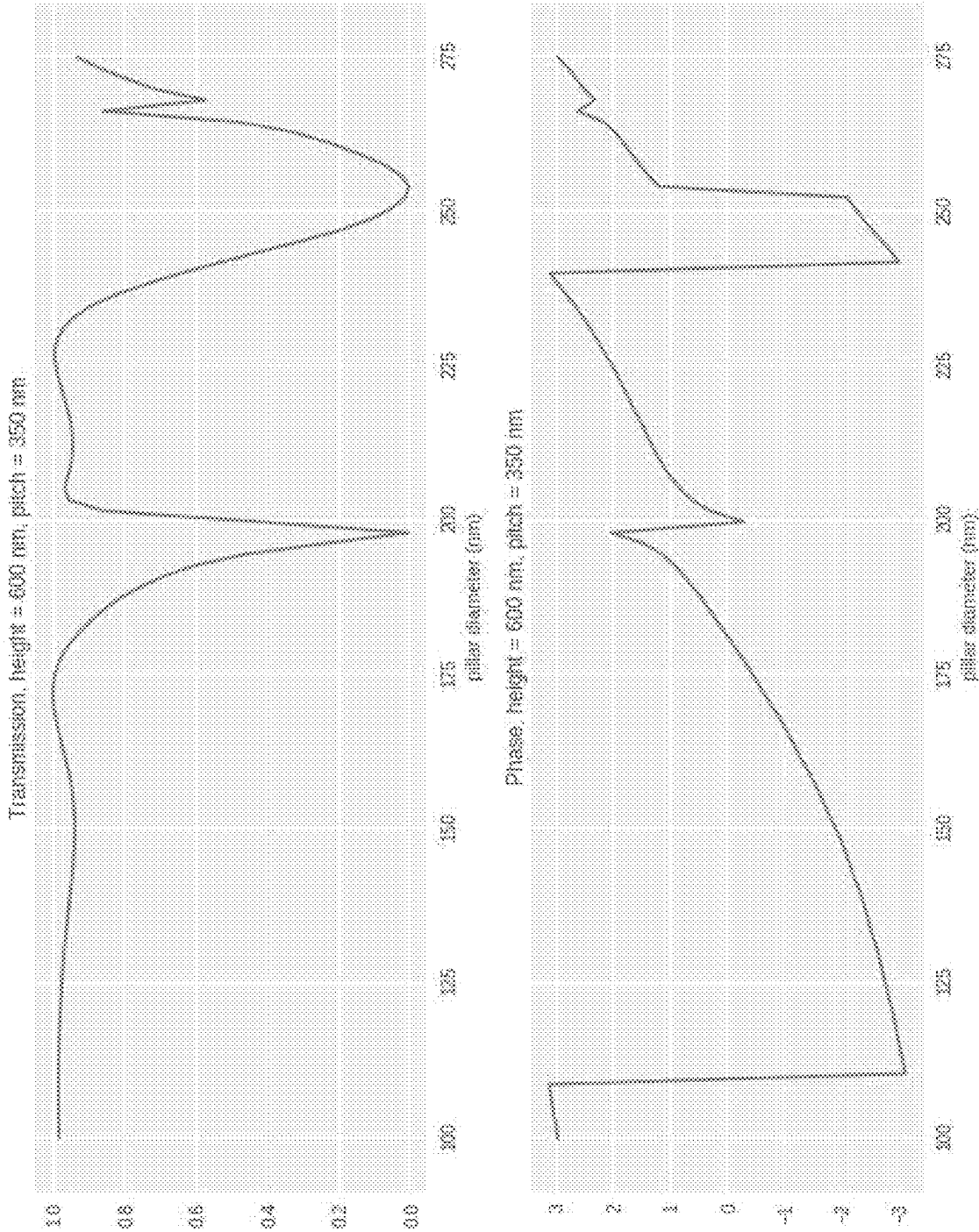


FIG. 24



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TRANSMISSIVE METASURFACE LENS INTEGRATION

CROSS-REFERENCE TO RELATED APPLICATIONS

The current application is a continuation of U.S. patent application Ser. No. 17/643,091 filed Dec. 7, 2021, which is a continuation of U.S. patent application Ser. No. 15/931,184 filed May 13, 2020, which is a continuation of U.S. patent application Ser. No. 16/120,174 filed Aug. 31, 2018 and issued on Oct. 6, 2020 as U.S. Pat. No. 10,795,168, which application claims priority to U.S. Provisional Patent Application No. 62/552,455, filed Aug. 31, 2017, the disclosures of which are incorporated herein by reference.

FIELD OF THE INVENTION

The current disclosure is directed optical arrangements of metasurface elements, integrated systems incorporating light sources and/or detectors with such metasurface elements, and methods of the manufacture of such optical arrangements and integrated systems.

BACKGROUND OF THE INVENTION

Metasurface elements are diffractive optics in which individual waveguide elements have subwavelength spacing and have a planar profile. Metasurface elements have recently been developed for application in the UV-IR bands (300-10,000 nm). Compared to traditional refractive optics, metasurface elements abruptly introduce phase shifts onto light field. This enables metasurface elements to have thicknesses on the order of the wavelength of light at which they are designed to operate, whereas traditional refractive surfaces have thicknesses that are 10-100 times (or more) larger than the wavelength of light at which they are designed to operate. Additionally, metasurface elements have no variation in thickness in the constituent elements and thus are able to shape light without any curvature, as is required for refractive optics. Compared to traditional diffractive optical elements (DOEs), for example binary diffractive optics, metasurface elements have the ability to impart a range of phase shifts on an incident light field, at a minimum the metasurface elements can have phase shifts between $0-2\pi$ with at least 5 distinct values from that range, whereas binary DOEs are only able to impart two distinct values of phase shift and are often limited to phase shifts of either 0 or 1π . Compared to multi-level DOE's, metasurface elements do not require height variation of its constituent elements along the optical axis, only the in-plane geometries of the metasurface element features vary.

BRIEF SUMMARY OF THE INVENTION

The application is directed to optical arrangements of metasurface elements, integrated systems incorporating light sources and/or detectors with such metasurface elements, and methods of the manufacture of such optical arrangements and integrated systems.

Many embodiments are directed to methods for fabricating one or more metasurface elements or systems including: depositing a hard mask material layer on at least one surface of a substrate, wherein the substrate is transparent to light over a specified operational bandwidth; depositing a pattern material layer on the hard mask material layer;

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patterning the pattern material to form an array pattern atop the hard mask layer, the array pattern comprising one of either a positive or negative reproduction of a metasurface feature array, the metasurface feature array comprising a plurality of metasurface features having feature sizes smaller than the wavelength of light within the specified operational bandwidth and configured to impose a phase shift on impinging light within the plane of plurality of metasurface features;

etching the hard mask layer using an anisotropic etch process to form a plurality of voids and raised features corresponding to the array pattern in the hard mask; and removing any residual pattern material from atop the hard mask layer.

In many other embodiments, the substrate is formed of a material selected from the group consisting of: fused silica, sapphire, borosilicate glass and rare-earth oxide glasses.

In still many other embodiments, the hard mask material layer is formed of a material selected from the group consisting of: silicon, silicon nitride of various stoichiometries, silicon dioxide, titanium dioxide, alumina, and is disposed using a deposition process selected from the group consisting of: sputtering, chemical vapor deposition, and atomic layer deposition.

In yet many other embodiments, the pattern material layer is formed from one of either a photoresist patterned using a lithographic process, or a polymer patterned using a nano-imprint process.

In still yet many other embodiments, the array pattern is etched using a reactive ion etching process selected from the group consisting of: SF_6 , Cl_2 , BCl_3 , C_4F_8 or any static or multiplexed mixture thereof.

In still yet many other embodiments, the residual pattern material is removed using a process selected from the group consisting of: chemical solvent, chemical etchant, and plasma etchant.

In still yet many other embodiments, the patterned hard mask material is a dielectric and forms the metasurface features of the metasurface element.

In still yet many other embodiments, the methods further includes:

depositing a dielectric metasurface material layer on the patterned hard mask material layer such that the metasurface material layer fills the voids in the hard mask material layer and extends above the raised features of the hard mask material layer forming an over-layer of metasurface material atop the hard mask layer; and planarizing the over-layer such that the metasurface material layer and the hard mask layer terminate at a uniform height above the substrate.

In still yet many other embodiments, the metasurface material layer is formed from a material selected from silicon, silicon nitride of various stoichiometries, silicon dioxide, titanium dioxide, alumina, and is deposited using a conformal process selected from the group of: chemical vapor deposition, and atomic layer deposition.

In still yet many other embodiments, the planarization uses a process selected from an etch process selected from the group consisting of wet etch and a plasma etch, or a chemical-mechanical planarization technique.

In still yet many other embodiments, the metasurface material disposed in the voids forms the metasurface features of the metasurface element, and wherein the hard mask material is configured as an embedding material having a lower index of refraction at the specified operational bandwidth than the metasurface material.

In still yet many other embodiments, the hard mask material has negligible absorption over the specified operational bandwidth and has an index of refraction at the specified operational bandwidth between about 1 and about 2.4

In still yet many other embodiments, the method further includes removing the hard mask material layer using a selective etch such that the metasurface material layer disposed in the voids of the patterned hard mask remains on the surface of the substrate after removal of the hard mask material layer to form a plurality of isolated metasurface features separated by a plurality of air gaps.

In still yet many other embodiments, the method further includes depositing an embedding material layer on the isolated metasurface features such that the air gaps between the features are filled and such that the embedding material layer extends above the surface of the metasurface material layer, wherein the embedding material layer has a lower index of refraction at the specified operational bandwidth than the metasurface material.

In still yet many other embodiments, the embedding material is a polymer selected from the group consisting of poly(methyl methacrylate), SU8, and benzocyclobutene.

In still yet many other embodiments, the embedding material is a solid film selected from the group consisting of silicon dioxide, aluminum oxide, titanium dioxide, silicon nitride, hafnium oxide, zinc oxide, and spin-on-glass.

In still yet many other embodiments, the method further includes planarizing the embedding material layer such that the metasurface material layer and the embedding material layer terminate at a uniform height above the substrate.

In still yet many other embodiments, the method further includes depositing an anti-reflective coating atop one or both the embedding material layer and the side of the substrate disposed opposite the metasurface element.

In still yet many other embodiments, the antireflective coating is composed of alternating layers of any combination of materials selected from the group consisting of silicon dioxide, titanium dioxide, aluminum oxide, silicon nitride, aluminum nitride, and amorphous silicon, wherein each of the alternating layers has a thickness less than the wavelength of light within the operational bandwidth.

In still yet many other embodiments, the substrate is one of either disposed atop an illuminator or sensor, or is itself an illuminator or sensor.

In still yet many other embodiments, the substrate has a substrate thickness unsuitable for use with a target optical system at and further comprising at least one of the following:

- removing at least a portion of the backside of the substrate through one or both grinding or chemical etching; and
- aligning and fusing an additional substrate to the substrate of the metasurface element.

In still yet many other embodiments, the additional substrate itself has a metasurface element disposed on one surface thereof, and wherein the substrate and additional substrate are fused along surface opposite the surfaces on which the relative metasurface elements are disposed.

In still yet many other embodiments, the method of fusing uses a bonding process having a thermal budget below 600° C.

In still yet many other embodiments, the bonding process is a wafer bonding process using an adhesive selected from the group of an optical epoxy, benzocyclobutene, a UV cured polymer, SU8, and a plasma activate silicon dioxide film.

In still yet many other embodiments, the method further includes removing at least a portion of the backside of one or both of the substrates prior to fusing.

In still yet many other embodiments, the method further includes forming at least a first metasurface element on a first side of a first substrate, and forming at least a second metasurface element on a first side of a second substrate, and fusing the first and second substrates together along sides opposite the first sides of said substrates using a bonding process having a thermal budget below 600° C.

In still yet many other embodiments, the plurality of metasurface features are inhomogeneous.

In still yet many other embodiments, the plurality of metasurface features diverge from an ideal shape by a pre-determinable amount based on the dimensions of the metasurface features.

In still yet many other embodiments, the metasurface element is embedded and planarized and comprises two layers of metasurface features offset from each other by a distance smaller than or on the same order as the wavelength of light within the specified operational bandwidth such that the two layers of metasurface features operate in conjunction to impose a phase shift on impinging light.

In still yet many other embodiments, the plurality of metasurface features are inhomogeneous and diverge from an ideal shape by a pre-determinable amount based on the dimensions of the metasurface features, and wherein the ideal shape is a square, and where the ideal square has a side dimension of less than 200 nm the metasurface features are formed as circles, and where the ideal square has a side dimension of less than 300 nm the metasurface features are formed as squares having rounded edges.

In still yet many other embodiments, the method further includes:

- forming a plurality of identical or unique first metasurface elements;

- providing a plurality of identical or unique illumination sources disposed in a planar array and integrating at least one of the plurality of first metasurface elements with each of the plurality of illumination sources in the array such that light from each of said plurality of illumination sources passes through at least one of the first metasurface elements and an angular deflection is imposed thereby;

- disposing a first spacer layer between the planar array of illumination sources and the first metasurface elements, the first spacer layer being configured to create divergence in light emitted from each of the illumination sources of the planar array prior to impinging on the respective first metasurface element;

- disposing a second metasurface element at a distance from the plurality of first metasurface elements, the second metasurface element configured to imprint a far-field illumination pattern onto a light field formed by the emission of all of the plurality illumination sources; and

- disposing a second spacer layer between the first and second metasurface elements such that an offset distance is formed therebetween.

Various embodiments are directed to methods of forming a multi-metasurface element comprising forming at least a first metasurface element on a first side of a first substrate, and forming at least a second metasurface element on a first side of a second substrate, and fusing the first and second substrates together along sides opposite the first sides of said substrates using a bonding process having a thermal budget below 600° C.

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In various other embodiments, the bonding process is a wafer bonding process using an adhesive selected from the group of an optical epoxy, benzocyclobutene, a UV cured polymer, SU8, and a plasma activate silicon dioxide film.

In still various other embodiments, the method further includes removing at least a portion of the backside of one or both of the substrates prior to fusing.

In yet various other embodiments the method further includes:

embedding and planarizing at least one of the first and second metasurface elements;

forming at least a third metasurface element on a first side of a third substrate; and

fusing the side of the third substrate opposite the first side to the planarized first or second metasurface using a bonding process having a thermal budget below 600° C.

In still yet various other embodiments, the planarization further comprises embedding at least one of the first and second metasurface elements in one of either a polymer or a solid-state bonding agent.

In still yet various other embodiments, the method further includes iterating the steps of forming, embedding, and fusing to form a layered stack of four or more metasurface elements.

In still yet various other embodiments, at least one of layers of at one end of the layered stack is one of either an illuminator or a sensor.

In still yet various other embodiments, the method further includes:

inserting a spacer substrate between the sides of the first and second substrates opposite the metasurface elements, the spacer substrate having at least one aperture disposed therethrough; and

fusing the spacer substrate to the first and second substrates using a bonding process having a thermal budget below 600° C., such that the at least one aperture forms an air gap between the first and second substrates.

In still yet various other embodiments, the spacer substrate is formed of a low-index of refraction material selected from the group of: polymer, SiO₂, and glass.

In still yet various other embodiments, the spacer material is coated in black chrome.

In still yet various other embodiments, the method further includes iterating the steps of forming, inserting, and fusing to form a layered stack of three or more metasurface elements.

In still yet various other embodiments, at least one of layers at one end of the layered stack is one of either an illuminator or a sensor.

In still yet various other embodiments, the plurality of metasurface features are inhomogeneous.

In still yet various other embodiments, the plurality of metasurface features diverge from an ideal shape by a pre-determinable amount based on the dimensions of the metasurface features.

Further embodiments are directed to methods of forming a compound metasurface element comprising forming two layers of metasurface features atop a substrate, wherein the two layers are offset from each other by a distance smaller than or on the same order as the wavelength of light within the specified operational bandwidth such that the two layers of metasurface features operate in conjunction to impose a phase shift on impinging light.

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Additional embodiment are directed to methods of forming a metasurface element including:

forming a plurality of identical or unique first metasurface elements;

providing a plurality of identical or unique illumination sources disposed in a planar array and integrating at least one of the plurality of first metasurface elements with each of the plurality of illumination sources in the array such that light from each of said plurality of illumination sources passes through at least one of the first metasurface elements and an angular deflection is imposed thereby;

disposing a first spacer layer between the planar array of illumination sources and the first metasurface elements, the first spacer layer being configured to create divergence in light emitted from each of the illumination sources of the planar array prior to impinging on the respective first metasurface element;

disposing a second metasurface element at a distance from the plurality of first metasurface elements, the second metasurface element configured to imprint a far-field illumination pattern onto a light field formed by the emission of all of the plurality illumination sources; and

disposing a second spacer layer between the first and second metasurface elements such that an offset distance is formed therebetween.

In additional other embodiments, at least the first spacer layer comprises a solid-state material.

In additional other embodiments, at least the second spacer layer comprises an air gap.

In still additional other embodiments, the plurality of metasurface features are inhomogeneous.

In yet additional other embodiments, the plurality of metasurface features diverge from an ideal shape by a pre-determinable amount based on the dimensions of the metasurface features.

In still yet additional other embodiments, the method further includes:

forming a plurality of identical or unique first metasurface elements;

providing a plurality of identical or unique sensor elements disposed in a planar array and integrating at least one of the plurality of first metasurface elements with each of the plurality of sensor elements in the array such that light impinging on each of said plurality of sensor elements passes through at least one of the first metasurface elements and an angular deflection is imposed thereby;

disposing a first spacer layer between the planar array of sensor elements and the first metasurface elements, the first spacer layer being configured to create convergence in light impinging on each of the first metasurface elements of prior to impinging on the respective sensor elements of the planar array;

disposing a second metasurface element at a distance from the plurality of first metasurface elements, the second metasurface element configured to imprint a far-field illumination pattern onto a light field impinging thereon; and

disposing a second spacer layer between the first and second metasurface elements such that an offset distance is formed therebetween.

Numerous other embodiments are directed to methods of forming a metasurface element including:

forming a plurality of identical or unique first metasurface elements;

providing a plurality of identical or unique sensor elements disposed in a planar array and integrating at least

one of the plurality of first metasurface elements with each of the plurality of sensor elements in the array such that light impinging on each of said plurality of sensor elements passes through at least one of the first metasurface elements and an angular deflection is imposed thereby;

disposing a first spacer layer between the planar array of sensor elements and the first metasurface elements, the first spacer layer being configured to create convergence in light impinging on each of the first metasurface elements of prior to impinging on the respective sensor elements of the planar array;

disposing a second metasurface element at a distance from the plurality of first metasurface elements, the second metasurface element configured to imprint a far-field illumination pattern onto a light field impinging thereon; and

disposing a second spacer layer between the first and second metasurface elements such that an offset distance is formed therebetween.

Several embodiments are directed to metasurface elements including:

an array of metasurface features disposed atop a substrate transparent to light over a specified operational bandwidth, the array comprising a plurality of metasurface features having feature sizes smaller than the wavelength of light within the specified operational bandwidth and configured to impose a phase shift on impinging light within the plane of plurality of metasurface features;

wherein the plurality of metasurface features are inhomogeneous, and diverge from an ideal shape by a pre-determinable amount based on the dimensions of the metasurface features.

In several other embodiments, the ideal shape is a square, and where the ideal square has a side dimension of less than 200 nm the metasurface features are formed as circles, and where the ideal square has a side dimension of less than 300 nm the metasurface features are formed as squares having rounded edges.

Many embodiments are directed to metasurface enabled illumination or sensor arrays including:

a plurality of identical or unique illumination sources or sensor elements, arranged in a planar array;

a first spacer layer disposed above the planar array of illumination sources and configured to create divergence in light emitted from each of the illumination sources of the planar array or convergences in light impinging on each of the sensor elements;

a plurality of identical or unique first metasurface elements disposed above the first spacer layer, at least one of the plurality of first metasurface elements being associated with each of the plurality of illumination sources or sensor elements in the array such that light emitted from each of the plurality of illumination source or impinging on each of said plurality of sensor elements passes through at least one of the first metasurface elements and an angular deflection is imposed thereby;

a second metasurface element disposed at a distance from the plurality of first metasurface elements, the second metasurface element configured to imprint a far-field illumination pattern onto a light field impinging thereon; and

a second spacer layer between the first and second metasurface elements such that an offset distance is formed therebetween.

In many other embodiments, the plurality of metasurface features on each of the metasurface elements are inhomogeneous, and diverge from an ideal shape by a pre-determinable amount based on the dimensions of the metasurface features.

In still many other embodiments, the plurality of metasurface features on at least the first or second metasurface element are configured to have an asymmetric cross-section and are disposed at least two different angles of rotation such that the metasurface element is configured to imprint at least two patterns having orthogonal polarization and being linearly offset one from the other on the illumination sources or detect such patterns from the impinging light prior to illumination of the sensor elements, the array being configured such that three-dimensional information is obtained from a scene by the array in a single-shot.

In yet many other embodiments, the illumination sources are polarized or unpolarized, and selected from the group consisting of: VCSELs, solid-state laser, quantum cascade laser, LED, and superluminescent LED.

In still yet many other embodiments, the two patterns are unique.

In still yet many other embodiments, the two patterns have at least 50,000 combined points.

In still yet many other embodiments, at least a first pattern is configured to obtain a measurement of the foreground of the scene, and wherein at least a second pattern is configured to obtain a measurement of the background of the scene.

In still yet many other embodiments, the two patterns are diagonally polarized relative to the laser polarization.

In still yet many other embodiments, more than two patterns having more than two different polarizations are used.

Various embodiments are directed to metasurface element enabled sensors including:

at least one sensor element;

at least one first and at least one second metasurface element disposed at an offset distance above the at least one sensor element, and having a first spacing layer disposed therebetween;

wherein each of the at least one first and second metasurface elements comprise an array of metasurface features disposed atop at least one substrate transparent to light over a specified operational bandwidth, the array comprising a plurality of metasurface features having feature sizes smaller than the wavelength of light within the specified operational bandwidth and configured to impose a phase shift on impinging light within the plane of plurality of metasurface features; and

wherein the arrays of metasurface features on each of the at least one first and second metasurface elements are configured to gather light of a specified operational bandwidth across a specified field of view and shift the incoming light such that it impinges the sensor element at a zero or near-zero degree critical ray angle.

In various other embodiments, the first spacing layer is one of either a solid-state spacer material or an air gap.

In still various other embodiments, the field of view is ± 44 degrees.

In yet various other embodiments, the sensor further includes a narrow bandwidth optical filter disposed between the metasurface elements and the sensor element.

In still yet various other embodiments, the narrow bandwidth optical filter is comprised of alternating layers with a low index of refraction and high index of refraction selected

from the group consisting of silicon dioxide, titanium dioxide, amorphous silicon, silicon nitride and aluminum oxide.

In still yet various other embodiments, the sensors further include a plurality of identical microlenses disposed between the metasurface elements and the sensor element.

In still yet various other embodiments, the at least one first metasurface element and at least one second metasurface elements are disposed on opposite sides of the same substrate, and wherein the substrate comprises the first spacing layer.

In still yet various other embodiments, the two metasurface elements on either side of the substrate have the same height.

In still yet various other embodiments, the two metasurface elements are formed from a film deposited simultaneously on the front surface and back surface of the same substrate using a conformal deposition process selected from the group of pressure chemical vapor deposition and atomic layer deposition.

In still yet various other embodiments, the at least one first metasurface element and at least one second metasurface elements are disposed facing inward toward each other on separate substrates separated by an air gap.

In still yet various other embodiments, the sensors further include an optical bandpass filter integrated into the outward facing surface of the substrate of the at least one second metasurface.

In still yet various other embodiments, the sensors further include at least third metasurface element disposed between the first and second metasurface elements and the CMOS sensor and configured to angularly diverge the path of the incoming light such that the light impinging on the CMOS sensor has a non-zero chief ray angle.

In still yet various other embodiments, the at least three metasurfaces are configured to minimize grid distortion to less than 5% over the specified field of view.

In still yet various other embodiments, the sensor element is a CMOS sensor.

Further embodiments are directed to a metasurface element enabled single platform imaging/sensing system including:

- at least one sensor element and at least one illumination source;

- at least one separate metasurface element disposed at an offset distance above each of the at least one sensor element and at least one illumination source, and having at least one spacing layer associated with each respectively;

- wherein each of the metasurface elements comprise an array of metasurface features disposed atop a substrate transparent to light over a specified operational bandwidth, the array comprising a plurality of metasurface features having feature sizes smaller than the wavelength of light within the specified operational bandwidth, and wherein the at least one illumination metasurface element disposed in association with the illumination source is configured to impose a radiation pattern on a light field emitted therefrom within the plane of plurality of metasurface features, and wherein the at least one sensor metasurface element disposed in association with the at least one sensor element is configured to detect the radiation pattern of the light field after the illumination of a scene.

In still further embodiments, the system further includes a plurality of separate metasurface elements and spacer layers associated with each of the illumination source and the sensor element.

In yet further embodiments, the metasurface elements associated with the illumination source imprint two orthogonal polarizations on the light field to produce at least two patterns having orthogonal polarization and being linearly offset one from the other on the light field illuminating the scene, and wherein the metasurface elements associated with the sensor element are configured to detect the at least two patterns such that three-dimensional information about the scene can be gathered.

Numerous embodiments are directed to methods for fabricating a metasurface element for imprinting a desired far-field intensity on an illumination source including:

- calculate an illumination source far field;

- calculate a target far field, wherein the target is a metasurface element;

- calculate a least-squares fit to the target far field to obtain a pseudo far field such that the convolution of the pseudo far field and the illumination source far field yields the target far field;

- set the initial metasurface feature array grid and phase to an initial condition;

- determine one or more objective cost functions and calculate a gradient function for each of the one or more cost functions for each of a plurality of pixels of the metasurface element;

- input a result from the one or more cost functions and the gradient function into an optimization algorithm;

- update the phases for each of the plurality of pixels of the metasurface element and repeat gradient calculation and optimization until the objective cost function converges; and

- output a calculated metasurface element phase profile.

In numerous other embodiments, the cost function is selected from the group consisting of: squared distance from target, nearest neighbor distance, squared error of the far field projection of the metasurface element under illumination, and smoothness of calculated far field.

In still numerous other embodiments, the optimization algorithm is one of either conjugate gradient or L-Broyden-Fletcher-Goldfarb-Shannon.

Several embodiments are also directed to methods of forming a metasurface element on a substrate comprising a plurality of metasurface features having feature sizes smaller than the wavelength of light within the specified operational bandwidth and configured to impose a phase shift on impinging light within the plane of plurality of metasurface features, wherein the substrate has a substrate thickness unsuitable for use with a target optical system at and further including at least one of the following:

- removing at least a portion of the backside of the substrate through one or both grinding or chemical etching; and
- aligning and fusing an additional substrate to the substrate of the metasurface element.

In several other embodiments, the additional substrate itself has a metasurface element disposed on one surface thereof, and wherein the substrate and additional substrate are fused along surface opposite the surfaces on which the relative metasurface elements are disposed.

In still several other embodiments, the method of fusing uses a bonding process having a thermal budget below 600° C.

In yet several other embodiments, the bonding process is a wafer bonding process using an adhesive selected from the group of an optical epoxy, benzocyclobutene, a UV cured polymer, SU8, and a plasma activate silicon dioxide film.

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In still yet several other embodiments, the methods further include removing at least a portion of the backside of one or both of the substrates prior to fusing.

In still yet several other embodiments, the methods further include forming at least a first metasurface element on a first side of a first substrate, and forming at least a second metasurface element on a first side of a second substrate, and fusing the first and second substrates together along sides opposite the first sides of said substrates using a bonding process having a thermal budget below 600° C.

In still yet several other embodiments, the plurality of metasurface features are inhomogeneous.

In still yet several other embodiments, the plurality of metasurface features diverge from an ideal shape by a pre-determinable amount based on the dimensions of the metasurface features.

In still yet several other embodiments, the methods further include:

embedding and planarizing at least one of the first and second metasurface elements;

forming at least a third metasurface element on a first side of a third substrate; and

fusing the side of the third substrate opposite the first side to the planarized first or second metasurface using a bonding process having a thermal budget below 600° C.

In still yet several other embodiments, the planarization further comprises embedding at least one of the first and second metasurface elements in one of either a polymer or a solid-state bonding agent.

In still yet several other embodiments, including iterating the steps of forming, embedding, and fusing to form a layered stack of four or more metasurface elements.

In still yet several other embodiments, at least one of layers of at one end of the layered stack is one of either an illuminator or a sensor.

In still yet several other embodiments, the methods further include:

inserting a spacer substrate between the sides of the first and second substrates opposite the metasurface elements, the spacer substrate having at least one aperture disposed therethrough; and

fusing the spacer substrate to the first and second substrates using a bonding process having a thermal budget below 600° C., such that the at least one aperture forms an air gap between the first and second substrates.

Numerous other embodiments are directed to a metasurface element including:

an array of metasurface features disposed atop a substrate transparent to light over a specified operational bandwidth, the array comprising a plurality of metasurface features having feature sizes smaller than the wavelength of light within the specified operational bandwidth and configured to impose a phase shift on impinging light within the plane of plurality of metasurface features, and wherein the plurality of metasurface features are formed from one of the group consisting of:

amorphous-Si metasurface features embedded in SiO₂ having a pillar height from 500 to 1000 nm and a pillar diameter from 100 to 300 nm;

amorphous-Si metasurface features embedded in SiO₂ having a pillar height of 600 nm and a pillar diameter from 100 to 300 nm;

amorphous-Si metasurface features having air gaps disposed therebetween and having a pillar height from 1 to 500 nm and a pillar diameter from 100 to 350 nm;

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amorphous-Si metasurface features having air gaps disposed therebetween and having a pillar height from 480 nm and a pillar diameter from 100 to 280 nm;

TiO₂ metasurface features having air gaps disposed therebetween and having a pillar height from 300 to 1000 nm and a pillar diameter from 100 to 350 nm;

TiO₂ metasurface features having air gaps disposed therebetween and having a pillar height of 975 nm and a pillar diameter from 100 to 300 nm;

amorphous-Si metasurface features embedded in benzocyclobutane and having a pillar height of 590 nm and a pillar diameter from 100 to 300 nm;

amorphous-Si metasurface features embedded in SiO₂ and having a pillar height of 600 nm and a pillar diameter from 100 to 275 nm;

amorphous-Si metasurface features embedded in SU8 and having a pillar height of 675 nm and a pillar diameter from 100 to 300 nm; and

amorphous-Si metasurface features in air and having a pillar height of 600 nm and a pillar diameter from 100 to 300 nm with an element spacing of 450 nm.

In numerous other embodiments where amorphous-Si is the material of choice, the amorphous Si may be hydrogenated resulting in higher optical transmission as compared to amorphous silicon where hydrogen is not present in the structures.

Additional embodiments and features are set forth in part in the description that follows, and in part will become apparent to those skilled in the art upon examination of the specification or may be learned by the practice of the disclosure. A further understanding of the nature and advantages of the present disclosure may be realized by reference to the remaining portions of the specification and the drawings, which forms a part of this disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

The description will be more fully understood with reference to the following figures, which are presented as exemplary embodiments of the invention and should not be construed as a complete recitation of the scope of the invention, wherein:

FIGS. 1A to 1G provides schematics illustrating a fabrication process for metasurface elements in accordance with embodiments of the invention.

FIG. 2A provides a schematic illustrating an embedded metasurface element with anti-reflection coatings in accordance with embodiments of the invention.

FIG. 2B provides a schematic illustrating a planarized-embedded metasurface element with anti-reflection coatings in accordance with embodiments of the invention.

FIG. 3 provides a schematic flowchart illustrating a process for fabricating metasurface elements in accordance with embodiments of the invention.

FIG. 4A to 4C provide schematics of a metasurface element having features of varying cross-section in accordance with embodiments of the invention.

FIG. 5 provides a schematic illustrating the combination of multiple substrates having metasurface elements in accordance with embodiments of the invention.

FIG. 6 provides a schematic illustrating the combination of multiple substrates having multiple metasurface elements in accordance with embodiments of the invention.

FIG. 7 provides a schematic illustrating the combination of multiple substrates having multiple metasurface elements incorporating an air gap in accordance with embodiments of the invention.

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FIG. 8A provides a schematic illustrating metasurface elements incorporating spacers integrated with a sensor/illuminator component in accordance with embodiments of the invention.

FIG. 8B provides a schematic illustrating metasurface elements incorporating spacers in accordance with embodiments of the invention.

FIG. 9 provides a schematic illustrating a multiple metasurface element substrate in accordance with embodiments of the invention.

FIG. 10A provides a flowchart for using metasurface elements to produce an arbitrary radiation pattern out of a VCSEL or VCSEL array in accordance with embodiments of the invention.

FIGS. 10B and 10C provide diagrams of phase (10B) and intensity (10C) obtained using the process of FIG. 10A in accordance with embodiments of the invention.

FIG. 11 provides a schematic illustrating an array metasurface element coupled with a set of pixelated sensor elements or illumination sources combined with a second metasurface into an integrated package in accordance with embodiments of the invention.

FIGS. 12A to 12C provide schematics illustrating a polarization splitting metasurface element producing two unique radiation patterns out of a VCSEL array in accordance with embodiments of the invention.

FIG. 13 provides a schematic illustrating a two metasurface element combined with a second element such as a cut filter where the chief-ray-angle of the focused light is 0 degrees with respect to the filter plane in accordance with embodiments of the invention.

FIG. 14 provides a schematic illustrating a two metasurface element system where each metasurface element is formed on a unique substrate in accordance with embodiments of the invention.

FIG. 15 provides a schematic illustrating a three metasurface element system designed to correct for multiple monochromatic aberrations in accordance with embodiments of the invention.

FIG. 16 provides a schematic illustrating the chief ray angle at the image sensor plane as a function of image height for the metasurface element system of FIG. 15 in accordance with embodiments of the invention.

FIG. 17 provides a schematic illustrating an example of grid distortion from the metasurface element system of FIG. 15 in accordance with embodiments of the invention.

FIG. 18 provides a schematic illustrating an integrated system with metasurface elements on the illuminators and detectors in accordance with embodiments of the invention.

FIG. 19 provides a schematic illustrating an integrated system with metasurface elements on the illuminators and detectors where the polarization of light provides an extra degree of freedom in accordance with embodiments of the invention.

FIG. 20 provides data graphs showing the phase and transmission response of metasurface features comprising cylindrical pillars composed of amorphous silicon embedded in SiO_2 in accordance with embodiments of the invention.

FIG. 21 provides data graphs showing the phase and transmission response of metasurface features comprising cylindrical pillars composed of amorphous silicon in air in accordance with embodiments of the invention.

FIG. 22 provides data graphs showing the phase and transmission response of metasurface features comprising cylindrical pillars composed of amorphous TiO_2 in air in accordance with embodiments of the invention.

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FIG. 23 provides data graphs showing the phase and transmission response of metasurface features comprising cylindrical pillars composed of amorphous silicon encapsulated in polymer for a wavelength of 850 nm in accordance with embodiments of the invention.

FIG. 24 provides data graphs showing the phase and transmission response of metasurface features comprising cylindrical pillars composed of Si encapsulated in SiO_2 for a wavelength of 850 nm in accordance with embodiments of the invention.

DETAILED DESCRIPTION OF THE INVENTION

Turning now to the drawings, metasurface elements, integrated systems incorporating such metasurface elements with light sources and/or detectors, and methods of the manufacture and operation of such optical arrangements and integrated systems are provided. Many embodiments are directed to systems and methods for integrating transmissive metasurface elements with other semiconductor devices or additional metasurface elements, and more particularly to the integration of such metasurfaces with substrates, illumination sources and sensors. In some embodiments, the metasurface elements may be used to shape output light from an illumination source or collect light reflected from a scene to form two unique patterns using the polarization of light. In such embodiments, shaped-emission and collection may be combined into a single co-designed probing and sensing optical system.

In many embodiments, metasurface elements may incorporate multi-layer metasurface elements comprising combinations of two or more metasurface optical elements. In various such embodiments the multi-layer metasurface elements may be free-standing (i.e., not directly integrated with a specific illuminator or sensor into a system). In some such embodiments, the optical system may consist of a single physical component or substrate having metasurface elements disposed on either side thereof. In some embodiments, multiple substrates having multiple metasurface elements may be combined to make more complex systems. In such embodiments, the thickness of the substrate may be determined by the requirements of the optical system, manufacturing constraints and the specific designs of the two metasurfaces. In various embodiments, the multi-layer metasurface elements may be formed by patterning each individual metasurface element on unique substrates and subsequently fusing the substrates together through a suitable technique, e.g., wafer bonding, optical adhesive. In general, however, in accordance with embodiments any number of metasurface elements may be combined through any number of steps using CMOS or related processes.

In many embodiment, the metasurface elements may be free standing or may be embedded within another material. In various such embodiments, the selection of the embedding material includes the appropriate selection of refractive index and absorption characteristics. In many such embodiments, the embedding material may provide mechanical stability and protection as well as an additional design degree of freedom that enables the metasurface to perform a desired optical function.

In various embodiments, the metasurface elements may be directly mounted or fabricated on an LED, VCSEL facet or each facet of a VCSEL in an array to minimize device thickness and optimize metasurface-illuminator/sensor alignment. In some such embodiments the resultant system can be used to convert a natural Lambertian or some

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arbitrary light distribution to a broad range and essentially arbitrary light distribution including, for example, a so-called top hat, a so-called bat-wing profiles, or any other desired structured light pattern.

In some embodiments, a spacing layer of a defined thickness (e.g., the working distance) may be deposited on the CMOS image sensor, LED, VCSEL, etc., to implement an optical distance appropriate for a desired camera design, illuminator design or optimal system performance. In various such embodiments, the spacing layer material may be organic or inorganic and may have a lower refractive index than the dielectric elements comprising the metasurface. In some such embodiments, the thickness of the spacing layer may be modified to provide appropriate optical spacing for the specific optical system.

Various embodiments are also directed to methods of fabricating metasurface elements. In some such embodiments, methods are directed to the manufacture of metasurface elements on a wafer incorporating other devices, such as sensors or illuminators, thereby avoiding, in some embodiments, expensive manufacturing processes, such as, for example, the mechanical assembly of small dimension elements, or the active alignment of optics with sensors. In some such embodiments, metasurface elements may be integrated with the sensor (or the illuminator) in a series of operations at a semiconductor fab. In many such embodiments a sequence may include: (i) sensor or illuminator, (ii) optional microlens array/collimator, optional filter, optional spacing layer, optional metasurface element(s), optional additional spacing layer, optional metasurface element(s), optional anti-reflection (AR) layer, optional protection layer. In many such embodiments a sequence of elements may include: (i) sensor or illuminator, (ii) optional microlens array/collimator, optional filter, optional spacing layer, optional metasurface element(s), optional additional spacing layer, optional metasurface element(s), optional anti-reflection (AR) layer, optional protection layer.

Embodiments for Manufacturing Metasurface Elements

Currently the manufacture of metasurface elements requires the use of specialized processes and systems that are incompatible to mass manufacturing limiting the implementation and adoption of such metasurface elements in CMOS devices. An exemplary description of a conventional process for forming metasurface elements can be found, for example, in U.S. Pat. No. 8,848,273, the disclosure of which is incorporated herein by reference. The capability to produce metasurfaces via standard semiconductor processes, would enable direct integration of metasurface optics with functional elements such as light emitting diodes (LEDs), vertical cavity surface emitting laser (VCSEL), complimentary metal-oxide-semiconductor (CMOS) image sensors, micro-electromechanical (MEMs) devices, etc., where direct integration means combination of a metasurface element and sensor/illuminator using unit processes identical or similar to those used to make the functional CMOS elements.

Accordingly, many embodiments are directed to methods for the fabrication of metasurface elements and systems, and more particularly to methods that are capable of being implemented within a conventional semiconductor foundry. In various embodiments, conventional processes adapted for the manufacture of metasurface elements may include photolithography, nanoimprinting, various chemical vapor deposition (CVD), atomic layer deposition (ALD) and

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physical vapor deposition (PVD) mass transfer processes, and chemical and plasma etching (and CMC), among others. Turning to the figures, an exemplary set of fabrication processes tailored for the manufacture of various aspects of embodiments of metasurface elements is presented in the schematic of FIGS. 1A to 1G.

Metasurface optical elements are comprised of dielectrics with feature sizes from 10's of nanometers to micron scale, or generally smaller than the wavelength of light at which the metasurface is being used. Referring to FIGS. 1A to 1C, in many embodiments, an initial step to manufacturing metasurface elements comprises the patterning and formation of an array of metasurface features. In many such embodiments, as shown in FIG. 1A, this metasurface feature formation process is accomplished by depositing a patterning material (14) on top of an appropriate hard mask material (12) of thickness t (t being the thickness of the film and the height of the final metasurface) that itself is disposed atop a suitable substrate (10). Any suitable deposition technique may be used to form these layers, including, for example, sputtering, chemical vapor deposition (CVD) or atomic layer deposition (ALD).

Although throughout this disclosure exemplary materials will be discussed in relation to specific embodiments, it will be understood that any suitable combination of patterning material, hard mask material, and substrate may be used for these purposes. For example, in various embodiments, the substrate material is selected to provide suitable structural support and to be transparent to light over a desired bandwidth. Exemplary substrate materials that have been successfully implemented using the processes described in embodiments include, for example, fused silica, sapphire, borosilicate glass and rare-earth oxide glasses. Similarly, the hard mask material may be chosen from any readily available material suitable for use in semiconductor foundries. Exemplary hard mask materials include, for example, silicon, silicon nitride of various stoichiometries, silicon dioxide, titanium dioxide, alumina etc. In various embodiments, such as, for example, where the hard mask material forms an embedding material (as described in greater detail below) the hard mask material may be chosen to have a low (e.g., from 1 to 2.4) refractive index at a specific wavelength of light. Finally, the patterning material in accordance with embodiments may be formed of any suitable photoresist, such as, for example, a photoresist tuned for a lithographic wavelength capable of forming metasurface features of a desired size. Exemplary lithographic processes include, for example, ultraviolet and deep ultraviolet (DUV) lithography. In other embodiments, the patterning layer may be a polymer material suitable for use in nanoimprint lithography. Independent of the specific material used, the patterning material must be capable of reproducing a desired pattern of features from 10's of nanometers to micron scale, and must suitably protect the underlying hard mask film in selective areas in subsequent steps.

In particular, as shown in FIG. 1B, once the substrate (10), hard mask (12) and patterning material (14) layers are in place, the patterning material is patterned to reproduce an array of features (16) corresponding to either a negative or positive reproduction of the final intended metasurface feature array structure. The process of producing this feature array pattern can take any form suitable to produce the desired feature sizes. For example, in embodiments of metasurface elements for use in visible or near-IR applications UV lithography (e.g., where the wavelength of operation of the UV lithography step is below 193 nm) may be

used. In still other embodiments, patterns may be imprinted physically by a master stamp in a nanoimprint lithography process.

As shown in FIG. 1C, once the desired feature array pattern (16) is in place, an anisotropic etch process is used to transfer the desired feature pattern into the hard mask layer (12). An exemplary anisotropic etch process for use in accordance with embodiments is reactive ion etching process. It will be understood that a number of possible chemistries may be used in the reactive ion etching process including, for example, SF₆ gas, Cl₂ gas, BCl₃ gas, C₄F₈ or any mixture of these gases. Further, the gas mixture may be static or timed in a multiplexed manner where one or more of the gasses is introduced followed after some set time period by a second unique set of one or more of the gases. Regardless of the specific anisotropic etch process used, once the pattern is etched into the hard coat layer the remaining photoresist may be removed using any suitable method (e.g., chemical solvent, chemical etchant, plasma etchant, etc.). Note, in various embodiments it may be desirable to use the feature array (16) formed in the hard mask material (14) as the final metasurface element. In such embodiments, the process can be stopped here, or combined with the deposition of suitable AR coatings or mechanical protection embedding layers as described, for example, in FIG. 1G.

As shown in FIG. 1D, where a specific metasurface material is to be used in the final metasurface element, the feature pattern (16) formed in the etched hard mask (12) (as described above in FIG. 1C) can act as a template for the final metasurface structure. In such embodiments, a separate metasurface material (18) is deposited using a suitable conformal coating method such as chemical vapor deposition (CVD), atomic layer deposition (ALD), etc., to fill the mask negative and produce the metasurface element. As shown the metasurface material (18) overfills the spaces formed by the feature pattern (16) etched in the hard mask (12) to completely fill the voids. In addition to filling the voids (20), this process leaves an over-layer of the metasurface material above the remaining hard mask. Again, although specific metasurface materials will be discussed throughout, it will be understood that metasurface materials in accordance with embodiments may be chosen from any readily available dielectric material having a desired refractive index and suitable for use in semiconductor foundries. Exemplary metasurface materials include, for example, silicon, silicon nitride of various stoichiometries, silicon dioxide, titanium dioxide, alumina etc.

Referring to FIG. 1E, once the overfill of metasurface material (18) is deposited, an etch or chemical-mechanical planarization may be performed in accordance with embodiments to remove the overfill layer providing a uniform height to the patterned hard mask (12) and metasurface material (12). In embodiments where an embedded metasurface is desired, and where a suitable hard coat material has been chosen to act as the embedding material (as previously described) the process may be stopped and the resultant metasurface material embedded with hard coat material structure used as the final metasurface element. This metasurface element may then be optionally coated with suitable AR coating layers or mechanical protection layers as described below.

In various embodiments, as shown in FIG. 1E, the hard mask material (12) is removed leaving behind free-standing metasurface elements (20). In such embodiments, the hard mask may be removed using a selective etch chemistry which etches the hard mask material (12) at a much higher

rate than the metasurface material (18) (e.g., 100:1 or 1000:1 or greater). As will be understood by those skilled in the art, such processes are dependent on the specific selection of metasurface material and hard mask material. For example, in embodiments where the hard mask is silicon and the metasurface is TiO₂, a XeF₂ etch chemistry will selectively remove the silicon while leaving the metasurface material essentially unaltered. Note, in embodiments where the metasurface element is designed to be freestanding, i.e., the metasurface features protrude from the end of the substrate with only air gaps (22) separating them, the process is complete at this step.

Finally, as previously discussed, in certain embodiments where it is desirable for the metasurface to have an AR coating or mechanical protection layer, additional steps are required to complete the final metasurface element. Referring to FIG. 1G, in various embodiments an AR coating or mechanical protection or planarizing layer (24) may also be deposited to fill the voids (22) between the metasurface features (20) and extend above the surface of the metasurface material layer (18). It will be understood that any material having optical properties suitable for a specific optical system design, such as, for example, a suitable refractive index and minimal optical absorption at a desired wavelength or over the bandwidth of interest (the planarizing layer can allow for multiple levels of metasurface elements for complex optics systems) may be used in this process. As described above, in order to protect the metasurface and provide improved functionality the metasurface constituent elements and substrate faces in many embodiments are coated in one or more materials or layers of material. Referring to FIG. 2A, a schematic of an embedded metasurface is shown. As illustrated, the metasurface features (20), which can be any material with desired optical properties as described above, are embedded in an embedding medium (24) typically having a lower-index of refraction. Exemplary embedding materials include, for example, poly(methyl methacrylate), SU8, benzocyclobutene, and/or a solid film such as: silicon dioxide, aluminum oxide, titanium dioxide, silicon nitride, hafnium oxide, zinc oxide, or spin-on-glass. The low-index embedding medium encapsulates the metasurface features and may extend some thickness above the metasurface features. In such embodiments, the low-index medium may act as a protective barrier to the metasurface elements (i.e., provide mechanical stability). The embedding material may also provide an additional design degree of freedom for the system that allows for certain properties to be optimized, e.g., to improve the overall transmission or efficiency of the metasurface and in some instances obviate the need for a separate AR coating. It is shown here that the embedded metasurface is fabricated on a substrate but the metasurface could also be extending from a sensor/illuminator, as will be described in detail below. The combined element (metasurface, embedding medium and substrate) may also be coated with a suitable anti-reflection coating on the side of the substrate containing the metasurface (26') and/or on the backside of the substrate (26). In many embodiments, the AR coating may comprise alternating layers of any combination of silicon dioxide, titanium dioxide, aluminum oxide, silicon nitride, aluminum nitride, or amorphous silicon, where each has a thickness less than the wavelength of light within the operational bandwidth of the metasurface. Additionally, as described above the embedding medium itself can potentially be used as an anti-reflection coating.

Although certain embedded metasurface embodiments are described above, in various other embodiments the

metasurface can be embedded and planarized, as shown in FIG. 2B. In such embodiments, the metasurface elements may be embedded in a suitable low-index material (as described above), and in an additional step, the embedding medium (24) is then etched or planarized such that its height is commensurate with the metasurface elements (20). Optional anti-reflection coatings may also be included on either the bare substrate surface (26) or on the patterned metasurface side (not shown).

Embodiments for Manufacturing Metasurface Elements on Conventional Substrates

Although the above discussion has described in detail manufacturing processes capable of forming a variety of free-standing or embedded metasurface elements using conventional CMOS fabrication techniques, in practice it may not be possible to adapt conventional metasurface elements to allow for the economic production of metasurface elements using pre-existing equipment in foundries. For example, one of the design criteria used conventionally to tailor the optical properties of a metasurface element is the substrate thickness. Varying this substrate thickness provides the metasurface element designer another degree of freedom in obtaining desired optical properties. However, the pre-existing equipment in most foundries have limitations that lead to specific mechanical requirements on the substrate upon which the metasurface elements will be fabricated. For the case of standard substrate diameters within foundries, e.g., 200 mm and 300 mm, the thickness of the substrate is limited to 725 microns and 775 microns, respectively. These fixed substrate thicknesses, in turn, impose specific requirements on the optical function, and therefore the design, of a metasurface formed on the substrate or a system of multiple metasurfaces formed on such a substrate (in mass production there may be, for example, 5,000 metasurfaces formed on a single die, or more).

Accordingly, many embodiments are directed to processes for tailoring the fabrication of the metasurface element or system to the specific substrate thickness upon which the metasurface or metasurface system is being produced. For example, in various embodiments the phase shifts that need to be imparted by the metasurface elements to impart a specific function to the overall optical component(s) will be specific to the substrate thickness upon which the elements are formed. Accordingly, in some embodiments the procedure for designing and fabricating the metasurface element comprises: (1) considering the device specifications of the metasurface element, (2) considering the thickness and optical properties (index of refraction and absorption) of the substrate on which the metasurface is being formed, (3) optimizing the phase profile of the metasurface to obtain the desired specifications for the properties of the substrate, and (4) determining the thickness and in-plane dimensions of the metasurface elements required to reproduce the phase profile.

Referring to FIG. 3, an exemplary process in accordance with producing metasurfaces using standard substrate thicknesses is provided. As shown, following deposition of the metasurface material and lithographic patterning and etching (as described in FIGS. 1A to 1G, above), if the metasurface layer is designed for the standard substrate thickness an additional protective layer or AR coating may be disposed on the metasurface layer before being sent for further back end processing. In many such embodiments, back end processing may comprise singulating the thousands of metasurfaces formed across the substrate using a dicing process.

Further, in embodiments where it is desirable to produce a metasurface that has a final substrate thickness different than the standard thicknesses discussed above an additional step to alter the thickness of the underlying substrate may be performed. In such embodiments, the metasurface would initially be formed on a standard substrate diameter (200 or 300 mm) with a thickness standard in semiconductor processes (725 or 775 micron, respectively), again to coincide with industry standards. After initial definition of the metasurface on the standard thickness substrates, subsequent back end processing would then be performed to alter the substrate thickness. Suitable methods for altering the thickness of the substrate according to embodiments include, for example, grinding or combination of grinding and chemical processing to gradually remove substrate material. In such embodiments, the final substrate thickness upon which the metasurface has been formed can be any value less than the starting standard thickness. Alternatively, if thicker substrate is required, two independent wafers each comprising any number of metasurfaces (1,000 to 10,000 individual metasurfaces) can be combined in accordance with embodiments to achieve desired thickness via wafer-bonding processes such that the final monolithic unit, with a set of metasurfaces on either side, has a total thickness as required by the final design. Processes according to such embodiments may be carried out for either two substrates with standard thicknesses, if the final thickness required is 2 times the standard thickness, or it can be carried out for two substrates that have been thinned so that the final thickness of the combined unit has any desired thickness. In such embodiments, the "back end processing" may comprise an additional wafer bonding step where each of the individual substrates can be aligned to each other before being combined.

Embodiments for Manufacturing Metasurface Elements with Non-Ideal Features

In conventional processes for designing metasurfaces, shape fidelity from the designed metasurface to the fabricated is often assumed to be either a 1 to 1 correspondence or to be maintained to within some error range. This approach leads to metasurface arrays often being comprised of a single set of shapes where one feature of that set of shapes changes, e.g., circles with varying diameters across the metasurface. However, fabrication techniques used for potential mass production of metasurfaces e.g., UV lithography as detailed above, are generally unable to carry-out faithful reproduction of certain geometrical shapes. As such, many embodiments are directed to metasurface elements and metasurface fabrication processes where the function of the metasurface is reproduced using non-ideal and inhomogeneous shapes.

For example, FIG. 4A provides a cross-sectional schematic of an exemplary section of a metasurface where an inhomogeneous set of shapes is distributed across the metasurface. In this specific embodiment, square pillars are desired. However, after fabrication what is actually formed within a given metasurface are an array of squares with varying side lengths (e.g., s_1), squares with rounded corners of varying radii, r_c and circles with varying radii r_1 or r_2 . Specifically, the larger features here are designed to be squares or squares with rounded corners; however, as the side lengths of the squares are reduced below some minimum side length the squares become circles. In processes according to embodiments, manufacturing limitations are simulated on each desired metasurface feature shape, and

these non-ideal or inhomogeneous feature elements then used to determine the final metasurface element array structure.

For example, FIGS. 4B and 4C, provide diagrams illustrating the variation of printed and designed patterns from a metasurface element mask reticle. As shown, in embodiments for a designed square feature of side 200 nm and period of 450 nm, the printed fabrication technique will actually replicate a circle of diameter 200 nm (FIG. 4B). By contrast, for a square feature of side 296 nm and spacing of 450 nm, the fabricated feature is a square with rounded corners (FIG. 4C). Accordingly, many embodiments of metasurface elements where square metasurface features are designed may be substituted with rounded squares below ~300 nm and circle below ~200 nm to allow for the use of industrial standard CMOS replication techniques.

Embodiments for Manufacturing Multiple Metasurface Elements

As previously discussed, various embodiments are directed to methods for wafer bonding two substrates incorporating metasurface elements together. Such embodiments may be modified to allow for the facile fabrication of multiple metasurface elements, such as, for example, doublets and triplets (e.g., metasurface elements comprising two or three separate metasurface feature arrays). In particular, although many wafer bonding processes exist each imposes a specific thermal budget on the substrates being joined. Since many embodiments of metasurface elements use amorphous Si as the metasurface material, excess heating of the substrates can result in the crystallization of the Si. Accordingly, embodiments are presented that allow for the formation of metasurface doublets and triplets using low temperature processes, such as, for example, using UV cured polymers (such as benzocyclobutane (BCB) or the like), or plasma activated SiO₂ to allow for wafer bonding of two or more metasurface elements at low temperature.

Referring to FIG. 5, a schematic for forming a metasurface doublet in accordance with embodiments is shown. As illustrated, in many such embodiments, a plurality of unique metasurface elements (30 & 32) are fabricated on two distinct substrates (34 & 36). The metasurface elements are then made into a combined system by fusing the bottom (e.g., the portion of the face of the substrate without any metasurface elements) of each unique substrate. As discussed above, the substrates may be fused by wafer bonding techniques, optical epoxy, or any suitable method for combining the two unique elements into a single element within the allowed thermal budget of the metasurface material used. The bonding material (38) in many embodiments can be adhesives such as benzocyclobutene, cured polymers SU8 or a silicon dioxide film that facilitates a glass bond. In cases where the thermal budget of the metasurface material is low (less than 600° C.) the silicon dioxide bond can be a low temperature plasma-activated SiO₂ bond. Additionally, although not illustrated, the metasurfaces may be embedded as described in the embodiments illustrated in FIGS. 2A and 2B. In addition, as described in reference to FIG. 3, the thickness of the two substrates, which ultimately constitutes the total thickness of the space between the metasurfaces, may additionally be altered to optimize certain properties of the combined system.

Although the disclosure thus far has detailed embodiments incorporating only two metasurface elements, the process may be generalized to any number of metasurface elements. For example, certain applications may require

three or more metasurfaces to be combined into a monolithic unit. In such a case, two substrates comprising separate metasurface elements may form the initial uncombined units. An illustration of exemplary embodiment of such a process is provided in FIG. 6. As shown, in many such embodiments, at least one of the metasurface substrates (40) has only one side patterned with a metasurface element (42) while the opposite side of the substrate can be completely unpatterned or may also include a bandpass filter (44) for a specific wavelength of interest. In such embodiments the filter may be formed of one or more suitable materials, including, for example, alternating layers of low index of refraction and high index of refraction materials such as silicon dioxide, titanium dioxide, amorphous silicon, silicon nitride, and aluminum oxide. At least a second metasurface substrate (46) has two unique metasurface elements (48 & 50) on each face of the substrate (this second metasurface substrate may also have been formed through an intermediate bonding step of its own as described above in relation to FIG. 5). The metasurface substrates (40 & 46) are then made into a combined system by fusing the bottom (e.g., the portion of the face of the first substrate (40) without any metasurface elements with the portion of the face of the second substrate (46) containing one of the two metasurface elements (48 & 50)) of each unique substrate. As discussed above, the substrates may be fused by wafer bonding techniques, optical epoxy, or any suitable method for combining the two unique elements into a single element within the allowed thermal budget of the metasurface material used. The bonding material in many embodiments can be adhesives such as benzocyclobutene, cured polymers SU8 or a silicon dioxide film that facilitates a glass bond. In cases where the thermal budget of the metasurface material is low (less than 600° C.) the silicon dioxide bond can be a low temperature plasma-activated SiO₂ bond. This bonding material is disposed on or between the faces of the two substrates to be joined. In certain implementations, various metasurface elements can be optionally encapsulated (52 & 54) as outlined above. However, in many embodiments for the facilitation of a bonding process, at least the metasurface element (48) proximal to either the bare substrate face or bandpass filter (44) in the combined triplet device is embedded in a polymer and/or solid-state bonding agent (56).

While the above instances of combining metasurface elements have had each metasurface element separated by a solid substrate, in some embodiments each metasurface elements may instead be separated by an air gap. Referring to FIG. 7, a schematic of a metasurface doublet comprising an air gap is illustrate. As shown, in many such embodiments, two or more metasurface elements (60 & 62) are formed on unique substrates (64 & 66) using a suitable method, such as that described above in relation to FIG. 5. The unique metasurface elements are then combined with a third substrate or spacer substrate (68) comprising one or more apertures (70) etched to make a monolithic unit where the space within the aperture between the metasurfaces is unfilled (i.e., such that an air gap is formed between the metasurfaces elements (60 & 62)). The space between the metasurfaces provides an additional design tool for system-level optimization. For example, in many embodiments by adjusting the spacer wafer thickness one can implement a variety of different designs. In addition, in various embodiments, as also shown in FIG. 7, it is possible to add additional spacer substrates (72) to incorporate other system elements, such as illuminators and/or sensors (74).

In embodiments incorporating such spacer substrates, any suitable substrate material may be used. For example, in

many embodiments the spacer substrates may be any low-index material, such as, for example, polymer, SiO₂, glass, etc. In addition, in other embodiments the spacer material may be coated in black chrome. The metasurface elements may also be formed of any material, which has been optimized for a specific bandwidth, such as for example, silicon, TiO₂, alumina, metal, etc. The metasurface elements may also be fabricated using such methods as described in FIGS. 1A to 1G or using semiconductor fabrication processes in general.

The above embodiments described processes for combining two and three metasurfaces; however, such embodiments may be extended beyond just two or three metasurfaces. For example, by iterating on the steps described above in relation to FIGS. 5 to 7, embodiments allow for the stacking of any number of metasurface elements. Referring to FIG. 8A, in various embodiments a set of metasurfaces (80, 82, 84, etc.) and spacer layers (86, 88, etc.) may be directly integrated with an illuminator or sensor. In such embodiments, an optional spacer layer (90) is first formed on a sensor/illuminator (92) through a suitable deposition process, as described, for example, in relation to FIGS. 1A to 1G. Following the spacer layer (90), any number of metasurface elements (80 to metasurface n+1) may be fabricated as needed to carry out a desired function. Each subsequent metasurface may also be separated by a spacer layer (86 to spacer n+1) and the thickness of each spacing layer may be varied as required by the optical design. As previously discussed, the spacer layers in such embodiments may be any low-index material, e.g., polymer, SiO₂, glass. As also previously discussed, the metasurface elements in such embodiments may also be any material, which has been optimized for a specific bandwidth, e.g., silicon, TiO₂, alumina, metal, etc. The metasurface elements may be fabricated using such methods as described in FIGS. 1A to 1G or using other suitable semiconductor fabrication processes in general.

Although the above description assumes integration with a sensor or illuminator (92), a set of metasurface elements and spacer layers can also be iteratively fabricated on a substrate (90), as illustrated in FIG. 8B. As shown, in such embodiments, the process is as described in relation to FIG. 8A, but rather than integrating the metasurface/spacer stack onto a sensor/illuminator (92), the stack is produced on a stand-alone substrate (90). The combined substrate and stack in accordance with such embodiments may then be integrated into an optical system or be used as a stand-alone optical component.

Embodiments of Multilayer Metasurface Elements and Their Manufacture

While in the previously-detailed embodiments each metasurface element is designed to carry out a unique optical function in a larger optical system and the metasurface elements are typically separated by macroscopic distances (distances of 10 or more wavelengths), in various embodiments a plurality of two layers of patterned material may be provided at a distance to each other microscopic distances (e.g., at distances to each other smaller than or on the same order as the wavelength of light) such that the layer in combination form a single metasurface element performing a single optical function. This may be particularly advantageous where an optical function requiring very complex metasurface features is called for. Such complex features may be beyond the capability of standard CMOS fabrication techniques to fabricate. In such cases, combinations of

simple features disposed at microscopic distances in accordance with embodiments may be used to replicate the optical functions of the complex features shapes. Referring to FIG. 9, a schematic of an embodiment of a metasurface element comprising two layers of patterned materials separated by a distance, t_{offset} , is provided. Although only a schematic for a two-layer system is shown, it will be understood that any number of such layers may be provided so long as the distance t_{offset} is small enough to allow for a combination of the optical functions of the plurality of layers. These feature layers may be formed and combined using any suitable combination of the fabrication steps set forth in relation to FIGS. 1 to 8, above.

Embodiments for Incorporating Metasurface Elements with VCSELs

The techniques and processes of fabricating metasurface elements in accordance with embodiments also directly enable their integration with illumination sources. Of particular interest is the combination of metasurface elements with VCSELs and VCSEL arrays. In general, a transmissive metasurface element can imprint an arbitrary phase profile on an electromagnetic wave to produce any radiation pattern in the far-field. Fabrication techniques for the metasurface elements in accordance with embodiments enable direct integration with a VCSEL, solid state laser, quantum cascade laser, LED, superluminescent LED or any solid-state light source.

VCSELs can be conceptualized as single- (or few-) mode lasers which generate an approximately collimated beam of laser light at a single wavelength. Often to generate sufficient power or spatial extent a device will contain not one VCSEL, but a 2-dimensional array of VCSELs. This light has a distribution (or illumination) in real space and in angular space. Metasurfaces when properly co-designed and integrated with a VCSEL array have the ability to transform both the real and angular space distribution of either a VCSEL or VCSEL array. In particular, pairing a metasurface element with a VCSEL allows the metasurface element to imprint an arbitrary radiation pattern on the source (e.g., batwing, top hat, super-Gaussian, or other patterns known in the art).

FIG. 10A presents a flow-chart of a process, in accordance with embodiments, for fabricating a metasurface to create any desired far-field intensity from a VCSEL array. To obtain the illumination in real space, in accordance with embodiments, the VCSEL is assumed to operate in the far-field regime. The VCSEL's characteristic far field (VCSEL-FF), from experimental data, is propagated to the surface under illumination, which in this case is a metasurface element as defined above. In the case of a VCSEL array, the output from many VCSELs via (VCSEL-FF) is summed at the surface under an assumption of incoherence to generate the illumination. This surface illumination then gives the intensity distribution at the metasurface element (I-MS). Every point at this illuminated surface also has a distribution of angles of incident light on it, taken from the VCSEL far field angular distribution. In a simplifying case, it is possible in accordance with embodiments to consider where all the illuminated points have this same VCSEL far field angular distribution, though each point has a slightly different angular distribution.

To generate a metasurface element which takes into account the distribution of angles at the metasurface element in the design, from a target far-field distribution (TAR-

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GET_FF) embodiments of the process construct a pseudo-far-field (PSEUDO-FF) that has the property that the convolution:

$$((\text{PSEUDO-FF}) * (\text{VCSEL-FF}))(x,y) = (\text{TARGET-FF})(x,y) \quad (\text{EQ. 1})$$

That is, in accordance with embodiments, the pseudo-far-field convolved with the VCSEL far-field reproduces the target far-field. In such embodiments, the pseudo-far-field is calculated by fitting a curve to the target function. Then (PSEUDO-FF) is used as the target (or objective function) in the remainder of the process.

In various embodiments, the process proceeds by initializing the metasurface grid by discretizing it and setting the phases to some initial condition. In many embodiments, a cost function is decided upon. In various embodiments, this is chosen to be the squared error of the far field projection of the metasurface under VCSEL illumination to (PSEUDO-FF). In some embodiments, other objectives, such as the smoothness of the result in the calculated far-field, can also optionally be set. For each objective, a corresponding gradient function is also derived. This gradient function and the results of the calculation of the cost function are then used in embodiments as an input into an optimization algorithm in accordance with embodiment, and as summarized in FIG. 10A. The optimization algorithm, in accordance with embodiments, updates the results of the pixel phases and continues until the objective function meets some criteria (i.e., convergence). Upon convergence of the cost function, the required metasurface phase profile is output and a metasurface element design is chosen depending on the wavelength and desired material (e.g., as described in detail below), and a physical design for the metasurface element may be created.

Exemplary data plots showing the output of the process according to embodiments for phase (10B) and intensity (10C) after employing the algorithm described above for an example case. As shown, in such embodiments, the phase is encoded by the metasurface elements and the intensity profile on the right is produced after the laser source passes through the metasurface element and is projected into the far-field. Accordingly, using such a process it is possible to obtain predictive performance data for a proposed metasurface element under desired operating conditions and to illustrate ways of optimizing that performance by modifying aspects of the metasurface element design, such as, for example, size, height and spacing of elements, etc.

Embodiments for Incorporating Metasurface Elements with Illumination Arrays

While the above processes have focused on integrating metasurface elements with a single illumination source, e.g., as shown in FIGS. 7 and 8A, metasurface elements can also be integrated with a set of pixelated and distributed sources. As illustrated in FIG. 11, various embodiments may comprise a set of illumination sources, $p_1, p_2, \dots, p_n$ (while this is shown in 1D, it will be understood that the system can be extended to a 2D array and in general the array need not be periodically spaced) designed to illuminate a scene (110). Although each illuminator may be identical, in many embodiments the character of each illuminator may be generally unique. For example, each illuminator may have a different wavelength, bandwidth or may even output a uniquely driven optical waveform. In specific applications, the array in FIG. 11 may be an array of VCSELs. In other applications, there may be three colors (e.g., red, green and

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blue) that are then repeated periodically within the array. Because each illuminator comprising array may have unique properties, it may also be advantageous to have an array of metasurface elements, each with uniquely designed properties. In such embodiments, the metasurface array (100) may be offset from the illuminator array (102) by a spacer (104) (similar to FIG. 7 or 8B). As previously discussed, the thickness of the spacer layer (104) will depend on the specific design, but in many embodiments the thickness is configured to allow the light from the illumination sources to diverge sufficiently before interacting with the metasurface array (100). Again, the function of each metasurface element in the array may be generally unique but in certain embodiments, each metasurface element may provide collimation of the underlying illuminator pixel or each metasurface element may act to further mix each underlying illuminator pixel.

In addition to the first metasurface array (100), various embodiments may incorporate a second metasurface element (106) to further shape the emitted light from the illuminator array (102). In various embodiments, the second metasurface element (106) is also offset by a second spacer layer (108). Although this second spacer layer (108) is shown in FIG. 11 as an air gap spacer, it will be understood that this spacer can also be a solid state material as in other embodiments as described above. In many embodiments, the first metasurface array (100) is configured to introduce an additional angular divergence into the illuminator array (102) while the second metasurface element (106) imprints a specific far-field radiation pattern onto the light field. In other embodiments, the second metasurface element (106) may also be formed of an array of metasurface elements, each with unique functionality. In all such embodiments, the metasurface elements, and particularly the second metasurface elements in the system, may be designed using embodiments of the algorithm described in FIG. 10A. While in this particular case, the system has been described as an array of pixelated illuminators (102) shaped by the metasurface arrays/elements to illuminate a scene (110); the system can also be considered in the reverse. For example, the pixelated illumination sources may instead be pixels of a CMOS image sensor and instead of light being projected onto a scene, the system may be configured to collect light from a scene (110) and focus the light down to a pixel.

In all embodiments where the illumination source is a VCSEL, it is understood that the disclosures can also apply to an array of VCSELs (VCSEL array). In such a VCSEL array many individual aperture VCSELs with designable properties are combined on a single chip. Such VCSEL arrays are used to increase the total output power of the illumination source. The array may consist of a one-dimensional row of individual VCSELs or 2D grid of individual VCSELs where in each case the specific properties of the VCSELs (e.g., power, wavelength, aperture diameter, beam divergence etc.) and the arrangement of the individual VCSEL (e.g., center-to-center distance, periodic or aperiodic spacing, etc.) can all be freely chosen.

In the context of metasurface element integration, embodiments of metasurface elements with generally (but not necessarily) uniquely designed properties can be patterned on top of each individual VCSEL in the array, utilizing any of the techniques outlined herein. For example, a metasurface may be fabricated directly on the facet of each individual VCSEL in the array or a suitable dielectric spacer may be deposited on the VCSEL followed by the integration of the metasurface on top of the combined dielectric layer and VCSEL. In such embodiments, the metasurfaces may

provide a particular radiation pattern for each VCSEL and the entire system (VCSEL properties, geometrical parameters and metasurface-enabled radiation pattern) can be iteratively optimized for a specific set of performance parameters.

In various other embodiments, a dielectric material, with an index of refraction lower than that of the constituent VCSEL material may be deposited and planarized such that a single metasurface can be patterned on top of the dielectric material. This contrasts with embodiments where each VCSEL in the array has a unique metasurface patterned on its facet. Again, in such embodiments the combined system may be optimized to achieve a desired performance. Finally, in all of the above embodiments, integration of a metasurface with a VCSEL array may be accomplished using wafer level optics processes. In such embodiments, the spacer layer may be air rather than a solid dielectric, similar to the device shown in FIG. 7.

Embodiments for Incorporating Metasurface Elements into 3D Applications

In certain 3D structured light applications, a pseudo-random binary array (PSBA) is projected onto a scene. A typical PSBA is built by discretizing 2D space, for example, in a square grid. Each grid point in the x-y plane can be characterized by a unique index (i,j), where i and j are integers. At each point (i,j) a pseudo-random algorithm is used to determine whether the grid point has a dot (representing the binary value of 1) or no dot (representing a binary value of 0).

Typically, a diffractive optical element (DOE) is used to convert incident laser light, for example from a VCSEL or VCSEL array into a single dot pattern. Such conversion schemes rely only on two intensity values in the projected field (dot or no dot). However, in general it is desirable to impart multiple patterns onto a single scene and for each of the multiple patterns to exist on separable information channels (i.e., having two patterns projected from a single element onto a scene and having a method for uniquely identifying each pattern at a sensor plane). In some schemes of 3D imaging, multiple patterns are projected onto a scene at different time slices (temporal variation). These schemes use either multiple distinct illumination patterns or some active element, such as a spatial light modulator, which can be electrostatically tuned to alter the projected pattern at different points in time. These schemes, however, do not allow single-shot acquisition, increase the complexity and therefore the cost of the system and tend to be substantially larger than an integrated laser/DOE. Accordingly, many embodiments are directed to metasurface elements configured to provide single-shot acquisition for 3D structure light applications.

Referring to FIGS. 12A to 12C, exemplary embodiments of a metasurface element (120) composed of a plurality of metasurface features (122) with an asymmetric cross-section (e.g., rectangular or elliptical), a fixed height, and an axis of rotation are provided that are capable of imprinting two unique dot patterns with two orthogonal polarizations onto an illumination source (124). While this exemplary embodiment will be discussed in reference to a laser of fixed polarization, embodiments may also be configured to operate on an unpolarized source (e.g., a light emitting diode (LED)), where the function of the metasurface element would be to split the unpolarized light into two distinct polarizations (as shown schematically in FIG. 12B) and imprint any desired arbitrary pattern onto the projected light.

Whatever the illumination source, in various embodiments, the metasurface elements can be integrated with the illumination source, (e.g., LED, VCSEL, VCSEL array, etc.), in accordance with embodiments described in this disclosure.

For example, referring to FIG. 12A, the metasurface element (120) may be fabricated on a substrate (126) of desired thickness and then bonded, either with a spacer layer that is subsequently bonded to the illumination source (124), bonded directly to the illumination source or the substrate upon which the metasurface elements are formed can be subsequently diced into individual units and combined with the laser source through back-end packaging. For various embodiments of 3D imaging systems, the illumination source may be in the near infrared (NIR) (e.g., at wavelengths of 850 or 940 nm).

Regardless of the specific configuration and manufacture of the metasurface elements used, in such embodiments the metasurface elements operate by shaping not only intensity, but also the polarization of the light emitted from an illumination source. Specifically, in addition to intensity variations, light also has a vector quantity known as polarization. Given a polarized illumination source, it is possible to decompose the illumination polarization into a basis of two orthogonal polarizations or channels. Due to the orthogonality of these polarization bases, any pattern imprinted on these different polarization channels can also be independently sensed through a suitable detector configured to separate these polarization channels.

As a specific example, consider the following case. For an illumination source (in this example a polarized illumination source such as a laser) that emits light in horizontal polarization, $|H\rangle$, it is possible to decompose the output into two polarizations according to the equation:

$$|H\rangle = \frac{|A\rangle + |D\rangle}{\sqrt{2}} \quad (\text{EQ. 2})$$

where $|A\rangle$ and $|D\rangle$ in this example are diagonal and anti-diagonal linear polarizations (although it will be understood that any set of polarizations may be used in accordance with embodiments). In such a case, as shown in FIG. 12C, the pattern of metasurface features (122) of the metasurface element (120) disposed over the illumination source (124) is configured such that it imprints a separate dot pattern (as shown in FIG. 12B) onto each of these two polarizations. In accordance with embodiments these dot patterns may be unique for each polarization for clarity, as shown in the exemplary schematic of FIG. 12B, or they may be partially overlapping. Regardless of the specific configuration of the dots, however, in embodiments these patterns are projected on the same region of space (i.e., their grids have an identical spatial origin). In various embodiments, complimentary dot patterns may be used such that for every point that there is a dot in Pattern 1, there is the absence of a dot in Pattern 2 and vice versa, however, this is not essential. Such complimentary dot pattern configurations have the advantage that if a single dot (e.g., area) of a scene is lost during the capture, there is a complimentary dot of separate polarization that may capture it, thus providing a certain redundancy. In addition, while two patterns of dots on a square grid are described in relation to this specific exemplary embodiment, it should be understood that the projected patterns might be of any configuration having any suitable geometry and shape. Similarly, while a specific illumination polarization and number of polarizations upon which that illumination

polarization was decomposed have been described, it will be understood that any polarization and any number of different polarizations may be used in accordance with embodiments.

As described above, the embodiments operate as a result of the orthogonality of the polarization channels upon which each pattern is imprinted. Because of this orthogonality, the reflected light from a given scene can be separated, by a suitable detector, to create multiple images of the same scene, one with Polarization 1 corresponding to Pattern 1 and one with Polarization 2 corresponding to Pattern 2, as shown in FIG. 12B. The net result is that embodiments of such a system provide two nominally independent measurements of the distorted pattern reflected from the scene without the need for time multiplexing. Accordingly, embodiments of such a system can be used to provide single-shot, multiple measurements in 3D imaging systems, thus reducing ambiguities and increasing accuracy.

Typical pattern projection systems used in mobile devices, for example, have a limitation on the total number of points that they can project onto a scene. This limitation arises from a combination of the number of constituent VCSELs in a VCSEL array (which is not alterable by any operation of the optic producing the pattern) and the ability of the optic producing the structured light pattern to create multiple replicas of each VCSEL comprising the VCSEL array. In practical implementations, this limits the number of projected points in a pattern to a specific number, N (typically around 30,000). In accordance with embodiments of the polarization-dependent metasurface systems described above, since a single metasurface element has the ability to create multiple completely unique patterns for each orthogonal polarization, even within the limitations stated above, the total number of dots in a given pattern can be doubled (e.g., 2N), leading in a typical system to as many as 60,000 points in a single projected pattern. This doubling can be understood by examining the nature of the patterns. Conceptually, a typical projected pattern has a set of grid points separated by some period, p. At some distance away from the projector, the pattern spans a field of view given by a vertical and horizontal distance, H and Y. For traditional projectors, 30,000 grid points at a maximum will fill that field of view such that the product

$$\frac{H}{p} \left(\frac{Y}{p} \right) = 30,000.$$

In embodiments of the metasurface-polarization based solution, the optic projects one pattern with a period, p, and a second pattern, also at a period, p, but with a linear offset of p+p/2 such that a new grid point is projected at each half period. The net result is that within the same field of view, H and Y, the density of grid points can be doubled using the metasurface-enabled 3D systems in accordance with embodiments.

Finally, because two unique and distinguishable patterns can be generated from systems in accordance with embodiments, such systems can also be optimized for both short distance (<1 m) and long distance (>1 m) 3D imaging. For example, certain patterns can be configured to distinguish objects that are a short distance from the device, while different patterns may be configured to distinguish patterns that are at a greater distance from the device. In such embodiments, it would be possible to use a single device to create, e.g., pattern 1 with polarization 1 for short distance

measurement and pattern 2 with polarization 2 for long distance measurement in a single-shot.

Embodiments for Incorporating Metasurface Elements into Imaging Systems

In some embodiments, the integration of multiple metasurface elements (two or more, for example), using methods such as those described in FIGS. 7 and 8 allow the combined system to achieve functionality necessary for practical CMOS camera imaging. Specifically, CMOS cameras, (such as those used in cell phones, computers, tablets etc., for collecting images of a scene of visible light or in the infrared for biometric authentication), require the imaging system to have an increased field-of-view (FOV), independent control of the chief ray angle (CRA) as a function of field height at the CMOS image sensor, and minimal optical distortion of the scene being imaged. These terms will be understood to have a meaning conventional to those skilled in the art. For traditional imaging systems, comprised of refractive lenses, as many as five or six unique lenses must be combined to perform this function. Moreover, implementing one metasurface element in such imaging systems does not provide enough degrees of freedom to adequately control these parameters (CRA, FOV and minimizing distortion). However, by combining multiple metasurfaces, each with a unique and independent phase profile, an imaging system with a wide FOV, controllable distortion and controllable CRA can be realized in accordance with embodiments.

Referring to FIG. 13, a ray-tracing diagram through an exemplary embodiment of a system with two metasurface elements (130 & 132) combined on a single substrate (134) in accordance with embodiments is provided. In various such embodiments the metasurface elements on either side of the substrate are formed to have the same height. (Although not described in detail here, it will be understood that these metasurface elements could be fabricated using methods as described in any of the previous figures and combined using processes such as those described in FIG. 5, for example. In many such embodiments the metasurface elements may be formed from films simultaneously deposited on the two sides of the substrate using a suitable conformal deposition process, such as, for example, low pressure chemical vapor deposition or atomic layer deposition.) In this exemplary embodiment, the two metasurface elements have been configured such that in combination they are able to form a good image across a wide FOV (± 44 degrees in this example, however, it will be understood that this is not a limiting case). Embodiments of such a two metasurface system, as shown, have been surprisingly found to naturally produce focused rays at the plane of the filter and at the image plane that are telecentric (i.e., having 0 degree CRA). In short, while traditional refractive designs require complex, many-element systems to realize such telecentric designs, in accordance with embodiments only two metasurface elements are needed to achieve similar telecentricity. This telecentricity in turn leads to improved optical properties. In particular, the low (e.g., zero or near zero degree CRA) allows for a narrowing of the bandwidth of the optical filter (136) for narrowband applications. In traditional refractive designs, especially for compact mobile applications, CRAs are typically on the order of 15 degrees to 30 degrees. These larger CRAs in turn require the filter bandwidth to be significantly increased allowing for more ambient light to enter the detector. In narrowband applications (e.g., a near IR VCSEL array), such ambient light can be a persistent noise source. Thus, embodiments of a combined

metasurface/filter system such as that shown in FIG. 13 allow for better ambient light performance.

An additional attribute of embodiments of such a telecentric design is that the metasurface system provides a more uniform illumination at the image sensor (referred to by those in the art as “relative illumination”). Embodiments of the metasurface system also provide an additional design variation with respect to traditional refractive lens systems. Typical CMOS image sensors (CIS) require a microlens to be associated with each pixel. Because there is a large variation in CRA across a given sensor plane that is inherent to refractive optical systems, the microlens array on the CIS also requires a complex CRA specification. However, in embodiments of metasurface systems as described herein, the CRA of the microlens array may be configured to be a constant 0 degrees across the CIS allowing for greater simplicity in the design and fabrication of the microlens array. Alternatively, in certain implementations the microlens array may be completely removed from the CIS, saving a process step in CIS production.

Although the embodiments of metasurface systems for use with CMOS sensors have thus far been shown with two metasurface elements on opposite sides of a single substrate, in various other embodiments the two metasurface elements may be disposed on separate substrates. An exemplary embodiment of such a system is illustrated in FIG. 14. As shown, in many such embodiments the metasurface elements are disposed on two separate substrates (138 & 140) with an air gap (142) disposed between the two elements. One advantage of such embodiments incorporating an air gap is that the light rays can be bent farther over a shorter distance, d , in air than in a glass substrate allowing for a greater widening of the illumination zone with a shorter separation between metasurface elements, thus allowing for a decrease in the overall form factor of the metasurface optical system. As shown in FIG. 14, in various embodiments the metasurface elements are disposed on the air gap (142) facing surfaces of the substrates (138 & 140). Such an implementation allows for the protection of the metasurface elements from environmental contamination. Additionally, such embodiments allow for the outward face of the imager side substrate (140) to remain unpatterned, allowing for the direct integration of an optical filter (144) on the substrate. Although in the embodiments illustrated in FIG. 14 the metasurface elements are arranged to face inward with respect to the air gap between them, it will be understood that they may be disposed on either surface of the two substrates. Production of the metasurface system illustrated in FIG. 15 can follow processes described above, such as, for example, those associated with FIG. 7.

Although the above discussion has described metasurface systems configured to provide telecentric optical characteristics, in some instances (e.g., where distortion correction is required) it is necessary to introduce a non-zero CRA. Accordingly, embodiments are also directed to metasurface systems comprising at least three metasurfaces capable of controlling for FOV, distortion and CRA simultaneously. A ray-tracing diagram of an exemplary embodiment of a metasurface system comprising three metasurface elements having unique phase profiles is illustrated in FIG. 15. The introduction of the additional metasurface element or elements, each allowing for the realization of a separate arbitrary phase profile, provides more degrees of freedom to control the path of the light rays as compared to a typical system comprised of an equivalent number refractive elements. For example, to duplicate the optical function of a system composed of three metasurfaces, in accordance with

embodiments, may require 6-7 refractive optical elements in a conventional system. As such the comparative metasurface system may reduce the overall thickness of such an imaging system by at least 50% while achieving equivalent or even improved performance.

Turning to the metasurface system itself, as shown in FIG. 15 such an imaging system may comprise three or more metasurface elements (150, 152 & 154) disposed on two or more substrates (156 & 158). As previously discussed, these metasurface elements can be comprised of any suitable dielectric material, especially those that have minimal absorption at the wavelength of interest. As shown, in various embodiments the first two metasurface elements (150 & 152) may impart a telecentric optical character on the incoming light, while the third metasurface (154) (e.g., the closest to the filter (156) and imager) may impart a further spreading or bending to the light thus imparting non-zero CRA to the light impacting the imager. Although a specific arrangement of metasurface elements and substrates is shown in the system illustrated in FIG. 15, such a diagram is meant to serve as an example and does not limit the current disclosure to this exact system response. Regardless of the specific arrangement of elements, fabrication of such a metasurface element can follow the processes described above, such as, for example, that shown and described in relation to FIG. 6.

Using such three metasurface element systems according to embodiments, it is possible to control CRA and thus minimize grid distortion in CMOS image sensors. For example, FIG. 16 provides a data plot illustrating the resulting control of the CRA as a function of field height at the CMOS image sensor for an imaging system based on the embodiment shown in FIG. 15. This is an exemplary case and the control of the CRA as a function of field height can take on other functional forms rather than the linear case shown below. Similarly, FIG. 17 provides a grid distortion plot for an imaging system based on the embodiment shown in FIG. 15. As shown, embodiments of such an imaging system allow for the minimization of grid distortion to less than 5% over the entire FOV of the imaging system.

Embodiments for Incorporating Metasurface Elements into Imaging/Sensing Systems

Given the advantages described of using metasurface elements according to embodiments on both sensing optics and on projection optics, various embodiments are directed to metasurface systems configured for use in a combined illumination-sensing module. Referring to FIG. 18, a schematic of an integrated illuminator and sensor system according to embodiments is provided. As shown, in such embodiments an illuminator (160) having a plurality of spacers (162 & 162') and metasurface elements (164 and 164'), either alone or in combination with refractive elements configured therewith to provide a specific radiation pattern is used to illuminate some object or scene (166). A sensor (168) (e.g., a CMOS image sensor) with a corresponding metasurface system is used to detect the radiation or from an image of the scene. In such embodiments, the overall system—metasurface elements (162 & 162'), sensor (168), illuminator (160)—may be configured to operate over some particular bandwidth or at a specific wavelength of interest and may be combined onto a single platform (170). The illuminator and sensor metasurface elements (162 & 162') may be configured to work on any polarization of the electric field. Embodiments of such combined systems may be used on a

computer, a cell phone, a television monitor, a wall-mounted unit, a credit card, tablet, mirror etc.

As discussed above in relation to FIGS. 12A to 12C, metasurfaces also allow unique functionality to be imprinted on two orthogonal polarizations. Thus, various embodiments of metasurface illuminator-sensing systems can also be co-designed taking polarization as an additional optimization variable. Referring to FIG. 19, a schematic of an integrated illuminator system that also acts on the polarization of the radiation field is provided. As shown, in one such exemplary embodiment, an illuminator (172) with one or more metasurface elements (174 & 174') along with suitable spacers (176 & 176') an optional refractive elements is used to illuminate a scene or object (178). The metasurface elements in such embodiments have been designed such that for any two orthogonal polarizations of light, two unique and independent radiation patterns can be produced. A sensor (180) with a corresponding set of metasurface elements is used to collect the light reflected from the scene. As shown, the illuminator and sensor metasurface elements have been configured to work cooperatively such that the two orthogonal polarizations used to produce the radiation pattern form two unique images on the sensor. The system—metasurfaces (174 & 174'), sensor (180), illuminator (172)—according to embodiments may be optimized for operation over some bandwidth or at a specific wavelength of interest and may be combined onto a single platform (182). It will be understood that the illuminator and sensor metasurface elements in accordance with embodiments may be configured to work on any polarization of the electric field. The combined system according to embodiments may be used on a computer, a cell phone, a television monitor, a wall-mounted unit, a credit card, tablet, mirror etc.

Embodiments of Metasurface Element Material Systems

As previously discussed, each individual metasurface element in any optical system, whether there be one or many metasurface elements comprising the system, has some specific 2D phase and transmission function, $\phi(x, y)$ and $t(x, y)$, that it carries out. While in general each metasurface element may have a unique distribution of phase and transmission, the nanostructure features that comprise any metasurface element embedded in the same material, with the same base composition, and at a specific wavelength are identical. In most practical single wavelength applications, the transmission is desired to be maximized (near 1) and uniform across the metasurface while the phase need only take on values between 0 and 2π . In summary, for some wavelength of interest, material system (metasurface material and embedding material), fixed thickness and element spacing, one need only find a set of in-plane dimensions of the comprising nanostructure features such that phase delays from 0 to 2π can be imprinted on an incident light field. Thus for various embodiments of metasurface element designs at the fixed material and wavelength conditions, the only variable from design to design is the distribution of those nanostructure features across the metasurface element. Accordingly, various embodiments of metasurface element material conditions suitable for performing specific optical functions over desired wavelength ranges are described. Although the following discussion sets forth embodiments of metasurface elements and systems, and processes for fabricating such metasurface elements and systems. It will

be understood that the following embodiments are provided only for exemplary purposes and are not meant to be limiting.

Referring to FIG. 20, phase and transmission maps for embodiments of metasurface elements comprising silicon pillars embedded in SiO_2 are provided. The top left diagram provides a heat map of transmission as a function of pillar diameter and height, color scale shown to the right. The top right diagram provides a phase map as a function of pillar diameter and height. The bottom left provides a diagram of a line scan of transmission as a function of pillar diameter at a fixed height of 600 nm. The bottom right provides a diagram of a line scan of relative phase as a function of pillar diameter for a fixed height of 600 nm. It will be understood that using these diagrams it is possible to determine a specific set pillar diameters and heights for a specific transmission and phase across all suitable wavelengths in accordance with embodiments, and also specific diameters for a height of 600 nm. In many embodiments, the pillar height may vary from 500 to 1000 nm and the pillar diameter from 100 to 300 nm. In various other embodiments, the pillar diameter may vary from 100 to 200 nm, and the pillar height from 500 to 800 nm. In various other embodiments at a pillar height of 600 nm, the pillar diameter may vary from between 100 to 300 nm. The specific heights and diameters represent a local optimum for the transmission of the elements but other pillar heights may be used in embodiments as required by the design of the specific optical system.

Referring to FIG. 21, phase and transmission maps for embodiments of metasurface elements comprising silicon pillars in air are provided. The top left diagram provides a heat map of transmission as a function of pillar diameter and height, color scale shown to the right. The top right diagram provides a phase map as a function of pillar diameter and height. The bottom left provides a line scan of transmission as a function of pillar diameter at a fixed height of 480 nm. The bottom right provides a line scan of relative phase as a function of pillar diameter for a fixed height of 480 nm. It will be understood that using these diagrams it is possible to determine a specific set pillar diameters and heights for a specific transmission and phase across all suitable wavelengths in accordance with embodiments, and also specific diameters for a height of 480 nm. In many embodiments, the pillar height may vary from ~1 to 500 nm and the pillar diameter from 100 to 350 nm. In various other embodiments, the pillar diameter may vary from 100 to 250 nm, and the pillar height from 150 to 500 nm. In various other embodiments at a pillar height of 480 nm, the pillar diameter may vary from between 100 to 280 nm. The specific heights and diameters represent a local optimum for the transmission of the elements but other pillar heights may be used in embodiments as required by the design of the specific optical system.

Referring to FIG. 22, phase and transmission maps for embodiments of metasurface element comprising TiO_2 pillars in air are provided. The top left diagram provides a heat map of transmission as a function of pillar diameter and height, color scale shown to the right. The top right diagram provides a phase map as a function of pillar diameter and height. The bottom left diagram provides a line scan of transmission as a function of pillar diameter at a fixed height of 975 nm. The bottom right provides a line scan of relative phase as a function of pillar diameter for a fixed height of 975 nm. It will be understood that using these diagrams it is possible to determine a specific set pillar diameters and heights for a specific transmission and phase across all suitable wavelengths in accordance with embodiments, and

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also specific diameters for a height of 975 nm. In many embodiments, the pillar height may vary from 300 to 1000 nm and the pillar diameter from 100 to 350 nm. In various other embodiments, the pillar diameter may vary from 100 to 300 nm, and the pillar height from 300 to 400 nm and/or 700 to 1000 nm. In various other embodiments at a pillar height of 975 nm, the pillar diameter may vary from between 100 to 300 nm. The specific heights and diameters represent a local optimum for the transmission of the elements but other pillar heights may be used in embodiments as required by the design of the specific optical system.

Referring to FIG. 23, phase and transmission maps for embodiments of metasurface elements comprising amorphous silicon pillars embedded in a benzocyclobutane (BCB) polymer are provided. The top diagram provides a line scan of transmission as a function of pillar diameter at a fixed height of 590 nm and element period of 400 nm. The bottom diagram provides a line scan of phase as a function of pillar diameter at a fixed height of 590 nm and element period of 400 nm. It will be understood that using these diagrams it is possible to determine a specific set pillar diameters and heights for a specific transmission and phase across all suitable wavelengths in accordance with embodiments, and also specific diameters for a height of 590 nm. In many embodiments, at a pillar height of 975 nm, the pillar diameter may vary from between 100 to 300 nm. In various other embodiments, the pillar diameter may vary from 100 to 225 nm. The specific heights represent a local optimum for the transmission of the elements but other pillar heights may be used as required by the design of the specific optical system.

Referring to FIG. 24, phase and transmission maps for embodiments of metamaterial elements comprising amorphous silicon pillars embedded in a silicon dioxide are provided. The top diagram provides a line scan of transmission as a function of pillar diameter at a fixed height of 600 nm and element period of 350 nm. The bottom diagram provides a line scan of phase as a function of pillar diameter at a fixed height of 600 nm and element period of 350 nm. It will be understood that using these diagrams it is possible to determine a specific set pillar diameters and heights for a specific transmission and phase across all suitable wavelengths in accordance with embodiments, and also specific diameters for a height of 600 nm. In many embodiments, at a pillar height of 600 nm, the pillar diameter may vary from between 100 to 275 nm. In various other embodiments, the pillar diameter may vary from 100 to 175 nm. The specific heights represent a local optimum for the transmission of the elements but other pillar heights may be used as required by the design of the specific optical system.

In other embodiments tests were also conducted on amorphous-Si metasurface features embedded in SU8, and it was found such surfaces having a pillar height of 675 nm and a pillar diameter from 100 to 300 nm are suitable for use. In addition, amorphous-Si metasurface features in air having a pillar height of 600 nm and a pillar diameter from 100 to 300 nm with an element spacing of 450 nm may be suitable in accordance with various embodiments.

Although specific combinations of metamaterials and embedding materials are described above, it will be understood that similar maps of metasurface feature, transmissivity and phase may be made in accordance with embodiments of the invention.

DOCTRINE OF EQUIVALENTS

Accordingly, although the present invention has been described in certain specific aspects, many additional modi-

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fications and variations would be apparent to those skilled in the art. It is therefore to be understood that the present invention may be practiced otherwise than specifically described. Thus, embodiments of the present invention should be considered in all respects as illustrative and not restrictive.

What is claimed is:

1. An illumination device comprising:

a light source emitting light comprising a first polarization and a second polarization; and

a plurality of metasurface elements arranged in a planar array such that the light emitted from the light source passes through the planar array of metasurface elements,

wherein the plurality of metasurface elements are disposed on a substrate,

wherein the light from the light source having the first polarization which passes through the planar array of metasurface elements produces a first light pattern,

wherein the light from the light source having the second polarization which passes through the planar array of metasurface elements produces a second light pattern, wherein the first light pattern and the second light pattern are complementary dot patterns,

wherein the first light pattern and the second light pattern are projected on the same region of space, and wherein the first light pattern and the second light pattern are different patterns that correspond to different measurements.

2. The illumination device of claim 1, wherein the light source comprises one or more laser light sources.

3. The illumination device of claim 2, wherein the one or more laser light sources comprises a vertical cavity surface emitting laser (VCSEL) array.

4. The illumination device of claim 1, wherein the first light pattern and the second light pattern are located in the far field.

5. The illumination device of claim 1, wherein the first light pattern is denser than the second light pattern.

6. The illumination device of claim 1, wherein the first light pattern is for short distance measurements and the second light pattern is for long distance measurements.

7. The illumination device of claim 1, wherein the first light pattern and the second light pattern are on a square grid.

8. The illumination device of claim 1, wherein the first polarization is orthogonal to the second polarization.

9. The illumination device of claim 8, wherein the first light pattern and the second light pattern are produced simultaneously and the first light pattern and the second light pattern are distinguishable from each other by polarization.

10. The illumination device of claim 1, wherein the light source emits light comprising the first polarization before emitting light comprising the second polarization.

11. The illumination device of claim 1, wherein the light source emits more than two polarizations.

12. The illumination device of claim 1, wherein the light source comprises one or more LEDs.

13. The illumination device of claim 12, wherein the light source emits more than two polarizations at the same time.

14. The illumination device of claim 1, wherein the first light pattern and the second light pattern are generated at the same time.

15. The illumination device of claim 1, wherein the first light pattern has the first polarization and the second light pattern has the second polarization.

16. The illumination device of claim 1, wherein the first pattern is configured for a first distance and the second

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pattern is configured for a second distance such that the different measurements include different distances.

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